NEUROLINGUISTICS

GIOSUÈ BAGGIO

THE MIT PRESS ESSENTIAL KNOWLEDGE SERIES

NEUROLINGUISTICS

GIOSUÈ BAGGIO

THE MIT PRESS ESSENTIAL KNOWLEDGE SERIES

Neurolinguistics

The MIT Press Essential Knowledge series

A complete list of the titles in this series appears at the back of this book.

Neurolinguistics

Giosuè Baggio

The MIT Press | Cambridge, Massachusetts | London, England

The MIT Press would like to thank the anonymous peer reviewers who provided comments on drafts of this book. The generous work of academic experts is essential for establishing the authority and quality of our publications. We acknowledge with gratitude the contributions of these otherwise uncredited readers.

This book was set in Chaparral Pro by New Best-set Typesetters Ltd.

Library of Congress Cataloging-in-Publication Data

Names: Baggio, Giosuè, author.
Title: Neurolinguistics / Giosuè Baggio.
Description: Cambridge, Massachusetts : The MIT Press, 2022. | Series: The MIT press essential knowledge series | Includes bibliographical references and index.
Identifiers: LCCN 2021003137 | ISBN 9780262543262 (paperback)
Subjects: LCSH: Neurolinguistics. | Language acquisition—Physiological aspects. | Brain—Physiology.
Classification: LCC QP399 .B34 2022 | DDC 612.8/2336—dc23
LC record available at https://lccn.loc.gov/2021003137

10 9 8 7 6 5 4 3 2 1

d_r0

Contents

Index

Series Foreword

The MIT Press Essential Knowledge series offers accessible, concise, beautifully produced pocket-size books on topics of current interest. Written by leading thinkers, the books in this series deliver expert overviews of subjects that range from the cultural and the historical to the scientific and the technical.

In today's era of instant information gratification, we have ready access to opinions, rationalizations, and superficial descriptions. Much harder to come by is the foundational knowledge that informs a principled understanding of the world. Essential Knowledge books fill that need. Synthesizing specialized subject matter for nonspecialists and engaging critical topics through fundamentals, each of these compact volumes offers readers a point of access to complex ideas.

Preface

Neurolinguistics is the study of the neural bases of human language. It describes the anatomical structures (networks of neurons in the brain) and physiological processes (ways for these networks to be active) that allow humans to learn and use one or more languages. Neurolinguists may study how particular languages—English, French, American Sign Language, or any other—are processed in the brain, but the goal is ultimately to describe the neural foundations of the human capacity for language as such.

As its name suggests, neurolinguistics is the result of the confluence of neuroscience and linguistics (and several other disciplines, as we will see). Alternative denominations are occasionally used, including "cognitive neuroscience of language" and "neurobiology of language." These labels are largely equivalent, but they emphasize different disciplinary contributions to research on language and the brain—from linguistics, cognitive psychology, or neurobiology. The role of theoretical linguistics is especially important. In order to study *how* the brain represents and processes language, it is often useful to know first *what* is being represented and processed: the formal structures of language at the levels of sound, grammar, and meaning. Neurolinguistics is also closely related to psycholinguistics: the study of language processing and acquisition using the full range of methods of theoretical and experimental psychology.

This book presents the essentials of modern neurolinguistics in a concise, engaging, yet precise form. It is intended primarily for students who are approaching the subject for the first time, for the scientist next door, and for the curious reader who wants to learn what is going on in our head when we acquire, use, or lose our precious language skills. A book of this kind should perhaps keep a safe distance from the frontiers of current research, where theory is put to the test and new results are generated and debated. However, that is not always possible or desirable. Essential knowledge of a field may also include knowledge of what is currently unknown: questions that can be posed confidently and rigorously but cannot yet be answered. It is the responsibility of the author to flag any contradictions between studies, open problems, or current controversies, along with more or less established methods and results.

By writing this book, I have taken on the challenge of deciding what is worth including in it and what is not. I feel therefore obliged to remind readers (and myself) that all scientific knowledge is invariably accessed and presented from some particular viewpoint. This book is no exception. Neurolinguistics is too diverse and too dynamic to capture from a single perspective accurately and completely (and to the full satisfaction of all of its practitioners). All I can do is encourage readers to read this book and then move on to other literature on the topic. The endnotes and the further readings at the end of the book are a good place to start.

1
Historical Introduction

You may not have heard about neurolinguistics before, but you may have heard of neuromania: the recent fascination with anything neuroscientific, especially if accompanied by colorful pictures of the brain "in action."[1] Is neurolinguistics yet another case of neuromania? The short answer is no. So far as one can tell, there is no general fad for language in the brain, and whatever interest there is in it has a long and rich history, predating many developments in cognitive neuroscience (the study of human brain structures and functions) and linguistics (the study of human languages). Research on language in the brain did not need a mature linguistics and a mature neuroscience to get started. In fact, some of the early results in neurolinguistics were instrumental in launching the enterprises of modern linguistics and modern cognitive neuroscience.

A comprehensive history of neurolinguistics has yet to be written, and the origins of the term *neurolinguistics* are still largely undocumented.[2] We may identify three major turning points in the history of speculation and, eventually, research on language in the brain. The first was an empirical breakthrough: the discovery by Paul Broca around 1860 that damage to certain parts of frontal cortex can be accompanied by impaired speech production. The second was a theoretical breakthrough: the proposal by Noam Chomsky and others around 1960 that language is a computational system, formally defined by a finite vocabulary plus a finite set of rules that can be used to generate an infinite set of sentences. The third was a technological breakthrough: the introduction and diffusion around 1980 of methods for in vivo neural measurements in humans and of computers for the analysis and simulation of experimental data.

For better or worse, neurolinguistics today is shaped by those theoretical and methodological choices. We still view language as a computational component of the human mind, we still try to map specific linguistic processes to specific areas of the brain, active at particular times, and we still rely on high-tech measurements and data processing techniques for getting, analyzing, and modeling experimental results. Let us review in more detail how neurolinguistics became what it is.

The Pioneers and the "Diagram Makers"

The idea that speech and language are manifestations of specific capacities of the human organism dates back to antiquity.[3] Descriptions of cases of speech disorders in the presence of traumatic head injuries became relatively more frequent during the Renaissance and early modern period,[4] but it was only in the nineteenth century that a connection between speech disorders and brain lesions was proposed. Franz Joseph Gall (1758–1828) was likely among the first who made that connection explicit. His version of "faculty psychology" was original and influential.[5] Gall argued that the mind is organized as a hierarchy of faculties that (1) are defined by their content (e.g., language, mathematics, music), (2) may be innate in humans or other species, (3) are distinct from and independent of each other, and (4) are localized in parts of the brain distinct from and independent of each other. Gall also noted that there are disorders limited to the "faculty of speech," and that would be impossible if speech was not localized in a particular part of the brain.[6] Gall put forward a simple idea that—in virtue not of being right but of being testable using the methods available at the time—contributed to the emergence of early empirical research on the brain bases of language.

The neurological doctrines circulating in the early nineteenth century included the thesis that any region of the hemispheres can take up a given function (equipotentiality), proposed by Jean Pierre Flourens (1794–1867), and the law of symmetry by Xavier Bichat (1771–1802), implying that if a brain area for speech or language exists, it should be an area in both hemispheres—a "bilateral area."[7] At that time, the question was not so much where precisely in the brain language and speech would localize. The issues were rather whether they would localize at all and whether they would localize in areas of one hemisphere but not the other. Different positions were possible, depending on how one would frame and answer each of those two questions. For example, Jean-Baptiste Bouillaud (1796–

1881), on the basis of analyses of the site of brain lesions in patients with impaired speech, proposed that speech would be localized in the frontal lobe of the brain and concluded that Gall was correct. However, Bouillaud assumed that the human brain was symmetrical. Therefore, if speech could be localized, it would be in the frontal lobe of both hemispheres.

Paul Broca (1824–1880) is usually credited with the discovery that aspects of speech or language are not just localized in the frontal lobe; they are also lateralized to the left hemisphere. Others before Broca, including Marc Dax (1770–1837), had argued that spoken language was in the left hemisphere. But Broca had the data, and he published them at what retrospectively seem to have been the right time and the right place. In a series of communications to the Anatomical Society in Paris in 1861, he described the cases of two patients, Leborgne and Lelong, who had enormous difficulty producing speech, except for a very few "articulated sounds that are always the same and always produced in the same manner."[8] Broca reported that both men could "hear and understand" spoken language, could easily produce "vocal sounds" and move their tongues and lips more "expansively and energetically" than is required to articulate speech, and were "fully intelligent."

These observations allowed Broca to conclude that, in both patients, what appears to be compromised are not input (auditory) and output (motor) systems but rather the "faculty of articulated speech."[9] Broca examined Leborgne and Lelong's brains on autopsy and he observed that both had extensive damage to the left frontal cortex in a region he referred to as the "third frontal convolution" (figure 1). "The integrity of the third frontal convolution, and perhaps of the second, seems indispensable to the exercise of the faculty of articulate language. In Leborgne and Lelong the lesion lies behind the middle third, opposite the insula and precisely on the same side" (the left), Broca concluded.[10]

Did Broca aim to test the hypotheses that speech is localized in the frontal lobes *and* lateralized to the left side? And did he actually claim he had evidence for both? This is not the place for historical controversy, but it may be worth mentioning that Broca's views evolved in the years between 1860 and 1865. Broca had never been a fervent supporter of the localization and lateralization theses: his "conversion" was "slow and effortful,"[11] and he detached gradually from the doctrines of equipotentiality and symmetry of Flourens and Bichat. For him, the Leborgne and Lelong cases were one opportunity to test Bouillaud's hypothesis that speech resides in the frontal lobes or the prediction that a patient who was unable to speak would present frontal lesions. It was not an issue whether the faculty of language could be localized frontally or at all, or whether the faculty of speech was left lateralized. Broca did claim that the frontal injury in Leborgne and Lelong was "the cause of loss of speech," confirming Bouillaud's view.[12] However, he was also careful to avoid general claims about the language faculty, and he only eventually accepted the possibility that speech could be lateralized to the left hemisphere.

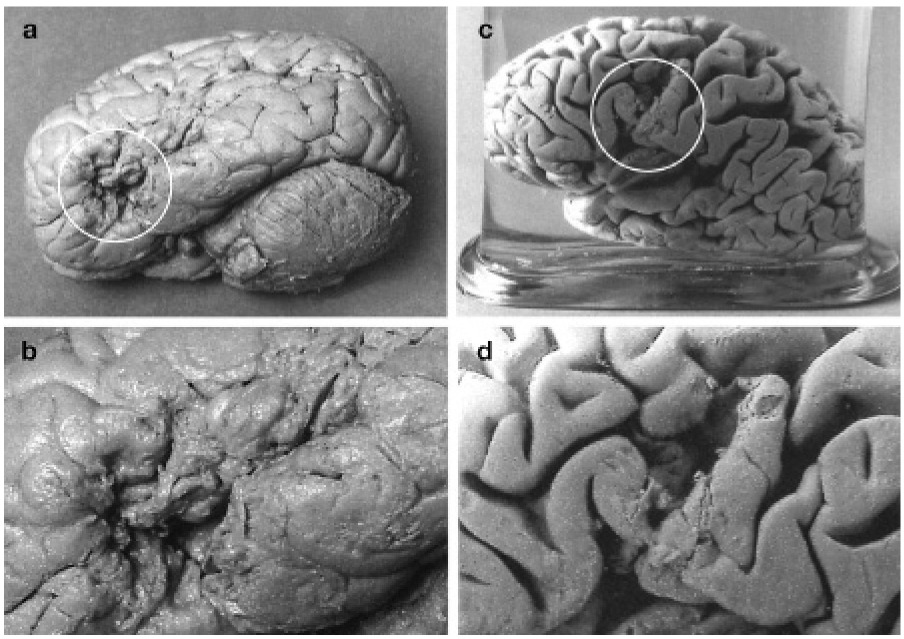

Figure 1 Photographs of the brains of Leborgne (a, b) and Lelong (c, d).

It was the task of the next generation of researchers to study other components of speech and language in the brain. Carl Wernicke (1848–1905) could examine cases of patients with different symptoms and lesions compared to Broca's patients, who presented limited ability to produce speech but had good comprehension—what became known as "Broca's aphasia." Wernicke described patients whose speech is rapid and effortless but may lack relevance and coherence, and whose speech comprehension abilities are impaired; this pattern became known as "sensory aphasia" or "Wernicke's aphasia." Wernicke also described patients who could still articulate and understand speech, but where repetition was impaired

—"conduction aphasia." As the title of his landmark 1874 publication indicates (*Der aphasische Symptomencomplex*), Wernicke was aware that there was a "symptom complex" of aphasia, not a single or uniform disorder, awaiting explanation.[13] Wernicke's answer was the first model of language in the brain (figure 2), which was bound to have a huge impact, lasting to this day.

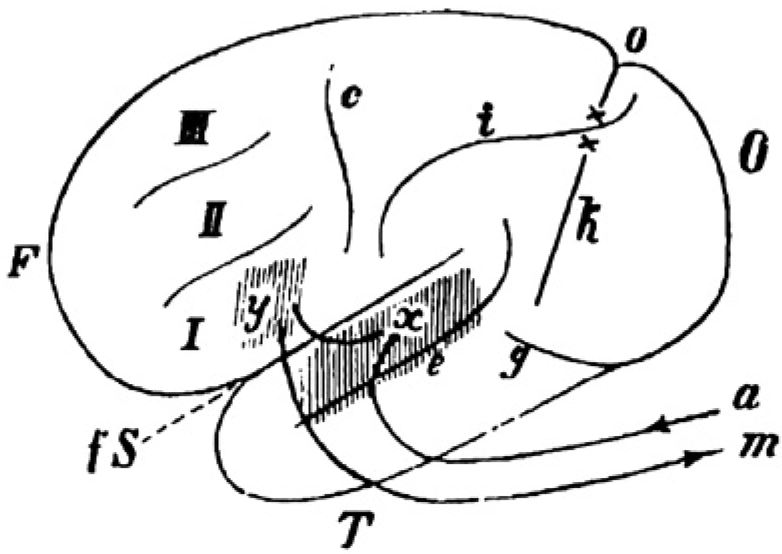

Figure 2 Wernicke's diagram model of speech processing in the brain.

For Wernicke, spoken language is supported by a set of interconnected centers: gone is the notion that a single "grand region" of the cortex is the basis for a single "mental faculty."[14] These are a center for "motor images" of words, corresponding to the left frontal cortex, as identified by Broca (figure 2, y), and a center for "auditory images" of words in the posterior superior temporal gyrus (figure 2, x). These centers are connected anatomically (Wernicke also put forward a hypothesis concerning the fibers that would provide the link), they receive inputs from auditory nerves (a), and they relay outputs through motor nerves (m).

These centers and connections can be damaged, and each lesion can cause a specific speech or language disturbance.[15] Disruption of acoustic nerves (a) can cause deafness, while disruption of motor nerves (m) can cause paralysis of speech—not aphasic symptoms. Damage to the motor images center (y) can cause Broca's aphasia, where speech is not correctly articulated. Damage to the auditory images center (x) can lead to sensory aphasia, where speech cannot be decoded and made intelligible. Finally, damage to links between the motor and auditory centers can cause conduction aphasia: if the transfer of information between these centers fails, the patient has difficulty expressing the intended thoughts or repeating words that are spoken to them.[16]

Wernicke is the last pioneer of neurolinguistics and the first of the so-called "diagram makers," as Henry Head (1861–1940) would later call the early aphasiologists.[17] His everlasting contribution is the first connectionist model of language in the brain, with both predictive and explanatory power. Today we view the language system effectively as a network of brain regions that collectively perform certain functions or tasks and where individual regions participate in several functions or tasks. Wernicke was responsible for an early shift toward this view, away from strict localization. He also believed that concepts would be distributed across the brain and so would not be localizable, as motor and auditory centers were. Ludwig Lichtheim (1845–1928) adopted and developed this insight.[18]

Versions of Wernicke's model—also known as the Broca-Wernicke-Lichtheim model or the Broca-Wernicke doctrine—were advanced until recently. Sometimes these were genuine and substantial updates of the model, as in a proposal by American neurologist Norman Geschwind (1926–1984).[19] But others were caricatures popularized as "the" standard neurological model of language. A recurrent simplification is that Wernicke's auditory images center is the brain's speech comprehension center, while the motor images center is the brain's speech production center. His view was instead that the two centers (plus the distributed concept centers) contribute different processes to speech production, which is always the result of the joint operation of several nodes in the network.

Much modern aphasiology was a sustained attempt at testing and revising Wernicke's model—a series of footnotes to Wernicke, one might say. But was nineteenth-century research on the brain bases of speech really neurolinguistics, or was it "just" neurology or aphasiology? The ideas and results of Broca, Wernicke, and others were not yet neurolinguistics, but they provided the initial impulse to its development before they could be integrated with linguistic theories during the twentieth century. Early attempts at integration were made with structuralism: the theory of language as a system of signs, such as sound-meaning pairings, formulated by Swiss linguist Ferdinand de Saussure (1857–1913).[20] But a leap forward came only with generativism, the theory that language is a computational system—a finite set of rules for generating sentences—made precise by linguist Noam Chomsky.

Neurolinguistics and the Cognitive Revolution

At the turn of the twentieth century, it must have seemed clear to discerning linguists that language was in some sense "in" the brain. For example, Saussure remarked in his *Cours de linguistique générale* (1916) that language exists as a "sum of imprints deposited in each brain."[21] But recognizing that it is so is an altogether different matter from proving it with the methods of the sciences. Broca, Wernicke, and others had shown that productive and receptive speech had their seat in the brain. But they left the much bigger issue of the localization of the general faculty of language untouched, as Broca readily conceded as soon as 1861. Approaching that question required a better understanding of language, specifically of syntax and semantics, than was available for most of the nineteenth century. Aphasiology and the neurology of language had to morph into neurolinguistics proper.

How that happened is a complex and untold story, but there were three critical junctures. First, John Hughlings Jackson (1835–1911), among others, noted that identifying the anatomical causes of a symptom is not the same as localizing a function. This paved the way for inclusion of psycholinguistic methods into aphasiology, in particular of controlled lab experiments outside the clinic. Second, it became clear that speech and language were about phrases and sentences, not just words. Heymann Steinthal (1823–1899) was among the first to distinguish word-level (aphasia) and sentence-level (acataphasia) disorders.[22] Adolf Kussmaul (1822–1902) introduced the term *agrammatism* for a spectrum of deficits relating to the syntactic organization of sentences. Third, linguistics began to penetrate aphasiology, for example, through the work of Roman Jakobson (1896–1982), who argued that output deficits in Broca's aphasia reflect the breakdown of the syntagmatic (sequencing) axis of language, while input deficits in Wernicke's aphasia reflect the breakdown of the paradigmatic (lexical selection) axis.[23] Jakobson's linguistic approach to aphasia, in particular its structuralist emphasis on language as a formal system, today seems much more sophisticated than that of most of his contemporaries.[24] But it is only with subsequent developments in linguistics, in the mid-twentieth century, that neurolinguistics could start to get a firmer grasp of syntax and semantics in the brain.

One breakthrough came in the 1950s in a series of papers by Chomsky, where he combined techniques from logic and mathematics with earlier analyses by Zellig Harris (1909–1992) on so-called syntactic transformations, such as the rules that transform active sentences ("The boy ate the candy") into passive ones ("The candy was eaten by the boy"). Chomsky proposed transformational-generative grammar as a theory of linguistic syntax—eventually, just generative grammar (GG). A GG is unlike the grammars we were taught in school. At bottom, a GG is a procedure for constructing sentences.[25] Initially, it was a finite set of rewrite rules; for example:

A. S → NP VP
B. NP → Det N
C. VP → V NP

where S is a sentence, NP a noun phrase (e.g., "the boy"), VP a verb phrase (e.g., "ate the candy"), Det a determiner, N a noun, and V a verb (rewrite rules where words are the terminal symbols may also be included—for example, Det → the, N → candy, N → boy, V → eat). A rule X → Y allows us to rewrite the left-hand symbol X as the right-hand symbol Y, for example, a sentence S as a sequence of a noun phrase and a verb phrase [NP VP], applying rule A. Let us gloss over the question whether this toy GG generates all and only the well-formed sentences of English (it doesn't), and consider how it generates the syntactic structure of "The boy ate the candy." We start from the symbol:

S

We apply rule A to S, and we obtain:

[NP VP]

Then we apply rule B to NP and rule C to VP, and we get:

[Det N] [V NP]

Finally, we apply rule B to NP again:

[Det N] [V [Det N]]

Each application of a rewrite rule leaves a trace in the final structure: the brackets [ ] are the fingerprints of the generative procedure. The final structure is *hierarchical*, as each phrase (e.g., [Det N]) can be embedded inside some other phrase (e.g., [V [Det N]]). The procedure is *recursive*, as each rule can be applied to the result of applications of itself or other rules: rule B is used twice, recursively, in the derivation. This formal approach can be extended beyond syntax—for example, to phrasal and sentential semantics.

Before the advent of generative grammar, logicians and philosophers were already developing formal systems and languages for various purposes. But it was Chomsky, in the 1950s, who designed formal systems that could be used to capture key properties of natural language syntax (it had not been tried before) and who, in the 1960s, wed the formal approach to a view of "generative grammars as theories of human linguistic competence" (earlier research on formal languages was decidedly antimentalistic).[26] This blend of formalism and mentalism in linguistics contributed to the emergence of the so-called computational theory of mind and eventually cognitive science: the interdisciplinary study of mind, embracing areas of philosophy, psychology, linguistics, and neuroscience. Some have dared call these developments a "cognitive revolution,"[27] while others have argued that it was in fact a return of linguistics and psychology to their (European) "mentalistic roots."[28] Many would agree that the computational theory of mind has changed how scholars and educators understand human cognition—its structure and its capabilities.[29]

What did that imply for neurolinguistics? In essence, it brought to completion and fruition the movement toward a mature neurolinguistics initiated several decades earlier. First, innovative experimental techniques were developed by psycholinguists, including George A. Miller (1920–2012) at Harvard, a close associate of Chomsky. Aphasiologists gradually adopted those new methods, for example, at the Boston Aphasia Research Center, established and led by Norman Geschwind and Harold Goodglass (1920–2002). Second, the emphasis shifted from words to sentences and from stimulus-response associations to the formal, hierarchical structure of complex cognition and behavior. Third, linguistic theories were actively used as predictive and explanatory tools in studies of language processing and acquisition. For the first time, in the late 1960s and early 1970s, it became possible to ask questions on the "psychological and neural reality" of specific operations posited by theoretical linguistics (such as transformations).[30]

In the 1960s, the term *neurolinguistics* emerged,[31] the first "grand synthesis" of linguistics and neurobiology was proposed by Eric Lenneberg (1921–1975),[32] and the community began to organize itself, with the founding of societies and the journal *Brain and Language* in 1974. The editorial of the first issue, by Harry Whitaker, reads: "The nascence of such fields as physiological psychology or, more recently, neurolinguistics, is simple evidence that many recognize the value of an enlarged conceptual base and new analytical tools in the complex area of language, the brain, and the interrelationships between them."[33] With much foresight, Whitaker saw that a radical technological transformation was coming to the field of neurolinguistics: "the development of new techniques and refinements of old ones which presage even more remarkable advances and insights in the future: event-related slow potentials of the brain, three dimensional X-ray analysis of brain tissue, mathematico-logical systems for linguistic analysis and psycholinguistic analyses of aphasia" are only a few examples of the technologies and methods that would shape neurolinguistics as we know it today.

Neurolinguistics and the Technological Revolution

During the 1970s, researchers in neurolinguistics began to use methods for recording electrical brain activity online in healthy participants (electroencephalography, EEG) and for measuring variations in the electrical currents (event-related potentials, ERPs) generated by the brain in response to external events (e.g., stimuli) or internal actions (e.g., decisions). In the 1980s and 1990s, scholars in Europe and the United States showed that some of these signals changed rapidly and in systematic ways in response to experimental manipulations of linguistic stimuli—for example, if a sentence ends with a word that is unexpected in that context (". . . we eat bread and socks") or if a sentence contains a grammar error (". . . they throws the toy on the ground").[34] It turned out that the brain produces different responses, at different times, to anomalies of linguistic form and content. This led researchers to speculate on the separability of systems for meaning and grammar in the brain: a topic of considerable debate even today (more in the next three chapters). EEG and ERP methods were extensively tested, perfected, and applied over the past three decades to several questions concerning language processing and acquisition in various populations and age groups. Other techniques for tracking neural responses to language have been adopted by many neurolinguists, including magnetoencephalography (MEG), electrocorticography (ECoG), and other direct measures of electrical brain activity, such as intracranial EEG (iEEG). These methods can provide detailed information on the timing of linguistic processes and, increasingly, on the underlying physiological mechanisms.

In the 1980s, it became possible to detect changes in the concentration of glucose or oxygen and hemoglobin in brain tissue with positron emission tomography (PET) or functional magnetic resonance imaging (fMRI) and to build maps of active brain regions, superimposed on anatomical three-dimensional scans. With the wider diffusion of these tools taking off in the mid-1990s, neurolinguists could finally "see" how the human brain responds to language, in vivo, noninvasively, and in exquisite spatial and temporal detail.[35] Reassuringly, the first imaging experiments on language comprehension were compatible with the neurolinguistic lore. For example, an fMRI study by Marcel Just and colleagues, published in 1996 in the journal *Science*, showed that Broca's area (in the left inferior frontal cortex) and Wernicke's area (in the left posterior superior temporal cortex) were active as a function of the syntactic complexity of the sentences that participants were processing. Such activation was overall stronger in the left hemisphere in these two regions than in their right-hemispheric counterparts.[36] Hundreds of fMRI experiments on language have been published since then, and fMRI remains a prime imaging method in many areas of human neuroscience, including neurolinguistics.

Most EEG and fMRI data sets contain millions of data points: millions of different numbers organized into a matrix or a higher-dimensional structure. Collecting and analyzing (and storing) so much data would be impossible without digital computers. The computer is the protagonist in the recent history of neurolinguistics: as a mathematical concept, it inspired the computational theory of mind, and as a physical object, it enabled experimental research on larger samples, longer laboratory sessions, and better-controlled conditions than was previously possible. Neurophysiology and computing have changed the whole research pipeline in neurolinguistics and, more generally, in psychology and neuroscience: data and ways to handle them today shape every step of inquiry, from study conception to publication.

The technological revolution has also changed how theory is developed and tested: it has become common to construct computer models of various linguistic processes, such as a program that learns to assign a syntactic structure to a given sentence, and to use such models to generate or "simulate" data that are then compared to experimental observations. The social structure of the field has evolved too: research on language in the brain today is performed by large, interdisciplinary teams of highly skilled specialists, working in different countries on short-term grants. Broca, Wernicke, and arguably even Jakobson would find today's computer-aided neurolinguistics alien in comparison to the solitary, low-tech efforts that characterized their daily work.

> The computer is the protagonist in the recent history of neurolinguistics.

We have come to the conclusion of our brief history of neurolinguistics. A few remaining gaps will be filled later. Consider now the picture that emerges from the narrative. The "neuro" aspect is a thoroughly experimental enterprise that uses the latest technologies to measure brain activity and synthesize these recordings in the form of models: spatiotemporal representations of activation waves in the brain. The "linguistics" aspect is an intensely formal project that produces logico-mathematical accounts of how complex linguistic structures can be derived from simple elements. Can these very different modes of inquiry and spheres of reality—biology and mathematics, matter and mind—be integrated instead of only juxtaposed? The adventure of contemporary neurolinguistics begins here.

2
Mapping Language in Brain Time

Language happens instantaneously. We do not experience time lags or gaps between intending to say something and actually saying it, and between hearing or seeing what our conversation partner says and understanding its meaning.[1] Yet everything that happens, happens in time. Producing and understanding language are no exception. However, language happens at timescales much smaller than what human observers normally experience—what we may call “brain time.” In the brain, multiple complex processes are typically carried out in parallel in timescales on the order of 10 to 100 milliseconds (1/100 to 1/10 of a second). That is why it is important to use methods with sufficient temporal precision to study language as it unfolds in real time.

What do neurolinguists want to know when they try to determine the timing of the neural processes underlying language comprehension and production? We can answer this question using an analogy: a sprint race. Sensors or cameras at the finishing line measure the time it took for each athlete to get there from the firing of the starter’s gun. That tells us how fast each runner has been. But what we (also) want to know is the order in which the runners have reached the finishing line. To do that, we have to compare individual times so as to rank them from first to last. Even if we add, say, 1 second to each athlete’s time, the order of arrival remains exactly the same, though individual records could of course change. We apply a similar logic when we study language in brain time. The relative temporal order of linguistic processes is more important than their absolute timing, although information on the latter is needed to draw conclusions on the former. Did process A happen before B, or the other way around, or did they occur simultaneously? In neurolinguistics, we often try to answer questions of this general form (examples to follow). But why do we care?

The human brain is quite a fast computing machine, fast enough for most tasks of adaptive or survival value for the organism—else presumably we would not be around to tell this story. But digital computers are often faster in many tasks amenable to algorithmic analysis, including aspects of language processing.[2] What is special about the brain, and about the human brain in particular, is its architecture: how different processes are woven together—sequentially or in parallel—so that information is handled efficiently and can be transferred or translated seamlessly across modules or components of the system, from perception to action and vice versa, often via commonsense reasoning. If I told you, "She poured herself a cup of coffee and left the room," you would probably understand that she has left the room *with a cup of coffee*. Not only are you translating speech signals into meaning; you are also inferring the intended meaning, even though I never explicitly said that the cup left the room with her. Humans are remarkably adept at interpreting the given input actively, even creatively, but machines much less so. This cannot be a matter of speed, or computing power, or data, since computers can harness more of those than we do. It is a consequence of the architecture that evolution has endowed our brains with. Obtaining experimental data that can reveal the actual sequence of linguistic processes is one way of peering into that most elaborate architecture.

Event-Related Brain Potentials

Until not too long ago, researchers could study the relative timing of cognitive processes only indirectly by measuring behavioral response times —an industry known as mental chronometry.[3] One could show, for example, that reading sentences involving grammatical transformations (active to passive, affirmative to negative) takes about 1 second longer than reading affirmative active sentences.[4] This may tell us something about language (some structures actually are more complex than others) or about mental grammars (that they deploy something like transformations, for example). But it does not tell us much about the architecture of language in the brain. In spite of the ingenuity and elegance of much chronometric research, behavioral response times are not the right tools to address core neurolinguistic questions.

Event-related brain potentials (ERPs), encountered in chapter 1, are a much more sensitive and powerful method for addressing questions about the time course of linguistic processes. In a typical ERP experiment, carefully controlled auditory or visual stimuli (speech or text) are presented to a participant while the EEG is being recorded from a number of electrodes placed on the surface of the person's head, usually in an elastic cap (figure 3). The EEG is amplified, and noise is removed by averaging over multiple repetitions of the same stimulus events. This procedure reveals voltage changes in the EEG that occur at fixed time latencies from the onset of the stimulus—so-called time-locked activity: these voltage changes are ERPs. The ERP signal appears as a sequence of peaks and troughs (figure 3), called ERP components,[5] whose temporal latencies (in milliseconds, abbreviated as ms or msec) and amplitudes (in microvolts, abbreviated as μV) are the main variables of interest. One could then ask whether modifying linguistic stimuli in particular ways (e.g., by introducing a grammatical error or an unexpected word in context; see chapter 1) affects the latency or amplitude of some ERP components. By means of clever experimental designs, one can isolate the ERP components that vary in response to changes in one aspect of the stimulus but not others. If one can also associate a single linguistic process to the stimulus manipulation used, then one can suggest a link between that linguistic process and the observed ERP signals, effectively "mapping" language in brain time.

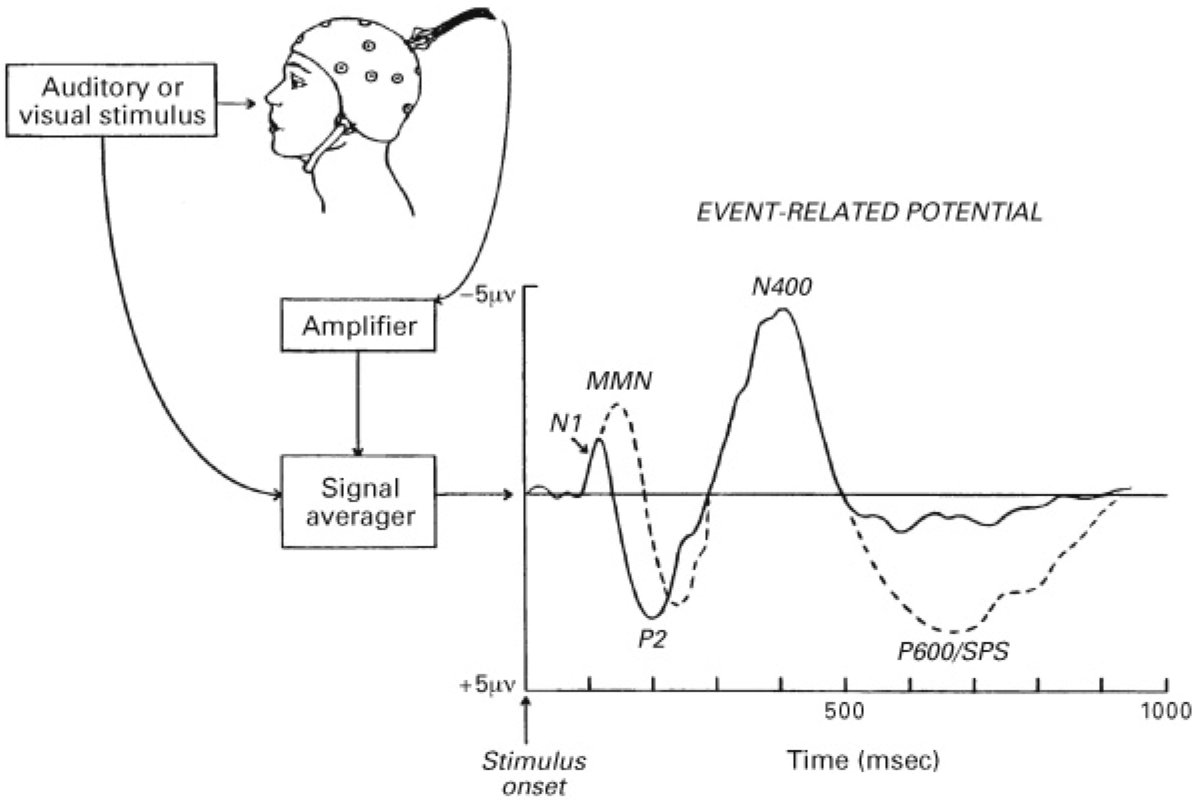

Figure 3 Event-related brain potentials (ERPs). Reprinted with permission from L. Osterhout, J. McLaughlin, and M. Bersick, "Event-Related Brain Potentials and Human Language," *Trends in Cognitive Sciences* 1 (1997): 203–209.

Much of what we know on the timing of linguistic processes comes from ERP experiments, although many more techniques are available now for measuring brain activity as it unfolds in real time—MEG, ECoG, iEEG, and others. Furthermore, we can now apply advanced data analysis procedures even to EEG data, extracting information that would be effectively lost in the ERP averaging process. In this chapter, we focus on ERP studies on language in brain time, with an eye to some of these other techniques. The reason for this choice is not—I hope—methodological conservatism ("old methods are better than new ones") but the fact that ERP research has produced a mass of results that still await explanation. Sometimes in science we like to move on not because we have explained "old" results, but because new technologies or methods become available.[6] Yet new tools won't make old (or different) data go away.

Processing Speech Sounds

All forms of language—spoken, written, signed, and so on—have a physical basis, and sound is the physical basis of spoken language. Speaking involves modulating sound waves with our vocal apparatus, and conversely understanding speech requires analyzing the complex, noisy sound waves that hit our eardrums.[7] Phonetics investigates the physical basis of speech, namely the acoustic signal as such (sound waves) and the organs that subtend its production and perception. Phonology describes instead the abstract categories which constitute the basic repertoire of spoken sounds of human languages. For example, the syllable /ba/, from the point of view of phonology, is the same unit of speech, irrespective of the volume, pitch, tone, and rate that different speakers, in different situations, may use to produce it. What matters is the invariant features of /ba/ and how these relate to the characteristics of other (phonetically similar) syllables, such as /pa/. Neurolinguistics is interested in two issues: (i) How does the brain transform sensory representations of sound into perceptual representations of speech, i.e., how does it translate phonetics into phonology? (ii) Are representations of speech sounds in the brain abstract and categorial?

Three ERP components are relevant to the study of phonological processing in the brain: the auditory N1 and P2 (or N100 and P200, so called because they appear as, respectively, a negative peak around 100 ms from stimulus onset and a positive trough around 200 ms from stimulus onset) and the phonological mismatch negativity (PMN, or mismatch negativity, MMN, shown in figure 3).[8] In general, these components reflect preattentive responses: they are modulated by properties of the stimulus even if we are not paying attention to them. The N1/P2 complex is automatically evoked by all sounds, not just speech sounds. However, the amplitude or latency of the N1 (and of its magnetic counterpart, M1) can also be modulated by features of speech. For example, these early components in EEG or MEG signals are sensitive to vowel category independent of pitch or speaker variation,[9] and they may be modulated by feedback of the speaker’s own voice during speech production.[10] These results show that speech perception is category based from the outset: the brain very rapidly extracts salient phonological information from the speech input, probably in less than 200 ms. Also, at such an early stage of speech perception, the brain can tell whether the inputs that the auditory system is receiving are self-generated or externally generated—whether one is hearing one's own voice or another's voice speaking.

We also know that speech processing—like virtually any other perceptual and cognitive process in the brain—is predictive: the brain is never waiting passively for new input to arrive; it is always trying to anticipate the next stimulus in a sequence and the outcomes of the organism's own acts. Moreover, there are different kinds of prediction, which may engage different predictive systems in the brain. Predictions may rely on knowledge of regular phonological properties of speech or, equivalently, acquired phonological patterns in a given language, which would make certain sequences of sounds more likely to occur, that is, more predictable, than others. Based on this prior knowledge, as speech unfolds in time, the brain actively generates a model of the current stimulus and its possible continuations. These internally synthesized representations of speech are then compared with the actual input, and the error (the difference between the internal model and the input) is fed back to the system for online correction.[11]

The brain may also generate a momentary model of a stimulus sequence, focusing on some salient acoustic or phonological dimensions while ignoring others. Studies on the MNN suggest that the brain identifies abstract speech sounds in artificial sequences, for example, in series of acoustically nonidentical /t/ sounds, and that it can quickly determine, within 300 ms at most, whether novel sounds in the same sequence do (or do not) belong to the same phonological category; for example, the consonant /t/.[12]

Finally, predictions may propagate across sensory modalities. In face-to-face conversation, we normally see articulatory movements on the interlocutor's face, and we may even see them before we hear the speech sounds the interlocutor is producing. The time lag may be tiny, but it is sufficient for the brain to use this visual information to try to predict speech. In ERPs, this is reflected by changes to the N1/P2 evoked by speech, when speech can be predicted based on the preceding visual input.[13] Speech processing is categorial, predictive, and multimodal from the get-go.

Processing Word Meaning in Context

Going back to the issue of the temporal order of linguistic computation in the brain, it is not surprising that in speech comprehension, the first processes to leave traces on ERP signals are phonological. But what is next in line? Individual sounds are strung together until a word is recognized; for example, the sequence of phonemes /bə'nɑːnə/ is recognized as the word *banana*. Within the stream of sounds that constitute a phonological word, there is always a point (also called its recognition point), which may precede the last phoneme, when it becomes clear which word it is (I discuss word recognition in reading in chapter 7). When a word is found in the speech stream, very different processes kick in: the perceptual phase of language processing gives way to its cognitive phase. Lexical information is activated about the internal structure of the word (Is it the singular *banana* or the plural *bananas*?), its syntactic properties (Is it a noun or an adjective?), and its meaning. Usually this information is accessed very quickly and in parallel. However, the goal of speech and language processing is to assign meaning to the input, not to carry out a complete, detailed grammatical analysis—unless the situation requires it. The brain attempts to compute meaning as soon as possible, even though only partial information is available about the internal structure of words and of the sentences in which they occur.

The amplitude of the N400 ERP component (figure 3) is sensitive to how well the meaning of a particular stimulus word suits the preceding linguistic or nonlinguistic context. Suppose you are hearing or reading the sentence fragment, "He spread his warm bread with," followed either by a word that fits the context well, such as *butter*, or by a word that does not, like *socks*. The amplitude of the N400 would be larger in the latter case, as if the brain had to do more work to process the less appropriate or less expected word. But the N400 is not the brain's own internal warning signal: it is not an all-or-none response produced by the detection of an anomaly or violation of semantic appropriateness or expectancy. The N400 is rather a graded neural response to continuously varying properties of the stimulus, such as semantic appropriateness, expectancy, and others.[14] For example, given the fragment, "He took a sip from the," the amplitude of the N400 would be larger for *transmitter*, a bit reduced for *waterfall*, and smaller still for *bottle*.[15] In general, the N400 does not seem to care about whether the preceding context is structured in any way—logically or syntactically, say—or even about whether the context is speech or language or something else entirely, so long as semantic information is present in it and is made available for processing incoming words. Research on the N400 has shown that meaning is activated as soon as a word is recognized and that this activation process is sensitive to information already available from the context.

> The goal of speech and language processing is to assign meaning to the input, not to carry out a complete, detailed grammatical analysis.

From the point of view of linguistics (semantics), the gradedness of the N400 response presents a puzzle: What is word meaning if it is activated dynamically, perhaps even probabilistically, and as a function of fit with the context? I will not try to answer this question here, except to note that not all semantic theories can explain this fact equally easily. Neurolinguistics admits that discoveries at the level of brain structures and processes can inform linguistic theories and perhaps assist in choosing among alternative theories.[16] In neurolinguistics itself, the debate is what exactly the N400 reflects: Does it only reflect activation ("semantic retrieval," to use the computer metaphor) or also integration of the word's meaning into a representation of the context?[17]

No single measure of human brain activity provides full access to the underlying neural dynamics. This implies that the N400 can offer only a partial picture of processing meaning in brain time. MEG research shows that the N400 is one result of the activity of several distinct neural sources and subcomponents: what appears simple is in fact complex.[18] Moreover, other reliable neural responses, related to semantics, have been observed that precede or follow the N400 in MEG or EEG signals.[19] The emerging picture suggests that from the moment of word recognition for at least half a second, the brain is engaged in activating meaning, relating it to stored knowledge, and determining what a word refers to in the given context.

Processing Grammatical Structure

EEG and MEG studies of spoken language comprehension show that phonological properties of words are processed very rapidly in the service of lexical recognition. As soon as the brain identifies each word in the input, it proceeds with activating their meaning and using that information to infer the intended message. These semantic processes tend to be complex, slower, and more diluted in time, and they engage different mechanisms in multiple brain regions. What is the role of grammar in this? Linguists believe that the grammar of the language—morphology and syntax—is necessary to compose meanings in the right way. Even if we soften this claim somewhat and admit that grammar is at least often useful to recover meaning, the question remains: How do grammatical processes unfold in brain time?

The first grammar-related ERP wave to be identified was the P600 (P stands for the positive polarity of its peak, and 600 is its approximate peak latency in milliseconds) or SPS (syntactic positive shift; figure 3).[20] A larger P600 was observed in response not just to grammatical anomalies or violations (structures that no grammar would generate or license) but also to syntactically ambiguous or complex sentences at specific points during processing. A fragment like, "The broker persuaded," can continue either as a simple active sentence ("The broker persuaded the man to sell the stock") or as a reduced relative clause construction ("The broker persuaded to sell the stock was sent to jail"), where the relative pronoun *who* and the auxiliary verb *was* are not expressed ("The broker *who was* persuaded to sell the stock was sent to jail"). Syntactic analysis in the latter case seems more difficult to pursue, and indeed a larger P600 is found in response to the infinitival marker *to*, comparing similar constructions with transitive or intransitive verbs (e.g., "The broker persuaded to" vs. "The broker hoped to"). One can complete these two fragments in a way that vindicates the reduced relative analysis: "The broker persuaded to sell the stock was sent to jail" versus "The broker hoped to sell the stock was sent to jail," which is obviously ungrammatical. A larger P600 was observed at the word at which it becomes clear that the second sentence is not well formed: readers could try guessing which word it was as an exercise.[21] The P600 seems to reflect the brain's own internal perceptions of well-formedness (what is grammatically viable and what is not) and of grammatical complexity (what requires extra processing). These perceptions change dynamically as the stimulus sentence unfolds.[22]

Is it at all correct to talk about "internal perceptions" of grammar? Isn't the grammatical structure of words and sentences given with the input? Grammaticality is defined as an objective property of sentences. In theories of syntax, once the formalism and the basic representations are fixed, one can (objectively) determine whether a given sentence is grammatical. The sentence, "The broker hoped to sell the stock was sent to jail," should come out ungrammatical on any adequate theory of syntax. But the brain does not always agree with the way syntactic theory classifies sentences: it may perceive certain grammatical sentences as ungrammatical, and vice versa. That is partly because the form and meaning of sentences are not given with the input; what is given is just sound waves hitting the tympanic membranes or photons bouncing off the retinas. Grammatical structures are assigned by the brain in response to sensory stimuli with a specific goal: recovering the most plausible intended meaning.

A working hypothesis is that the brain *can* generate word-by-word interpretations of the input string, guided at each step by knowledge of grammar. However, it does not necessarily do that as a default. Instead, the system processes each content word (nouns, verbs, adjectives, and so on) in an effort to reconstruct the most probable meaning of the clause or sentence, given background or contextual information. This process is partly manifested by the N400: content words that do not fit semantically with the context, prior knowledge, or the preceding sentence fragment can increase the N400's amplitude. The brain may then compare the resulting semantic representation to an analysis of the input string, checking whether the reconstructed meaning is consistent with the syntactic structure of the sentence. If it isn't, a P600 effect may be elicited. It follows that a P600 can also be produced when the sentence is grammatically well formed but inconsistent with the meaning suggested by the content words. For example, "The hearty meal was devouring the kids" is an acceptable English sentence: no theory of syntax would stamp that as ungrammatical. The problem here seems to be that the verb *devouring* does not sit well with the inanimate subject: hearty meals don't devour kids or anything else. If the brain took syntax as a given, then it would process this sentence as semantically anomalous, and we should see an N400 at *devouring*. It turns out that a P600 is found instead.[23] Why? What may be happening here is that the brain computes a probable meaning for the sentence where the meal is devoured by the kids, temporarily ignoring grammatical cues (the suffix "ing"): this occurs in the N400 frame. However, the brain would soon detect a conflict between this "attractive interpretation" and the actual grammatical structure of the input; this is reflected by the P600. A process of semantic construction by association (N400) is followed by a kind of internal reality check that effectively uses the grammatical information derived from the input (P600).

If this view is correct, grammar does not necessarily guide semantic interpretation step by step: syntax may not precede and feed into semantics. Turning a famous dictum on its head: "Semantics proposes, syntax disposes."[24] One role for grammatical processes is to constrain the semantic representations that the brain computes autonomously based on the meanings of content words. On this account, grammar would be a latecomer to the language processing party, as the latency of the P600 indicates. This is the kind of conclusion we can draw from ERPs on the relative order of linguistic operations. Of course, there may well be early grammatical processes. In fact, ERP research shows that violating word-level grammatical constraints (e.g., of word form or word category) can modulate the LAN (left anterior negativity) component, approximately coinciding temporally with the rising flank of the N400.[25] But there is no evidence for early phrase- and sentence-level syntax, preceding and feeding into semantic interpretation.[26] Syntax may, however, play a prominent role in language production. There, it may serve as a necessary interface between meaning and sound, regaining the prime function it has in linguistic theory. As for many other neurolinguistic issues, for this one too, the jury is still out.

> One role for grammatical processes is to constrain the semantic representations that the brain computes autonomously based on the meanings of content words.

3
Mapping Language in Brain Space

The project of neurolinguistics is to trace language back to its roots in the brain. As a preliminary step, this may require mapping linguistic processes in brain space and brain time. What brain networks are causally responsible for particular linguistic processes? When do they get engaged in a given task? A popular metaphor in the early 2000s was that of a "movie" of cortical activity. If only we could reconstruct that kind of detailed spatiotemporal picture of what is going on in people's heads when they produce, understand, or learn language, we would be in a position to answer the deepest questions of neurolinguistics. Or would we?

Jerry Fodor famously wrote, "If the mind happens in space at all, it happens somewhere north of the neck. What exactly turns on knowing how far north?"[1] Indeed, it may not be clear what we actually learn when we learn that a certain part of the brain "lights up" when people carry out a mental task. But that is only because it is too easy to be impatient and assume that a spatiotemporal map of brain activity is itself the answer we are looking for. It isn't. The true aim of neurolinguistics is to address "how" questions—questions of architectures and mechanisms. It is doubtful that those can be addressed without answering "when" and "where" questions first. Mapping is a necessary first step.

In chapter 2, I explained why answering "when" questions is so important. If we know whether processes A and B happen in parallel or sequentially, and in what order, then we gain information about the underlying architecture of the system—a first step to understanding the system in causal, mechanistic terms. What about "where" questions? Take linguistic process A: What do we learn by discovering that brain area or network X, and not Y, Z, and so on, is activated by A? Not much, if little else is known about the functions of X, but possibly quite a bit if more is known on X's roles in other cognitive processes. So, for example, we might discover that nouns denoting concrete objects, like *ball* or *fork*, recruit partly the same areas engaged by visually perceiving or manipulating those objects. Perhaps word meaning builds on perceptual and pragmatic object knowledge? These are the kinds of explanatory links that can be put forward and tested using brain imaging data.

Each higher-order area of the brain is likely involved in several mental functions. That is what makes the spatial mapping approach to language particularly promising. We can learn a lot about language in the brain just by studying what other functions are carried out by cortical regions that are traditionally associated with speech and language, such as Broca's region or Wernicke's region (chapter 1). We can then try to establish a common denominator between the different functions carried out by area or network X, and try to explain linguistic processes in terms of those more basic or shared common denominator computations. This rules out the possibility that there are functionally specialized and language-dedicated areas of the brain: regions that only do language and nothing else. That is not the case. Language has emerged (relatively) recently in our evolutionary history, likely recycling preexisting neural circuits.[2] That is exactly what makes the spatial mapping enterprise so exciting and worthwhile. If there were one or a few brain areas fully and exclusively dedicated to speech or language, we would not learn much by discovering just where they are in the brain. Indeed, nothing would turn on knowing "how far north."

Functional Magnetic Resonance Imaging

Information about active areas of the brain can be gleaned from common neurophysiological measures such as EEG. However, in those cases, one often faces a damning inverse problem: there are mathematically infinitely many configurations of neuronal sources in three-dimensional brain space that can account for the electromagnetic effects—potentials or fields—observed over electrodes or sensors in two-dimensional surface space. Imagine a translucent paper dome, with some light bulbs beneath the surface coming on and off as time goes by. A small, round patch of light on the dome's surface can be either the projection of a diffuse light halo from a nearby bulb, or the result of a tighter beam from a stronger source farther away, or the combined effect of several converging beams. Identifying the physical source of the light patch is hard. The space of possible solutions is huge. That is what makes source localization of M/EEG effects difficult unless one has some prior expectations as to where those effects may be generated in the brain. For example, early auditory ERPs are generated in the auditory cortex. This information constrains the search space to fewer solutions. But clearly, the more information one has on possible cortical sources, the lesser the gain in showing that a given surface M/EEG effect actually localizes there. Discovery of spatial maps of cognitive processes, as opposed to perceptual and motor ones, is possible but generally hard with M/EEG data.

> We can learn a lot about language in the brain just by studying what other functions are carried out by cortical regions that are traditionally associated with speech and language.

Tools like fMRI and fNIRS (functional near-infrared spectroscopy) allow neuroscientists to measure directly in three-dimensional brain space, removing at the root the hard problem of mapping effects on 2D surfaces to their neuronal sources. What we can measure with fMRI is not neuronal activity as such, and not even its associated electromagnetic effects, like local or diffuse potentials or fields, but rather neuronal metabolism.[3] Brain cell activity is powered by glucose and oxygen, supplied via blood flow. Neuronal activity leads to increased local oxygen consumption. Even small changes in oxygen levels in blood flow can be detected in fMRI. In a typical experiment, the participant lies supine inside a magnetic resonance (MR) scanner (a computer-controlled magnet) while attending to the stimuli and a task (figure 4). Anatomical and functional images are derived from different MR signals in the course of an experimental session. Functional scans are obtained in an experimental task (involving some manipulation of the stimuli) versus a baseline task. Task-specific neural metabolism is calculated by subtracting the baseline activity from the experimental activity and can be visualized by overlaying the resulting functional images to the anatomical images. These activation maps are shown as colorful pictures in which some brain areas appear to be on and others off. In reality, fMRI data are smooth spatial maps that allow us to infer (indirectly and probabilistically) which regions have been more or less active in the few seconds following the stimuli. Increasingly diverse data types are pulled from MRI signals, providing clues not just on which brain regions are active and when, but also on how they are wired together and on how activation patterns encode information.

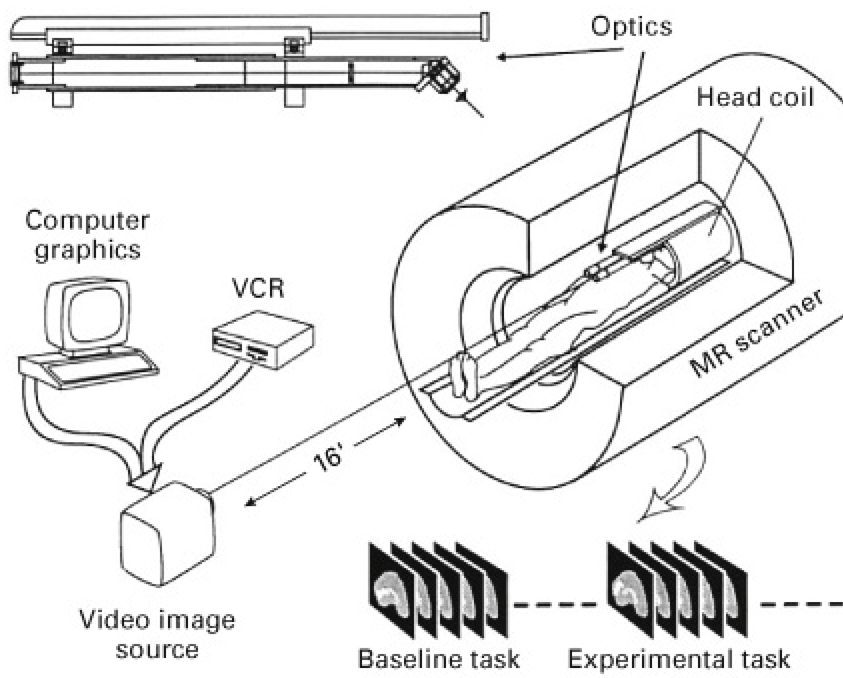

Figure 4 Functional magnetic resonance imaging. Source: Reprinted with permission from E. A. DeYoe, P. Bandettini, J. Neitz, D. Miller, and P. Winans, "Functional Magnetic Resonance Imaging (FMRI) of the Human Brain," *Journal of Neuroscience Methods* 54 (1994): 171–187.

Is Language a Whole-Brain Affair?

Philosopher and psychologist William James, in his classic book, *The Principles of Psychology* (1890), wrote that "the entire brain, more or less, is at work in a [hu]man who uses language." Was James correct, or is that an exaggeration? James was mounting a critique of the radical localizationist thesis that a "faculty of speech" exists in the mind and a "center of speech" in the brain—an idea we encountered in chapter 1, originally articulated by Gall and defended by Broca. James knew from the work of Wernicke and others that our capacities to produce and understand speech rest on networks of cortical regions, not on a single center. The brain's speech network comprises Broca's and Wernicke's areas, respectively, in the left inferior frontal and left superior temporal cortices, anatomical connections between them, and a number of other regions contributing in various ways to speech and language. Recall, for example, that Wernicke and Lichtheim understood that concepts or meanings were stored throughout the cortex—a distributed system in modern jargon. If that is true for spoken language, it is also true for language in general—written, signed, and so on.

One challenge for imaging research is precisely that language tasks activate large portions of the neocortex and even some subcortical structures, older than the neocortex in evolutionary terms. James was right, "more or less," and fMRI research confirms it at every turn. Language depends on large-scale networks, for two important reasons. One is that language is multimodal, as we saw in chapter 2. The input we receive during face-to-face conversation is hardly ever limited to auditory input (speech) and often includes visual input. In addition, speech is common but is far from being the only means of linguistic communication. Text and sign are two other systems we use to externalize thoughts for the purpose of communication. Several types of gestures can accompany speech, sometimes providing information that effectively adds content to the message being conveyed verbally. In general, discourse happens in rich sensory environments, and communicative interaction requires that both speakers and hearers undertake actions of various sorts; speaking, signing, and gesturing are three clear examples.[4] What does this imply for language in brain space? We cannot hope to reduce language in the brain to a single network that carries out abstract computations of the kind envisaged by formal linguistic theories. Language also engages sensory and perceptual systems (e.g., auditory, visual, somatosensory), action systems (e.g., executive and motor control, social cognition), and others.

The second point is that even considering only the cognitive side of things, language builds on computational systems beyond syntactic structure building (see chapters 1 and 2). Phonological and semantic processes play a key role, in addition to reasoning to and from interpretations (logical and commonsense inferencing; recall the coffee example from chapter 2), inspecting the contents of interpretations (to ask questions when we realize that information is missing, for example), various learning and memory processes (such as to encode and store the meaning of a new word), and more. These processes engage networks that do not belong to sensory or motor systems, but activations are broadly distributed across the frontal, temporal, and parietal lobes.[5] It is difficult—some might say pointless—to try to draw a line between "core" linguistic computations (phonology, syntax, and semantics) and supporting processes, like inference, learning, and memory. A working premise is that language builds on multiple brain networks. It is not a system (unlike vision) but rather a system of systems, recruiting different networks depending on the task, given information, and available internal resources.

On Perisylvian Language Networks

Is there a set of brain areas that are reliably engaged in language comprehension and production regardless of the task, given information, and available resources? A cortical network that is robustly involved across variable conditions in language experiments may deserve the title of "language network." Here, it is important not to think in terms of brain areas necessary and sufficient for language. This is rather a question of *invariance*: we are asking whether there is a set of brain regions that "light up" consistently across different experiments and that may also be expected to activate in the next experiment. It turns out that there is a large cluster of cortical areas around the Sylvian fissure, the deep sulcus that separates the temporal lobe from the frontal lobe, that meets this requirement. This part of the brain is sometimes called "perisylvian cortex" or the "perisylvian language network" (figure 5).[6]

Some of these regions are more active in response to speech, in particular the auditory cortex and portions of the superior temporal gyrus, including the classical Wernicke's area. Together, these regions form a system that translates auditory input into phonological representations of speech by using phonological words and their constituents as the relevant units of analysis and (abstract, categorial) linguistic representation.[7] The superior temporal cortex hosts circuits specialized for auditory speech processing, which, however, appear to be organized similarly in humans and some other primate species.[8] There is increasing evidence that different aspects of the auditory input, including (intelligible) speech, are analyzed in several subregions of the superior temporal gyrus bilaterally.[9] Wernicke's region—

the posterior part of the superior temporal gyrus—was originally believed to be crucial for speech perception and comprehension, that is, part of the receptive speech system. More recent work has cast doubt on this idea. Speech-specific activations are found in areas of the superior temporal cortex (figure 5) surrounding the auditory cortex and anterior to Wernicke's. The latter region seems to be more involved in one specific process leading to speech production: the retrieval of the phonological material to be articulated.[10]

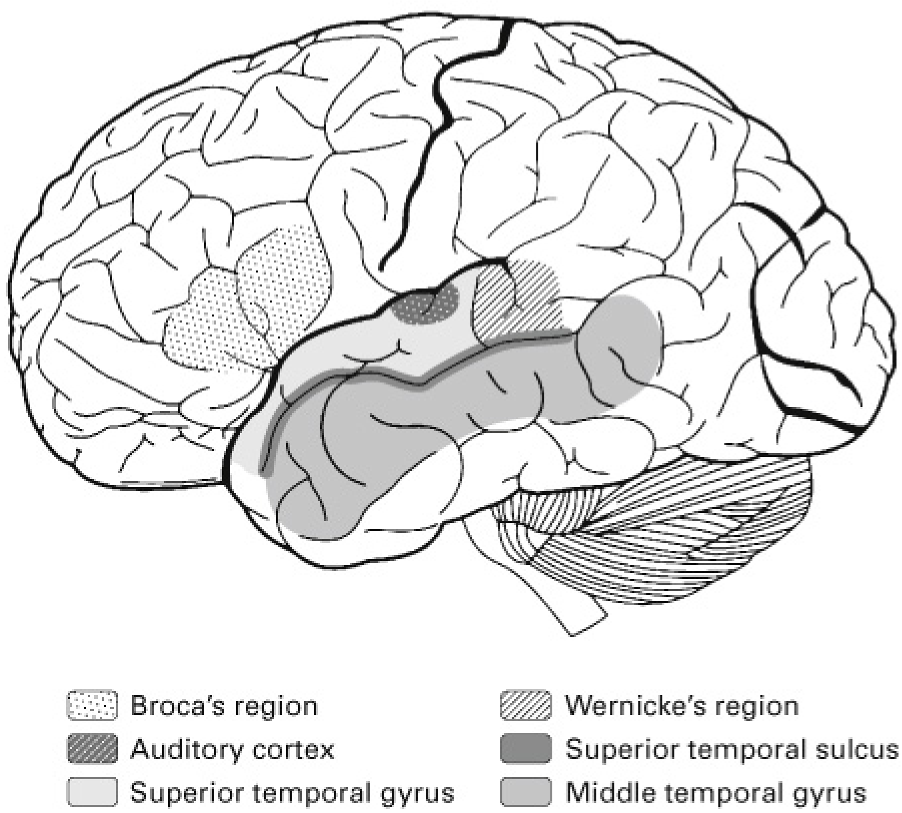

Figure 5 Speech and language regions in the left perisylvian cortex. Source: Reprinted with permission from P. J. Monahan, "Phonological Knowledge and Speech Comprehension," *Annual Review of Linguistics* 4 (2018): 21–47.

Mid and posterior parts of the middle temporal gyrus and of the superior temporal sulcus are often activated in response to auditory, written, and signed linguistic stimuli. Occasionally these activations extend posteriorly and superiorly to temporo-parietal areas, especially the angular gyrus and the inferior parietal lobule, and to the anterior portion of the temporal lobe. In general, the middle temporal cortex is activated as a function of the presence of content (semantics) or structure (syntax) in the stimuli, with stronger activations, for example, when words have multiple unrelated or related meanings (homonymy or polysemy).[11] This brain region most likely represents lexical information, specifically information about the syntactic and semantic properties of words, while neighboring dorsal and ventral portions of temporal cortex likely serve as interfaces with phonological and orthographic systems, respectively.[12] The left posterior middle temporal gyrus is also active when signers perceive sign language sentences, suggesting that the representations it contains are not tied to phonological or orthographic codes and are independent of modality of presentation.[13] This region is among the neural generators of the N400, which is consistent with the hypothesis that it represents lexical semantic information that can be quickly activated on word recognition.[14] Anterior sections of the temporal lobe, especially in the left hemisphere, have been implicated in the semantics of abstract words and in forms of conceptual combination between content words (e.g., a noun and an adjective).[15] These results are consistent with the role of the temporal lobe as part of the brain's semantic or declarative memory system, possibly with a direct role in computing aspects of linguistic structure and meaning.

Another functionally important area of the perisylvian language network —according to some, *the* most important region—is Broca's region, often taken to correspond to the posterior and middle portions of the inferior frontal gyrus in the left hemisphere: the posterior bit roughly coincides with the traditional Broca's area and overlaps with the site of lesion in Broca's two patients (see chapter 1). The terms *Broca's region* and *Broca's complex* are sometimes used to refer to a larger lump of the left inferior frontal gyrus than Broca's area. None of these labels is used particularly consistently in the neurolinguistics literature. This has been a source of some confusion and incongruity in published studies. In classical models of speech and language, from Wernicke to Geschwind and beyond (details in chapter 4), Broca's area was seen as a part of the speech production network and was argued to be involved in the articulatory stages preceding the preparation and execution of speech motor programs. Broca's area is a sweet spot for speech production, also by virtue of its proximity to and connections with the insula and with the premotor cortex, which provide inputs to primary motor cortex—the part of the brain that controls the execution of bodily movements.[16]

Imaging studies have largely confirmed this view for Broca's area specifically, but they have also broadened our understanding of the functions of adjacent segments of the left inferior frontal gyrus, that is, of "Broca's region" at large (figure 5). This work indicates that Broca's region is activated during speech production and comprehension, and indeed in production and comprehension of language, regardless of input and output modalities—signed, written, or spoken. Activation of this region seems to depend on the well-formedness or complexity of the inputs or outputs (along several dimensions, including structural and semantic), and it appears to be often coupled with activity in other regions of perisylvian cortex, in particular posterior temporal regions.[17] This frontotemporal coupling, and in general the functional interactions between Broca's and other regions, are largely supported by complex patterns of anatomical connections: white matter fiber bundles linking two distant areas of the cortex. It is via these connections that cortical areas can talk to each other and perform computations that give rise to cognition and behavior.[18]

Mapping Language or Linguistics?

Mapping language in brain space and time is a necessary preliminary step toward providing mechanistic explanations of how language is represented, processed, and learned by humans. The role of theoretical linguistics—the formal study of linguistic structures—in this enterprise is the subject of much debate. To study *how* the brain represents and processes language, we must specify *what* is being represented and processed. Linguistic theory should guide neurolinguistic investigations. If this is correct, the best approach to mapping language in the brain would be to try to map specific structures and operations posited by linguistic theories. Ideally, one should use structures and operations that are posited by all or most linguistic theories on the market, or structures and operations that have exact or approximate equivalents across theories.

This approach has been quite successful in phonetics and phonology, where there is significant agreement on the basic units of representation: the features that make up the abstract categories of individual consonants and vowels. In the standard analysis due to Roman Jakobson and Nikolai

Trubetzkoy, the elementary units of speech are defined by *distinctive features*, such as where air is compressed on its way out (e.g., the lips) and whether the vocal cords vibrate. A series of experiments using direct recordings from the cortical surface have shown that such phonetic features are encoded in several subregions of the superior temporal gyrus,[19] in a format that seems largely invariant with respect to many contingencies of speech perception (recall chapter 2). The building blocks of phonetics have now been roughly mapped in brain space, but not all details have been filled in yet. What about other areas of linguistics?

The attempt to connect formal linguistic operations to specific brain processes has been much less successful but nonetheless instructive. For example, studies have tried to link Merge in recent versions of generative grammar (the operation that takes two syntactic objects and forms a new one—for example, {red, balloon} from {red} and {balloon}) to activity in portions of the left inferior frontal gyrus.[20] But similar operations (if not formally, in terms of the processing predictions they make) are postulated in other frameworks, such as Unification, which also puts together fragments of linguistic structures into larger units depending on specific constraints. Unification in syntax and semantics has been shown to engage parts of the left inferior frontal gyrus.[21] In some proposals, this brain area is assigned a more specific syntactic function (e.g., movement),[22] while in other models it serves the function of a short-term memory crutch in support of sentence processing.[23] The "battle for Broca's region" is still raging, but it is worth noting that this part of the brain, like other regions, is not specialized for language or syntax and likely performs a range of functions.[24]

In semantics, researchers are looking for correlates of meaning composition—the syntax-driven operation that combines typically predicates and arguments into logically interpretable expressions: for example, "laugh(x)" and "ann" into "laugh(ann)," for the sentence "Ann laughs." But so far, composition has proved elusive and difficult to tease apart from other syntactic or semantic processes.[25] Logical and pragmatic aspects of language comprehension have been related to the function of frontoparietal networks, in some cases providing novel explanatory links between language and other cognitive domains—for example, quantifiers (like *all, some, most, few*) and magnitude and number processing, or pragmatic inferencing and social cognition.[26] Figurative language, such as metaphor, is another area that has been in focus recently, with a growing number of linguistic analyses and fMRI experiments on the topic.[27] Neurolinguistics is taking its first baby steps toward understanding how the brain runs the "software" described by theoretical linguists.

4
Models of Language in the Brain

Why do scientists build theories and models of the world? Are the results of observation and experiment not enough for us to understand how things work? Sometimes they can be, but often the systems we are interested in are too complex and dynamic for us to be able to figure them out just by crunching data, even if our powers of measurement and observation are augmented by the latest technologies. In science, understanding is usually achieved by means of some kind of theoretical construct, which can be anything between a compact set of mathematical equations and a computer simulation of a specific process. Empirical data can provide information for steering the theory in the right direction. But this works best when theory is used to guide observation and experiment, so that very specific questions can be answered. Scientists are primarily engaged in theory development and model building, and as philosopher Bas van Fraassen has eloquently put it, "Experimentation is the continuation of theory construction by other means."[1]

Neurolinguistics too is concerned with developing theories, and most neurolinguists recognize that empirical data should at least play the role described above. Yet the brain has so far frustrated our best efforts at modeling it. It is a peculiar computational environment, unlike anything humans have engineered (think of, or rather don't think of, a digital computer). This has two key consequences. First, exploratory research tends to be valued in neurolinguistics. Sometimes there just is no theory available that may guide experimentation and modeling, or perhaps extant theories are not applicable for various reasons. It then makes sense to run well-controlled experiments asking broad questions that emerging theories could build on. For example, the N400 and P600 (chapter 2) were discovered that way. On that approach, one can at least establish robust measures that can be used in future experiments. Fruitful discoveries can happen outside a theoretical frame: neurolinguistics has thrived on such discoveries for a long time.

The second effect of the brain's complexity is that it is easier to construct models or simulations that represent, often in a simplified and idealized way, the causal structure and the functional role of relevant neural processes than to summarize the workings of the system in a compact set of equations. In neurolinguistics, as in human neuroscience in general, concrete modeling tends to take precedence over abstract theory. This facilitates understanding by capturing brain processes at intermediate or low levels of analysis, at which cognitive architectures, algorithms, and mechanisms are specified. Neurolinguistic models are designed so as to remain close to data and be easily modifiable based on the evidence. Such models are ductile by design, which is both a feature and a bug: neurolinguistic models that are easier to repair are also more difficult to criticize and reject.[2]

Linguistics, Psycholinguistics, Neurolinguistics

How can one make models of language in the brain harder to change in the face of new data, so that we can learn not just from data as such but also from failures of the model? One approach is to embed neurolinguistic models—causal descriptions of what happens when and where in the brain during language processing—into formal theories spanning multiple levels of analysis. In language science, the relevant levels are fairly straightforwardly provided by linguistics (the level of the phonologic, syntactic, and semantic structures that the brain stores or builds online), psycholinguistics (the level of the architecture and algorithms effectively employed by the brain in structure building), and neurolinguistics (the level of the neurobiological machinery that runs those algorithms).[3] We need precise, cognitively plausible linguistic theories to describe the algorithms that the brain may be using and to guide the discovery of the underlying cortical mechanisms. Theories and models at each level must constrain theories and models at every other level: it all has to hang together. This makes piecemeal adjustment of theories and models by sparse data points much harder. Internal consistency is just as important as empirical adequacy, and theories may be criticized and rejected for failing to meet either criterion.

> In neurolinguistics, as in human neuroscience in general, concrete modeling tends to take precedence over abstract theory.

Let us consider some of the fundamental constraints that theoretical linguistics and psycholinguistics impose on neurolinguistic models before discussing in greater detail a few models currently on the market. In what follows, I use language understanding as an example. A key requirement (and here linguists of all persuasions would probably agree) is that the brain must assign meaning to the input, using at least knowledge of word meanings but usually also syntax and externally available information about the speaker, the context, and the goals of the exchange. Exactly what kinds of linguistic or nonlinguistic information can be used in the process of meaning construction is still an open subject of inquiry, but one can at least map the space of possibilities. Understanding can sometimes proceed based only on the meanings of content words in the input (nouns, verbs, and so on) and knowledge of plausible ways in which words can be combined (chapter 2): semantic systems may compute meaning autonomously.[4] On other accounts, grammatical information is obligatorily used to derive interpretations, as required by theories in formal semantics. The question then is how much syntax goes into the interpretation process.

Another question is how the brain can go beyond the meaning of given words and plausible semantic relations, if not by computing grammatical structure. Context is one of the prominent factors here. Linguistics has provided formal models of how certain classes of referring expressions pick out specific times and places (e.g., the here and now), thus anchoring meaning to the context of the utterance. Also, it is now well understood how different words can co-refer in discourse, such as noun phrases and anaphoric pronouns, as in, “A man owns a cat. He feeds it every day.” Moreover, logical or pragmatic inference is often required to recover the intended meaning: “Could you close the door?” should be interpreted as a request to close the door, and not as a question on the hearer’s capacity to do it. Finally, language is rife with figurative meaning, which exceeds the capacity of a compositional theory (constituent meanings + syntax): idioms, metaphors, and other forms of speech are ubiquitous in language use.[5] In what way are these insights from linguistics constraints on neurolinguistic models? Neurolinguistics should explain the neural mechanisms that support not just the syntactic and semantic composition of words in sentences, but also the interpretation of discourse in context and inference to the intended figurative or pragmatic meaning.

Neurolinguistic models must also be constrained by results from psycholinguistics, which has a rich tradition of theoretical and empirical inquiry, predating both modern (formal) linguistics and neurolinguistics as such.[6] It is a highly diverse field methodologically that has generated some of the most robust findings of all human psychology. Four sets of results are particularly relevant here: together, they illuminate the architecture and algorithms supporting language processing.

First, words and their meanings are not retrieved from memory as discrete bits of information. Instead, they are distributed structures whose activation can be sensitive to properties of the words themselves (e.g., their frequency in the language), relations with other words, and information already available in context. This makes the computation of phrase, sentence, or discourse meaning dependent on the situation and the long-term structure of human memory.

Second, language comprehension is incremental: it unfolds as the input comes in, in general word-by-word or even morpheme-by-morpheme. Consequently, the system can make decisions or commitments that might have to be revised at later stages; for example, "The horse raced past the barn fell," may be analyzed as a simple active sentence up to the noun *barn*, but *fell* implies a reanalysis such that "raced past the barn" is a relative clause.[7] Recomputation occurs routinely across levels of linguistic form and meaning.

Third, language processing is open to influences from internal or external sources of multimodal information that contribute to constructing or disambiguating meaning (e.g., visual perception; chapters 2 and 3). Such information is typically used as soon as it becomes available, exploiting interactive, parallel processing in the brain.

Fourth, meaning representations built online are good enough for understanding, but often they are not as rich and detailed as the structures assumed by theories of syntax and semantics. Also, phonological, morphological, and syntactic information may be used in the construction process but may be disposed of once meaning has been established.[8] At least these constraints from linguistic and psycholinguistic research, and possibly others, should be used to build and evaluate current models of language in the brain. Below, I briefly present three models that have been most influential in recent years.

Current Models of Language in the Brain

Classical models of language in the brain—by Gall, Broca (if early localizationist hypotheses count as models), Wernicke, Lichtheim, and others—are limited at least in that they lack one or more computational components combining words and meanings into syntactic and semantic representations of phrases, sentences, and discourse. For most of the nineteenth century, the requisite linguistic insights were largely lacking, and for most of the twentieth century and up to this day, despite advances in the formal sciences, it has proved arduous to use insights from theoretical linguistics to uncover how the brain computes linguistic form and meaning given sensory inputs. In other words, except for some restricted areas of linguistics (particularly phonetics and phonology; chapter 3), it has been unexpectedly difficult to transfer knowledge from linguistics into neuroscience and to use it to discover new facts about the organization of language in the brain. Still, at least three current proposals have tried to do just that: use some of the best available insights from linguistics to shed light on the neurobiology of language.[9]

The first is a model by Angela Friederici, in which syntax plays a pivotal role.[10] The theoretical starting point is the latest version of generative grammar, where syntactic structures are derived via recursive application of a single operation, Merge (chapter 3). Merge takes two syntactic objects, for instance, the words *blue* and *sky*, and puts them together into a new, more complex syntactic object, say the noun phrase “blue sky.” Merge requires additional machinery to produce syntactic representations of natural language sentences, such as algorithms for labeling each object with its part of speech (e.g., noun, adjective) and for ordering merged objects (merge produces only unordered sets). For instance, the order of the adjective and the noun in noun phrases may differ across languages ("blue sky" in English vs. "cielo blu" in Italian). In Friederici's own model, Merge is executed by the posterior portion of the left inferior frontal gyrus (Brodmann's area 44, or pars opercularis), while the required additional syntactic operations may involve other frontal and temporal regions.

The Merge area in left inferior frontal gyrus (IFG) is assumed to support structure building early on in the comprehension process. Friederici is one proponent of the syntax-first hypothesis, where local syntactic structure is built up before meaning (recall chapter 2 for a discussion). Anatomically, this core area for syntax is part of a broader network that also includes regions of the left posterior temporal lobe, connected by bundles of white matter extending dorsally, in the shape of an upsloping arc, spanning temporal and frontal areas. Friederici's model has tight links to other influential neurolinguistic hypotheses, for example, by Andrea Moro, Yosef Grodzinsky, and others on the role of the left inferior frontal cortex in syntactic processing.[11]

The second proposal is the MUC model by Peter Hagoort, which assumes three cortical components of the language system: a Memory component, containing lexical (phonological, syntactic, and semantic) representations of the building blocks of the language (morphemes, words, and constructions); a Unification component, which serves as an active workspace where such building blocks are put together in parallel across levels of linguistic representation; and a Control component, overseeing the joint operation of Memory and Unification in language use (e.g., in turn taking in conversation and language switching in bilinguals).[12] The Memory component is realized in (left) temporal lobe areas, particularly in middle and posterior regions. The Unification component is implemented in the left inferior frontal gyrus, with the posterior segment being involved in phonological unification, the middle portion in syntactic unification, and the anterior segment in semantic unification.[13]

The online assembly of a complex linguistic structure results from the dynamic interaction between Memory and Unification, temporal and frontal regions. Interpretation in a broader sense, also involving inference or the derivation of figurative and pragmatic meaning, further engages regions outside classical perisylvian language networks, such as medial frontal and inferior parietal cortices.[14] Functionally, Unification requires that lexical information is copied from Memory and maintained online over time, else Unification cannot generate a coherent representation of the meaning of the sentence or discourse, which typically spans several words and therefore several seconds or longer. The copy mechanism is particularly important. Language is made of tokens (instances or "copies") of lexical types stored in Memory. Words may reoccur in discourse, as in "The little star is beside the big star." The same word type, *star*, with the same meaning, is used twice, as two tokens, to refer to two different stars in the context of each noun phrase: "the little star" and "the big star." To avoid confusing these two referents, the brain must distinguish types and tokens and map such tokens to the appropriate referents.[15] Merge and Unification can operate directly on tokens. More research is needed to determine how those are generated from types and how they are mapped to referents.

Another important proposal in neurolinguistics is the dual-stream model by Gregory Hickok and David Poeppel. It is primarily a model of spoken language processing that has implications for language in the brain more generally.[16] The model posits two information processing streams in the brain that support speech and language. Each stream corresponds to a particular cortical network with a degree of functional autonomy. A bilateral ventral stream, running low through the temporal lobes, processes speech signals for comprehension. A left-lateralized dorsal stream, which connects superior temporal and parieto-temporal cortices to the left posterior inferior frontal cortex, possibly including classical Broca's area, supports instead spoken language production and speech development (more in chapter 5). The ventral stream links together representations of sound (phonology) and meaning (semantics), whereas the dorsal stream provides a sensorimotor interface between sound and articulatory motor programs. The dual-stream model has been fine-tuned and thoroughly tested experimentally and now enjoys considerable empirical support.

One difference with Friederici and Hagoort's models concerns the role of left inferior frontal gyrus, specifically of segments anterior to classical Broca's area. According to Hickok and Poeppel, this piece of cortex supports speech and language processing as a memory crutch that kicks in when information has to be maintained online over time. This is similar to one function of left inferior frontal cortex in Hagoort's model, but the analogy ends here. Both Hagoort and Friederici assume that this area puts together linguistic information into larger units. Friederici would restrict Merge to syntax, while Hagoort extends Unification to phonology, syntax, and semantics alike. Where are complex linguistic structures and meanings computed, according to Hickok and Poeppel? Not in the frontal cortex, but in the anterior temporal lobe, as part of the ventral stream.[17]

Correlational, Formal, and Mechanistic Models

The three models just presented, and in fact the majority of modern neurolinguistic models, have similar structures and goals. They assign different functions to different regions or networks of the brain, with the aim of explaining why those regions or networks are activated, in fMRI or MEG or other studies, in response to particular stimuli or tasks, and why loss of certain language functions may be associated with damage to specific brain regions or networks (more in chapter 8). This is a start, but no more. Many in the field would agree that these explanations are often largely correlational. If a model was built on observed correlations between stimuli or tasks and neural activations, no wonder it can now "explain" a similar correlation pattern in the next experiment. Getting correlations right is mandatory for a theory to be stamped as empirically adequate, but more is needed. We want our models to state what computation is carried out by the system—for example, whether it is Merge or Unification or something else—and how that computation unfolds neurally. In other words, correlational models must be augmented with formal and mechanistic analyses, respectively.

Formal models of what linguistic processes the brain may (or must) be carrying out, given a task, are required to guide the analysis of complex experimental data. Typically, it is difficult or impossible to extract even a sketchy picture of what the brain is doing from data. Current neuroscience methods fail also when applied to systems we understand well, in terms of both hardware and software, as has been recently shown for a microprocessor playing an Atari video game.[18] The ideal situation in neurolinguistics would be that theoretical linguists tell us that a certain operation (say, meaning composition) is necessary for understanding to occur; neurolinguists may then build models that include such operation and start using those models to look for its footprints in the brain. The trouble is that linguists disagree on what the basic representations or operations should be, which puts the project of neurolinguistics on hold or forces it to be largely exploratory. This is not to blame linguists. It is true and perplexing that human language can be formally reconstructed in different ways. The trick is to build on resemblances and convergence between theories and use shared insights to guide research: for example, it is not surprising that Merge and Unification have been suggested to engage partly overlapping cortical networks. Once basic operations have been identified—modulo slight differences between linguistic theories—the search for the underlying neurophysiological mechanisms can begin. Neurolinguists are currently focusing on mechanisms for lexical activation, prediction (chapter 2), composition, and reference. In some of these areas, the results are quite promising and give us a glimpse of possible biologically grounded answers to the big neurolinguistic questions.[19]

> Formal models of what linguistic processes the brain may (or must) be carrying out, given a task, are required to guide the analysis of complex experimental data.

5
Growth of Language Networks

Nearly all children acquire one or more languages, reaching essentially adult competence in the first five to seven years of life. What is most striking about language learning is not so much the speed at which it happens—a developmental process that takes several years to complete is not exactly fast—but the fact that it requires little adult supervision, let alone explicit instruction, and that the progression from no language to a language is not quite linear and incremental. In children's productive language, some error patterns may disappear while others emerge—for example, irregular verbs may be overregularized, as *breaked* for *broken*.[1] Some abilities may even develop abruptly—for example, the use of function words, such as articles or pronouns, which transform speech from telegraphic to structured.[2] Yet language learning follows a path that is largely similar across children and languages, as reflected in the developmental milestones familiar to many new parents.[3]

Research in psycholinguistics indicates that the child is never a blank slate prior to exposure to a language and that the process of language acquisition is not just a matter of brute-force learning from data. If the first thesis was true, we would see much variation among children in how they acquire language, in particular in what skills emerge when. Instead, one finds characteristic developmental trajectories, which suggests that maturation of brain systems determines what aspects of language infants or children can learn and when that can happen. If the second thesis was true—that language acquisition reduces to learning from data—then we would see largely linear growth curves, as if knowledge was stockpiled incrementally and monotonically, much like adding coins to a coin collection. Instead, we find U-shaped turns (correct-incorrect-correct patterns of use over time) and S-shaped transitions (rapid changes from incorrect to correct patterns of use) in development. This points to some discontinuous internal reorganization, where the acquisition of new knowledge may have an impact on how previously learned information is represented and used.

The idea reviewed in this chapter is that infants and young children are able to acquire specific aspects of language only in specific stages of cognitive development. The clock that sets the pace of this process is not an external one (linguistic input is there all the time, even before birth) but an internal one—the process of brain maturation and growth. This process is genetically constrained (details in chapter 9) and therefore largely the same for all children. Some believe that language is innately specified. Yet the idea that knowledge of grammar may be encoded in the human genome is biologically implausible. Others would soft-pedal the innateness thesis by saying that the human brain is "language ready." That is closer to truth but is not quite it. If we could upload a complete language system to a newborn's brain, we would certainly not magically create a competent, fluent speaker of that language.[4] The reason is intuitively clear: the hardware (an immature brain) would not be ready to run that software (a full-fledged language system). In the case of language acquisition, the software must be programmed in gradually, largely as a function of the stage of growth and organization of the hardware.[5]

The Quiet Phase: Cracking the Speech Code

When does language learning start, and how? The answer may surprise you: it begins as soon as the fetus can hear, several weeks before birth at term, however limited the sounds perceived in utero might be. Prenatal exposure to speech may give the baby clues about specific temporal characteristics of spoken language—properties that do not change much when distorted, as they may be in the womb. It is, however, postnatal experience—in particular, exposure to speech sounds, usually in a social context—that lets the baby "crack the speech code."[6] The capacity to detect and process rhythm and prosody in speech emerges in the first weeks of life, followed by the ability to categorize (that is, form increasingly abstract representations of) phonemes. During the first months, infants become sensitive first to vowels in the language they are exposed to (or what will be their first language, their L1), then to consonants, and eventually to combinations of consonants and vowels (syllables).[7] These emerging phonological representations are the result of the child's ability to detect patterns in adult speech by tracking statistical regularities, including co-occurrence frequencies and transition probabilities between input sounds, such as individual phonemes or syllables, and later between words or groups of words. Infants are capable statistical learners: this allows them to discover meaningful building blocks of language—morphemes and words—in an initially opaque speech stream.

The fact that advanced auditory processing can start so early, even before birth, and then at full speed in the first weeks of life, indicates that the baby's auditory system and probably neighboring regions of the superior temporal lobe are developing rapidly around the perinatal period. Indeed, the maturation of neural auditory circuits, together with other sensory systems in the brain (vision, touch, and so on), takes off quickly. But processing speech is more than just hearing speech. Nonauditory speech processing areas of the bilateral superior temporal cortex mature in the first three years of life, during which they become increasingly attuned to the native speech sounds the child is exposed to. There is no evidence for the initial speech specificity of the human auditory network, which is largely shared with our nearest evolutionary neighbors (chapter 10).[8]

> Infants are capable statistical learners: this allows them to discover meaningful building blocks of language—morphemes and words—in an initially opaque speech stream.

An important index of structural brain maturation is synaptogenesis: the process of temporary overproduction of synapses (the junctions between two neurons), followed by pruning to reach adult levels. Synaptogenesis peaks at around 6 months of age in many sensory areas, including auditory cortex, and at around 9 months in receptive speech areas. There is a window of opportunity in the first year of life for acquiring correct receptive representations of all the basic speech sounds of one's first language. During this period, the brain commits to (at least) one particular language phonologically, which, on the one hand, renders it easier to then learn the words (vocabulary) of that language but, on the other hand, may interfere with the acquisition of additional languages, especially those with a different phonology.[9]

The first year, give or take a few months, may well be called the quiet phase of language acquisition. During this period, production of spoken language sounds is limited to babbling. The average infant may be silent (that is what the Latin word *infans* means), but she is not at all idle. Finding words in the speech stream is one task she is facing in her waking hours. Another is to begin to attach meaning to the words she is extracting from the input. Recent work shows that as early as 6 to 9 months, infants "know" the meanings of many concrete nouns (e.g., *apple*, *shoe*). Moreover, at about the same age, they can relate words semantically (e.g., *milk* and *bottle* vs. *milk* and *ball*).[10] Their internal vocabulary is forming, supported by maturation of parietal and temporal cortical networks for semantic processing.[11]

First Words and Utterances: Learning to Speak

Let us go back for a moment to the dual-stream model of speech and language by Hickok and Poeppel (chapter 4). The model assumes a ventral stream that interfaces sound and meaning, engaging primarily structures of the temporal lobe bilaterally, and a dorsal stream, mapping phonological to motor representations of speech, in temporal and frontal cortices of the left hemisphere. As suggested by the results discussed, ventral stream function, such as learning to decode speech sounds and associate them with new meanings, begins to develop during the first 6 to 9 months of age and is soon followed by development of dorsal stream function.

Learning to speak is partly a motor learning task, but it is one that crucially builds on previously acquired sensory representations of speech sounds. Already in the babbling phase and later, the infant can use those representations to correct and fine-tune her productions, so that the sounds she tries to articulate will resemble more and more those she hears from others. The dorsal stream uses what the ventral stream has learned (how phonemes and words sound and what they mean) to learn something new (how phonemes and words should be pronounced). This model is consistent with much neurodevelopmental data, suggesting that brain maturation occurs earlier in sensory systems (in occipital-temporal regions, for instance) than in motor or control systems (the frontal cortex, for instance). Before information can be used by higher-order integrative systems, like the dorsal stream in the model by Hickok and Poeppel, it must first be encoded and organized within sensory and perceptual systems.[12]

When do children begin to produce their first words and utterances? Babbling develops around 9 months of age, when motor and adjacent prefrontal cortices begin to effectively connect with auditory and speech perception regions of the temporal lobes. Children produce their first words at around 10 to 12 months of age and two-word utterances toward the end of the second year of life. As with other aspects of language acquisition, there is some variability as to when individual children attain these skills. Productive child language exhibits little grammatical organization before that age, with the exception of basic word order and morphology in some languages. Between ages 12 and 18 months, children's utterances are typically short, and about half of all their utterances are nouns. The remaining half includes other single words, such as verbs and the first two-word utterances. Between 18 and 24 months of age, two-word utterances become much more frequent, and the total share of nouns now drops to around one-third of all utterances. Data from the Wordbank Project indicate that, at around 16 months of age, only about 25 percent of all children are reported by their parents to be combining words.[13] This fraction grows to about 50 percent by 20 months of age and to around 75 percent by 24 months, similarly across languages.

Syntax and Semantics: Toward Adult Competence

At the end of the second year of life, around age 18 to 24 months, most children can produce two-word utterances and can communicate effectively with adults using words, gestures (pointing and beyond), and other expressions. At this stage, a child's productive grammar is still rudimentary. For example, even such a basic set of syntactic properties of languages as word order (e.g., whether adjectives follow or precede the nouns they modify; whether verbs follow or precede their objects) is acquired relatively late, around 19 months according to one study.[14] Recent work has shown that even young infants track statistical patterns in speech that correlate with syntactic structure, such as cues to the placement of function words (determiners and pronouns, for example, which are often shorter and more frequent than most content or lexical words).[15] Statistical learning may indeed lead to discovery of syntactic properties of words, but that is an extended process that involves syntactic abstraction, whereby each individual word is represented as a member of one or more grammatical classes: noun, verb, adjective, and so on. Syntactic processing cannot happen unless words are categorized, and most categories take time to emerge. The first to appear is that of nouns, starting at 12 months, while most functional categories emerge only later, beginning at 14 to 16 months.[16] One-year-old infants are still figuring out the basic syntactic categories of their language.

Here is a puzzle: How do young children understand and produce multiword utterances (never mind how many and how long and complex) when their mental grammar is only beginning to take shape? Linguist Ray Jackendoff has coined the term *asyntactic integration* to denote the kinds of processes that may be at work here. In short, the child's brain can put together word meanings even if not all words are labeled syntactically or if knowledge of syntactic rules of meaning composition has not yet been acquired.[17] ERP studies, reporting modulations of the N400 effect in infants, support this view. Children can relate or integrate semantic information from coincident or overlapping sensory stimuli across modalities (an auditory word and a picture, for example) by 6 months of age. The ability to relate or integrate meaning over time within and across modalities arises at 9 months. The capacity to relate or integrate the meaning of spoken words in sequences or sentences appears at 18 months at the latest.[18] Behavioral and EEG studies show instead that grammatical processing skills, precisely of the sort required for syntax-driven composition of word meaning in phrases and sentences, develop later, between 18 and 32 months. Nonsyntactic meaning-building operations are available to children prior to the acquisition of a complete grammar.

From the second half of the third year of life, thanks to the emergence of syntactic and pragmatic skills and to significant vocabulary expansion, children's receptive and productive language begins to gradually converge toward adult competence. Utterances are longer and syntactically more complex and organized. Sentences containing more than two words are not uncommon even at 3 years of age, and most 4-year-olds frequently ask and answer questions. Note that the correct syntactic structure of questions is far from trivial in many languages. At this age, children begin to recount events and personal experiences (in the past) and to express clearly their desires, intentions, and goals (about the future). The ability to understand the intentions of others allows infants to read social cues from very early on. But from age 5, most children are able to actively exploit this information in communication: they will understand that what is said is not always what is meant. As any parent or older sibling knows well, a 5-year-old can be a delightful conversation partner, fully able to take turns appropriately and to contribute to the conversation informatively, often even creatively. None of this would be possible without the maturation of areas of prefrontal and parietal cortex subserving aspects of social cognition and abstract representations of properties of entities or events (such as numerosity, space, and time) and without functional integration of these regions with core perisylvian language networks. Throughout school age, the child needs only to fine-tune a system that is essentially in place, looking similar to adult language. But as we will find out in the next two chapters, additional language learning and acquisition of reading and writing skills can spice up things a little.

> The child's brain can put together word meanings even if not all words are labeled syntactically, or if knowledge of syntactic rules of meaning composition has not yet been acquired.

6
Bilingualism and the Brain

There is a greater-than-chance probability that you, or your kids, or someone otherwise close to you speaks more than one language. The majority of the world's population is at least bilingual, and around 10 percent of all people speak three or more languages. We may call these people multilinguals, or polyglots, if they speak several languages fluently. This leaves monolinguals in the minority, albeit a sizable one at around 40 percent of people. The challenge for neurolinguistics is to develop theories and methods for testing such theories that apply to all speakers, not just to a (large) minority. That is why bilingualism is such an important research topic, not only in neurolinguistics but in language science at large.

Studying bilinguals is difficult for many reasons. The "bilingual is not two monolinguals in one person" or, more precisely, a bilingual language system is not the sum of two competence and processing systems sealed off from each other.[1] Different languages may interact in systematic ways in the individual during acquisition and use. Some of these interactions can be studied experimentally, using methods from psychology or neuroscience.[2] The interplay between two languages in one brain can lead to subtle effects with opposite sign: learning and processing one language can either facilitate or interfere with learning and processing of the other language. These effects may depend on various factors, such as the similarity between the two languages (languages can resemble each other and differ from each other in multiple dimensions of phonology and grammar), the age at which the learner acquired each language, the level of proficiency attained in each language, concurrent factors like biculturalism (which can introduce aspects of semantic and cognitive variation), and many more. Being bilingual may well be a huge advantage for the individual: speaking (and reading and writing) several languages is a boon in our societies, whether or not it also has additional benefits for a person's cognitive functioning or well-being. Even so, for an individual's brain, attaining and maintaining a steady state of functioning bilingualism has definite costs. It is precisely the neural consequences of bilingualism that psycho- and neurolinguistics aim at understanding.

Sensitive Periods and Second Language Acquisition

In chapter 5, we saw that research points to a limited window of opportunity during which children may acquire the sound system of the target language. Trying to learn the same system of phonology later in life typically results in slower acquisition or lower levels of attainment, or both. At least superficially, this is compatible with our experience as learners of a second language. Everything else being fixed, someone who learns an additional language as a child will stand a better chance of achieving native-like competence in that language than someone who embarks on the same task as an adult. Why is that? Neurolinguists have explored for over fifty years now the hypothesis that there might be a critical or sensitive period for language acquisition as for several other systems in the brain. The idea is this. There is a correlation between the brain's developmental stage and the amount of learning that results if certain types of inputs are available during that stage. A sensitive period is defined by heightened sensitivity to those inputs, such that learning occurs with greater efficiency or better outcomes, or both.[3] Sensitive periods coincide with specific neurophysiological events, such as the formation of neurons, axonal projections (the long-distance connections between neurons), or synapses. The brain thus opens up to the environment in an attempt to capture as much relevant data as possible. During these periods of transitory neural exuberance, brain mechanisms may be specified and rendered efficient. Afterward, neural plasticity is reduced, and learning largely occurs within the constraints set up during the sensitive period.[4]

Is there a sensitive period for language acquisition? A key methodological point is that this question may be easier to address in bilinguals than in monolinguals. After all, none of us acquired our first language as adults, whereas some of us do learn an additional language early in life, during the putative sensitive period, and others acquire one later. It is precisely this contrast between early and late acquisition of additional languages that makes it possible to test ideas about sensitive periods. Indeed, as implied by the sensitive period hypothesis, there is a close relation between age of exposure to a second language and proficiency attained in that language. This type of observation could be explained by maturational changes affecting brain plasticity, such that learning a language is harder after the sensitive period. But it could also be an effect of restrictions imposed by the first language: one's acquired first-language phonology may be sufficiently deeply entrenched in the brain so as to interfere with or even prevent the acquisition of a new phonological system and eventually a new language. These types of accounts are difficult, or perhaps impossible, to disentangle. They share, however, the assumption that early experience, enabled by early plasticity, narrows the system's later learning abilities. This idea provides a springboard for explaining many of the effects reported in the neurolinguistics of bilingualism.

Before we look at those results, it is worth clarifying the idea of a sensitive period for language acquisition and the notion that early experience often restricts later learning capabilities. Such complex, high-level functions as language rely on several partly autonomous systems in the brain for visual, auditory, phonological, syntactic, semantic, and pragmatic information processing. These systems may well show different windows of enhanced plasticity—and, in fact, it seems they do—so that plasticity for first- or second-language learning, as a whole, will be a mixture of plasticity effects in each of these systems plus their interactions. It is then necessary to ask questions about sensitive periods in each of these systems independently: "sensitive period for language" is just a collective shorthand for those distinct phenomena. Several studies now support the view that the clearest effects of early functional commitment are seen for first-language phonology, that there are no sensitive period effects for lexical and for compositional semantics, and that sensitive period effects for grammar are still contentious.[5]

Proficiency and Age of Acquisition

When we peek inside the brains of bilinguals using methods such as M/EEG or fMRI, we find that the clearest anatomical and functional differences between participants are driven by two factors: age of acquisition of the second language and proficiency attained in that language. These two factors are correlated, so they are hard to tease apart. Someone who learns a second language early in life tends to attain higher proficiency in adulthood than one who has learned that language later in life, perhaps simply because the former has had more time to perfect her or his mastery of the second language than the latter. But there are ways to unyoke the effects of proficiency and age of acquisition (AoA) on experimental measures—for example, by carefully selecting participants and groups and by recruiting larger samples with sufficient variation in both variables.

Across individual studies and meta-analyses, three trends emerge. First, the same left-hemispheric perisylvian network (see chapter 3, figure 5) is comparably activated by L1 and L2, or by two native languages, in simultaneous and in early bilinguals, and when proficiency is matched between the two languages. Second, a weaker or lower-proficiency L2 results in more diffuse brain activations compared to L1 in speaking, reflecting added costs of language production in L2, and inversely in more focal activity in comprehension. These differences are more widespread for lexical semantic processing than for grammar and phonology, where only a few regions show different responses for either L1 or L2. A second language learned earlier in life leads to increased activity in networks adjacent to auditory and motor cortices, including traditional Broca's and Wernicke's areas: an early-acquired L2 may result in more automatic or sensorimotor processing than an L2 learned later in life. Third, the level of proficiency in the L2 is a more prominent factor than AoA in determining whether L1 and L2 recruit common or different brain regions in production and comprehension. In general, the higher one's proficiency level is in the L2, the more similar the active regions will be to those activated by the L1.[6]

Age of exposure and AoA are also known to have an impact on the way a second or an additional language are processed in the brain. Experiments using fMRI and EEG show similar responses in the two native languages in bilinguals. But if a second language is acquired later in childhood, from age 6 or older, activation patterns and evoked responses emerge that do not resemble, qualitatively and quantitatively, those for the L1. These differences persist until proficiency levels off. But in some domains, such as phonology or grammar, late exposure and acquisition can prevent the learner from attaining native-like proficiency. In those cases, the L1 and L2 may fail to converge on similar neural mechanisms and to share the same left-lateralized perisylvian networks. The neural organization of late-learned languages, when native proficiency is not attained in at least some domains, tends to show lesser lateralization and presents greater variability across individuals. Moreover, early and late acquisition will differ also in terms of the learning problem to be solved. In late L2 acquisition, meanings have already been acquired with the L1; hence, learning will amount to mapping novel phonological and grammatical forms to known semantics. Yet the same developmental progression from meaning to grammar attested in L1 is found in L2 acquisition too. For example, Lee Osterhout and colleagues showed that ERP responses to verbal person agreement in L2 French, if the contrast between inflectional morphemes is phonologically realized ("Tu adores/adorez* le français"), triggers an N400 effect after one month of instruction and a P600 after four months. This finding suggests that the costs of processing are gradually being shifted from the semantic system to the grammar system and that the L2 recruits neural resources increasingly similar to those engaged by the L1.[7]

Late acquisition of a second language can lead to persistent fine structural differences in the brain. Recent work shows that people who have learned an L2 during their school years (age 8 or older) have a thicker cortical sheet in subregions of the left inferior frontal gyrus (LIFG, recall chapters 3 and 4) than either monolinguals or early and simultaneous bilinguals. The latter groups did not differ in cortical thickness in any brain region.[8] A meta-analysis of imaging studies has found that the same piece of cortex, LIFG, exhibits different levels and patterns of activation in the L1 and L2, irrespective of AoA, in highly proficient L2 speakers, and across levels of linguistic structure: lexical semantics, phonology, and grammar.[9] This core node of the left-hemispheric language network may therefore be one of the few regions in which structural and functional differences between the L1 and a nonsimultaneous L2 persist in spite of variation in proficiency levels and AoA.

Language Control and the Bilingual Advantage

The upshot of our discussion so far is that two languages tend to share neural tissue and cognitive resources in the brain. This may be expected if the languages are learned simultaneously or with a temporal overlap, if they are used regularly in similar or contiguous contexts, and if they share representations (e.g., meanings and some syntactic forms) or are similar at the phonological or grammatical levels. All of these factors conspire to create conditions for potentially catastrophic interference between two or more languages in the same individual. Automatic online interaction effects between languages are now well documented, showing, for example, that the meanings of content words are accessed spontaneously and rapidly in both languages of fluent early bilinguals, even when participants are instructed to ignore words in one language or the other.[10] Indeed, it is a major computational problem for a multilingual brain to maintain control over all languages, ensure efficient processing in each, and reduce unwanted disruption due to reciprocal intrusion between them.

A clear example of the problem of language control is language switching. Bilingual speakers and listeners should have a degree of control over which language is accessed, activated, and used on each given occasion. This requires continuous investment of neural resources and applies to all types of bilingual speakers, including bimodal bilinguals (users of spoken and signed languages). Lateral areas of frontal cortex are involved in effortful control, also during language switching, while regions along the medial surface of the frontal lobes are essential for maintaining information about the chosen language. Frontal function and efficiency depend on the age of speakers: younger children and older adults have the most difficulty in language-switching tasks, and the least difficulty is seen around the peak of executive and cognitive control skills, from late adolescence to adulthood. Frontal areas may be joined by subcortical structures, such as the basal ganglia, and regions of parietal cortex in tasks that involve language choice, switching, or maintenance of the selected language. These brain areas differ from those involved in spoken, written, or signed language production and comprehension as such.[11]

The realization of the role of prefrontal functions in language control leads straight to the question of whether bilingualism can have beneficial effects on the normal wax and wane of executive and control functions—in particular, whether early exposure to multiple languages may confer a cognitive or executive advantage to bilinguals. The reality of the "bilingual advantage" has been a hot topic of discussion for some years now, with potentially important practical and educational consequences. Several empirical and theoretical studies have implied that bilinguals perform better than monolinguals in nonlinguistic tasks requiring or involving forms of cognitive and executive control, although recent studies have challenged this conclusion.[12] Research has also suggested that bilingualism has a larger impact in old age than in adulthood,[13] acting as a protection against cognitive decline. Bilingualism may contribute to building a cognitive reserve for individuals, but it is not clear to what extent these effects—and other putative illustrations of the bilingual advantage—are due to bilingualism as such or to learning experiences requiring enhanced cognitive control from earlier in life more generally. Perhaps mastery of a second language is all there is to the "bilingual advantage," leaving aside small effects that are detected only in the lab and may have little consequence in everyday life.

> It is a major computational problem for a multilingual brain to maintain control over all languages, to ensure efficient processing in each, and to reduce unwanted disruption due to reciprocal intrusion between them.

7
Literacy and the Brain

It may have come as a surprise to you that most of us are at least bilingual, but it will probably not shock you to learn that nearly all of us can read and write in at least one of the world's writing systems. In recent decades, literacy rates globally have soared to well over 80 percent for all adults,[1] although gaps remain with lower percentages, particularly for women in developing areas of Africa and Asia. Beyond educational and professional contexts, information is increasingly made available in textual form, and social interactions now occur much more frequently through various written media. Speech is still prominent, but language is as much a visual matter now as it is an auditory one. The importance of the visual modality for language is also emphasized by the fact that vision mediates two forms of linguistic communication: written and signed. One aim of neurolinguistics is to study known and emerging forms of visual language in the brain and how they differ from the auditory form. Understanding the literate brain sits at the top of this agenda.

Speech has a deep evolutionary origin that goes back several hundred thousand years in the history of the homo lineage (more in chapter 10). Reading and writing, however, are relatively recent cultural inventions, just a few thousand years old.[2] Alphabetic systems, in which graphemes, or the smallest distinct written signs, correspond to elements of sound (such as phonemes), are probably just four thousand years old. The transition from stylized pictorial representations of daily objects or events to alphabetic writing systems appears to have taken only a few thousand years. This may not strike us as particularly fast, given the rates of cultural innovation we are currently witnessing. However, it is impressive from the perspective of neuroscience, as the human brain could not have changed much either structurally or functionally in such a short period of time. It seems unlikely that any parts of the brain could have evolved specifically for reading and writing. A few thousand years is insufficient time for neural adaptations to become fixed in our genes. More plausibly, literacy is made possible by the existing macrostructure of the cerebral cortex: the human brain is reading-compatible. Yet reading and writing skills have to be acquired anew by each generation, harnessing both the fixed architecture of the visual and auditory systems and plasticity in language networks and beyond.

Historical Roots and Early Findings

Research on reading in the brain began in the early days of aphasiology, not long after the groundbreaking discoveries of Broca, Wernicke, and Lichtheim. The classical models of language in the brain discussed in chapter 1 were centered on speech and hearing, but vision was soon understood to be a major channel through which linguistic information in the brain could be accessed. Important early results were produced by Joseph Jules Déjérine (1849–1917). In 1891, Déjérine described the case of a patient who had lost the ability to read and write in consequence of a lesion to the angular gyrus (AG)—a region posterior to Wernicke's area and sitting approximately where the temporal and parietal lobes meet. Déjérine called this "a case of verbal blindness with agraphia." He conjectured that the lesion destroyed a "visual memory centre for words," analogous to Wernicke's auditory center (chapter 1), rendering the patient unable to read and write.

In 1892, Déjérine described a second patient who, in 1887, suddenly lost the capacity to read. The patient had a visual acuity of 8/10, he could recognize and name objects and pictures, but he presented with a right hemianopia: he could not see in his right visual hemifield.[3] The patient could speak fluently and understand spoken language. He could also write, spontaneously and upon dictation, but what he wrote, he could not read back. The patient could "read" by tracing the outlines of letters with his fingertips, and if one formed letters by moving the patient's hand through air, he would correctly name those letters. On January 5, 1892, a new cerebrovascular incident left the patient with impaired speech and unable to write. He died a few days later. When Déjérine examined his brain, he saw older lesions of (a) the medial and inferior aspects of the left occipital lobe, including calcarine cortex, part of the visual system, and (b) of white matter wiring up the right and left visual cortices to language areas of the left hemisphere, including portions of the corpus callosum, which connects the two hemispheres. He also found (c) a more recent lesion of AG and adjacent areas of temporal and parietal cortex (figure 6).

> Literacy is made possible by the existing macrostructure of the cerebral cortex: the human brain is reading-compatible.

It is instructive to see why this patient was unable to read and eventually incapable of writing. What mechanisms are disrupted by those lesions? In order for a visual word to be read, there must be functional connections between the visual system and language areas of the left hemisphere. In this patient, damage to left occipital cortex (a) caused right hemianopia. But damage to the left visual cortex alone may not compromise reading or writing skills. Such a patient would still see words in the left visual hemifield (figure 6) because the right occipital cortex is intact and visual information can be passed on to language areas of the left hemisphere by the corpus callosum and by other white matter tracts. These connections are severed in Déjérine's second patient (b; X in figure 6). It is difficult to say whether the loss of reading skills in this patient was due to the lesion in the corpus callosum, in occipital white matter, or both. What is quite certain is that these lesions together suffice to interrupt traffic between right occipital cortex and left hemispheric language regions, leaving the patient unable to recognize the words he sees.

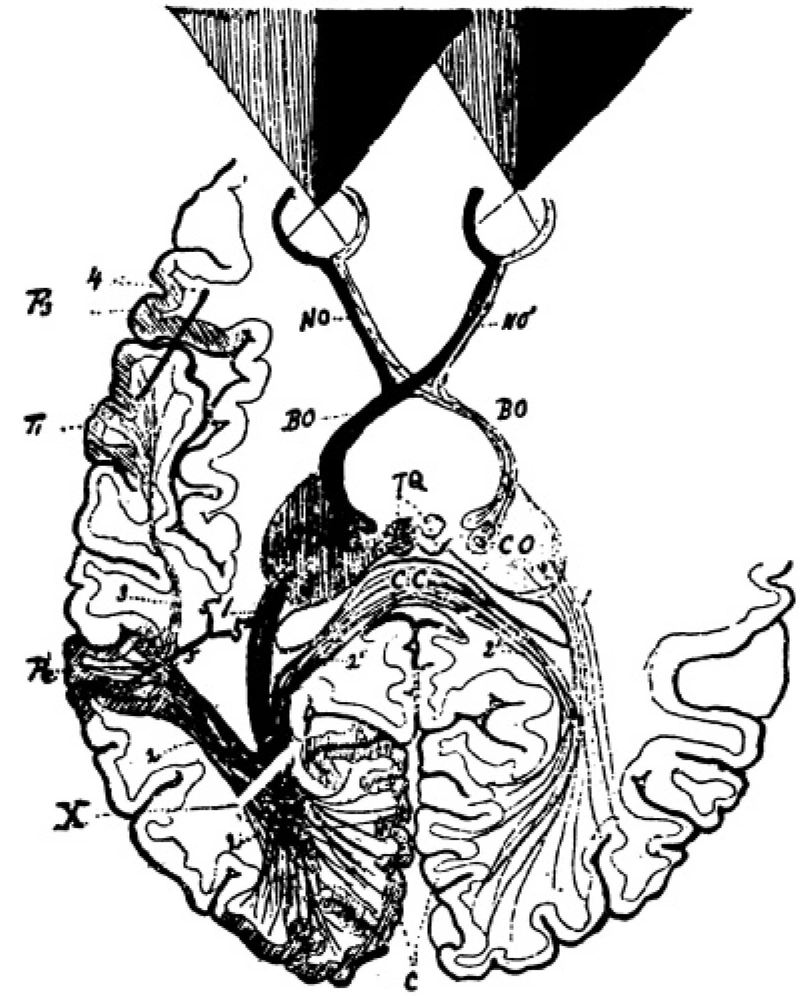

Figure 6 Horizontal cross-section of Déjérine's second patient's brain. Labels: NO, optic nerves; BO, optic tracts; C, calcarine cortex, destroyed on the left; Pc, left angular gyrus; X, lesion in white matter fibers connecting the left and right visual cortex to the left angular gyrus; CC, corpus callosum; F_3, third frontal gyrus (posterior LIFG); T_1, superior temporal gyrus. Source: J. Déjérine, "Contribution a l'étude anatomo-pathologique et clinique des différentes variétés de cécité verbale," *Mémoires de la Société de Biologie* 4 (1892): 61. The figure is also reproduced and discussed by N. Geschwind, "The Anatomy of Acquired Disorders of Reading," in *Selected Papers on Language and the Brain*, 1–17 (New York: Springer, 1974).

In 1891, Déjérine had argued that a lesion to the AG was associated with "word blindness" and impaired writing (alexia with agraphia), and he had proposed that AG stored visual memories of words. To save that hypothesis, he now had to argue that the lesions causing pure word blindness (alexia without agraphia) in the second patient were those disconnecting the right visual cortex from the left AG. Déjérine held that visual information could not flow from either the left or right visual cortex to the AG because of the injury to left occipital white matter (X; see figure 6). AG cannot be reached by residual visual signals, hence “visual memories of words” cannot be accessed. Before the recent lesion of AG (c), the patient was able to write: language and motor areas of the left hemisphere could communicate with AG. But when AG too was damaged, the patient also became agraphic, just as had happened for the 1891 case.

Where’s the Brain’s Letterbox? A Modern Take

Déjérine’s hypothesis that the AG is the brain’s center for visual images of words quickly came under attack. Critics rejected the idea that optical representations of words are stored in a single center, be it the AG specifically or some other region of the left hemisphere. For instance, Wernicke had proposed that visual memories of words are bilaterally localized, counter to Déjérine’s view that the left AG is the key functional node of the reading and writing networks of the brain. The debate persisted for some decades,[4] but no firm conclusion or significant new results emerged until the advent of neuroimaging technologies and their application to the study of the literate brain, about thirty years ago.

Since Déjérine’s early studies, we have learned that pure alexics (patients who cannot read but can write) fall into two broad categories: patients such as Déjérine’s 1892 case, who cannot read even single letters, let alone words, and patients who can recognize single letters and can, with considerable effort, read words letter by letter. It is therefore conceivable that there is a brain area or region that enables recognition of written characters—it has been aptly named “the brain’s letterbox”— whether they form meaningless letter strings or actual or possible words in a given language.[5] Studies of pure alexia using imaging data have discovered that the site of maximum lesion overlap in these patients is not the left AG but a region on the bottom surface of the left hemisphere, where the occipital and temporal cortices join. This region is referred to using either anatomical nomenclature (the left fusiform gyrus and adjacent occipito-temporal sulcus) or novel functional labels (e.g., the visual word form area, VWFA). Many fMRI and MEG studies have found increased activations of this region in tasks involving reading letters and words.

A key result is that this region displays a gradient of activation depending on the nature of the input, specifically the extent to which a given stimulus resembles a real word. All letter-like stimuli, including false font strings—stylized signs with obvious features such as lines, arcs, and angles, but no letters— infrequent letters (JZWYWX), frequent letters (QOADTQ), frequent bigrams (QUMBSS: the sequence QU is a frequent bigram), frequent quadrigrams (AVONIL), and real words, activate posterior occipito-temporal cortex. But only word-like letter strings—frequent quadrigrams and real words—reliably activate anterior portions of the same area: responses become more selective for actual words in more anterior segments of the fusiform gyrus (figure 7). A similar gradient-like organization was also observed in other brain regions, most clearly in the left inferior frontal cortex (figure 7). This area is not a reading or visual center but probably contributes to the reading process by keeping information online and by creating different "copies" of the same letter as it occurs in the same n-gram, letter string, or word.

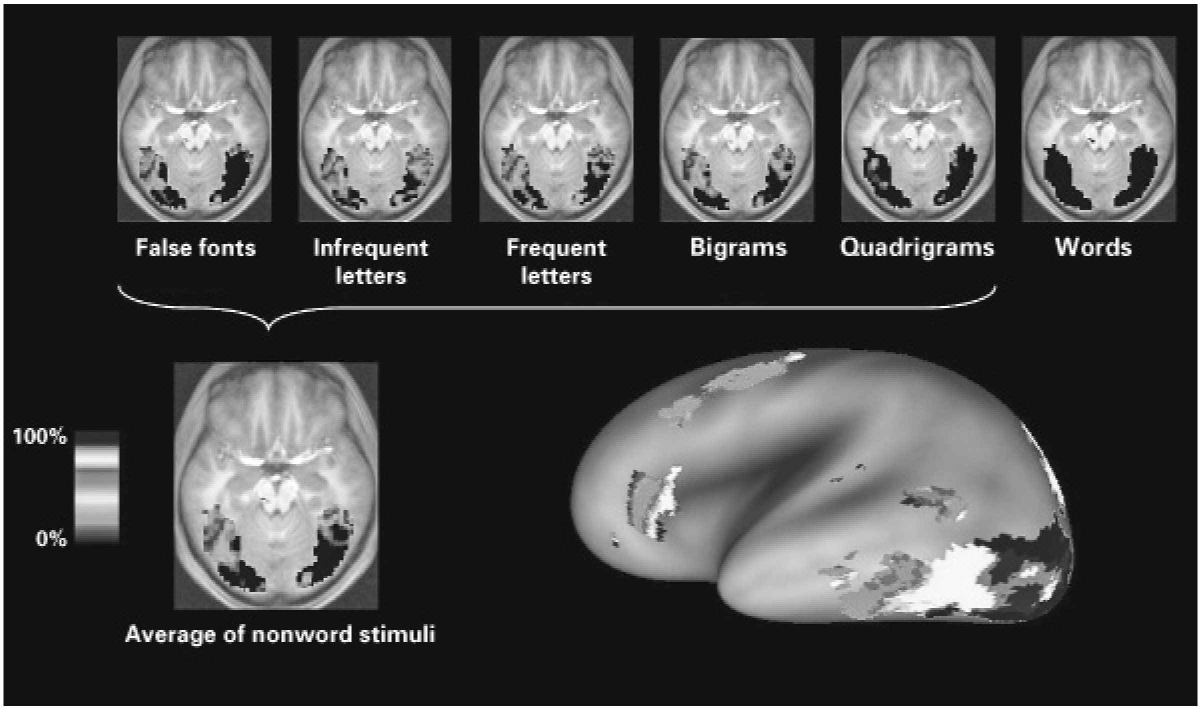

Figure 7 Gradient-like functional organization of occipito-temporal cortex. Top: Horizontal cross-sections of the brain showing percent activations (color bar) by five types of stimuli relative to real words: more anterior portions of the occipito-temporal cortex are recruited by stimuli increasingly similar to words. Bottom left: Gradient image showing averages over nonword stimuli. Bottom right: Left lateral view of the average gradient image. Source: Reprinted with permission from F. Vinckier, S. Dehaene, A. Jobert, J. P. Dubus, M. Sigman, and L. Cohen, "Hierarchical Coding of Letter Strings in the Ventral Stream: Dissecting the Inner Organization of the Visual Word-Form System," *Neuron*, 55 (2007): 143–156.

I invite readers to consider the implications of these results. Imagine two focal, well-localized lesions of fusiform cortex: one in the posterior portion and one in the anterior portion. Which one is more likely to produce an impairment in processing all kinds of letter-like stimuli? Which one may instead cause a selective deficit in recognizing real words? Check the endnote once you have found an answer.[6] In addition to lesions to the fusiform gyrus, one can consider the implications of lesions to connections to and from this region. Damage to white matter downstream of the brain's letterbox (roughly from the fusiform cortex to more anterior brain regions) may leave the patient able to recognize that what he sees is a written word but unable to tell what its meaning or pronunciation might be. This type of injury may break down communication between the brain's letterbox and other regions, including left frontotemporal language cortex. In contrast, lesions to white matter carrying neural signals to the fusiform gyrus would likely cause pure alexia: the patient might be unable to gain visual access to information in the intact letterbox but might be capable of tactile reading or forms of orthographic decoding that do not rely on visual inputs.

This brings us back to Déjérine's data. Modern work indicates that Déjérine was wrong about the location of the visual word center: it is not the left angular gyrus. We now know that the brain treats visual words, at least initially, just like other visual objects. In the ventral visual system or the inferior part of the temporal lobe, we find areas sensitive to various types of visual objects that are especially important to us, such as faces, tools, and buildings. Visual words end up being recognized where other visual objects are: the left fusiform gyrus is indeed part of the ventral visual stream. In Déjérine's 1892 patient, the lesion to left occipito-temporal white matter (b) was sufficient to disconnect bilateral visual cortices from the left fusiform gyrus, resulting in pure alexia. Recent work supports this notion.[7] But the bigger puzzle is still incomplete. For example, the (left) AG might play a role in writing and in agraphia, based on Déjérine's findings and on knowledge of the dorsal visual system—a set of parietal regions, including the AG, that represent spatial relations and locations and contribute to visual guidance and action, as required by skilled writing. Recent research has indeed found associations between reading and occipito-temporal (including fusiform) cortex and between writing and inferior parietal and temporal cortex (including the AG).[8]

Learning to Read, Brain Plasticity, and Dyslexia

How does this elaborate cortical network for reading and writing emerge and organize during childhood? How does becoming literate change the reading-compatible brain? A complete answer cannot be given here, but I illustrate a few important points with a familiar example: mirror writing. As children begin to read and write, they often write letters and whole words in the correct direction (say, left-to-right) and in the opposite direction freely, even randomly. Why is that? The child's visual system knows the general principle that objects stay what they are, no matter how one rotates them in space: object identity is invariant under rotation. If a cup is a cup when it's upright on the desk and when it's upside down on the dish rack, why should letters and words be any different? Indeed, letters and words are the exception, and all children eventually learn this fact against the general principle of rotational invariance. This is what it takes to realize that *b* and *d* and *M* and *W* (figure 8) are different letters despite the fact that they appear to be the same three-dimensional visual objects, only rotated along vertical and horizontal axes, respectively. But how does this unlearning occur? Dehaene suggests that initially, letters are learned by motor systems and by the dorsal visual stream as motor gestures in space and as three-dimensional objects that can be displaced and rotated.[9] Learning is gradually transferred from dorsal to ventral visual areas as the child learns to attend to other salient properties of letters, such as their unique orientation in two-dimensional space. The child then discovers that these properties disallow rotation and that the same restriction also applies to bigrams, n-grams, and eventually words. The dorsal and ventral visual streams would then coordinate their activities, so that the child will not only tell a *d* from a b when she sees one, but will also correctly produce distinct instances of *d* and *b* in writing.

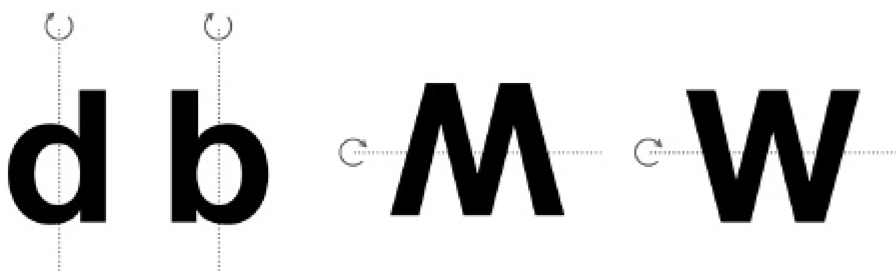

Figure 8 Letter identity is not invariant under rotation.

Becoming literate involves much more than training the ventral visual system to the idiosyncrasies of words as visual objects.[10] Recognizing real words and letter strings that correspond to pronounceable phoneme sequences in one's language requires interactions between the letterbox region and other temporal areas supporting phonology and semantics. Learning to read depends on plastic changes in and across the visual and language systems.[11] It unfolds in three main stages: an early stage, in which children learn visuomotor representations of a few whole words, such as their name; a phonological stage, where they learn to map graphemes to phonemes (this differs across languages and writing systems, depending on how transparent the interface between orthography and phonology is); and an orthographic phase, where the mapping to phonology is in place and reading is automatic and fast. These functional changes correspond to plastic changes in relevant cortical regions, across the language, visual, and motor systems.[12]

Literacy acquisition can be disrupted in a number of ways, leading to reading difficulties or dyslexia. Historically, the first reading disorders to be investigated were cases of adult patients who had lost their reading or writing skills as a consequence of brain lesions (see above). But it was not long after Déjérine's seminal discoveries that the first cases of congenital or developmental dyslexia were described.[13] The difference between acquired and congenital dyslexia is a real one: these are not two manifestations of similar underlying disruptive events in the brain. Localized cortical lesions can cause (pure) alexia in neurological patients, but one rarely finds evidence of local anatomical and functional insufficiencies in the brain of dyslexic children. Besides, the neurobiological signatures of developmental dyslexia seem to also involve auditory speech processing systems rather than just visual cortex or inferior temporal (fusiform) areas.[14] Dyslexia may be accompanied by a reduced ability to read words aloud, especially words whose pronunciation does not follow the regular pattern in a language or letter strings that do not sound as words in any language (nonwords). This is consistent with the behavioral pattern of dyslexia, in which reading (and reading acquisition) deficits relate to the dyslexic child's difficulties in converting written letter strings into spoken sounds (grapheme-phoneme conversion) and can be accompanied by speech processing deficits. These results, together with new work on the genetics of dyslexia (discussed in chapter 9), show that developmental dyslexia arises from atypical anatomical and functional organization of language cortex in early childhood, therefore preceding literacy acquisition as such.

8
Neurology of Language

Aphasiology is the study of acquired speech and language disorders. It has left a profound mark on neurolinguistics as a whole (chapter 1) and remains a live area of inquiry in the cognitive and brain sciences, with potentially significant societal impact. Brain injury, whether traumatic or not (caused by, for example, strokes, tumors, or infections), is a leading cause of disability or death, affecting millions of people every year worldwide.[1] Because language functions are distributed in the brain, chances are that cortical lesions will reduce, temporarily or permanently, a person's ability to use language. Prospects of successful rehabilitation depend on understanding the exact nature of the disorder in behavioral, functional, and neuroanatomical terms. This is the goal of modern clinical neurolinguistics or the neurology of language.

Aphasia is not a single disorder; it is a complex class of neurological conditions manifested as the impaired ability to produce or understand spoken language. There is much variation as to the type and severity of the impairment. The linguistic heterogeneity of aphasia makes the use of ideas from theoretical linguistics particularly important to come to a fine-grained understanding of aphasic disorders. Modern classifications and theories have not resulted in a coherent, complete explanation of aphasic syndromes.[2] However, in the process of discarding accounts of typical and atypical language function, a few important insights were acquired. One is that people with aphasia may make similar errors in language tasks (e.g., naming, repetition, paraphrasing), but the number of errors they make can vary: the disorder may be more or less severe. Or error rates may be similar, but error types may differ; for example, some patients may find it more difficult to name actions (verbs), and others may have more difficulty with naming objects (nouns). Linguistics can contribute much to the analysis of different error types and to connecting error patterns across tasks.

As noted in previous chapters, linguistic processes are executed by several centers distributed broadly across the cortex. Another insight coming from aphasiology is that information flow within these networks can be mediated by multiple connectivity routes. Some speech and language processes can be carried out in different ways in the brain, exploiting different regions and connections between them. If one route is damaged or fails functionally, the functions it performed may be taken over by another route, so long as its connections are preserved.[3] This is not the only way the brain responds adaptively to lesions. Some functions could be remapped from a damaged area to a spared one. But rerouting may be a more effective repair solution for widely distributed functions than remapping. Before we look into the details of language reorganization after brain damage, let us consider the nature of aphasic disorders.

Traditional and Emerging Models of Aphasia

Aphasic disorders are traditionally classified depending on whether speech production, comprehension, or repetition is impaired. Each of these skills may be damaged at least partly independently of the others. If skill X can be impaired while Y is spared or vice versa, we say that X and Y can be *dissociated*. When impairments of X and Y co-occur, we say that the two functions are *associated*. X and Y can be rather specific speech or language skills (such as repetition of phonemes vs. monosyllabic words or comprehension of words vs. sentences). Indeed, models of aphasic disorders have progressed to assess sufficiently finely grained speech or language skills for diagnostic purposes. But dissociation patterns can also be found between the broad functions of production, comprehension, and repetition.

The main bifurcation in traditional models is whether spoken language production is fluent or nonfluent. *Global aphasia*, the most severe form of aphasia, occurs when nonfluent speech is accompanied by difficulties with both (spoken) language comprehension and repetition. A patient can present global aphasic symptoms shortly after a stroke or other types of brain damage. Such symptoms can even persist beyond the acute phase of recovery if brain lesions extend to multiple cortical areas or white matter fibers, or if they compromise core language networks: the left inferior frontal cortex, posterior temporal cortex, parietal cortex, or connections between them (chapter 4).

If speech is disfluent, that is, production and repetition are impaired, but language comprehension is spared, then the disorder is classified as *Broca's aphasia* (chapter 1). In traditional models of aphasia, it was assumed that lesions to Broca's area (posterior LIFG) or to Broca's region (LIFG as a whole) were sufficient to cause Broca's aphasia. Now we know that lesions limited to Broca's area rarely result in complete Broca's aphasia and that damage to other areas of the language network can disrupt speech production or repetition. If speech is disfluent but repetition is intact, we speak of transcortical aphasia, of two types: *transcortical motor aphasia*, if comprehension is spared, and *mixed transcortical aphasia*, if comprehension is affected.

Four other syndromes result when speech is fluent but comprehension, repetition, or naming is compromised. *Wernicke's aphasia* (chapter 1) is usually characterized by poor comprehension and repetition. Naming too is often impaired. Wernicke's aphasia is not limited to semantics, but also involves phonological systems necessary for both comprehension and production. In these aphasic patients, speech is fluent but may lack structure and coherence. In addition, reading and writing can be affected. Traditionally, Wernicke's aphasia was associated with damage localized to the left posterior superior temporal cortex, but impaired comprehension can result from lesions to cortical areas or connections involving (bilateral) temporal ventral pathways (recall Hickok and Poeppel's dual-stream model from chapter 4). *Conduction aphasia* is defined by fluent speech, spared comprehension, but impaired repetition, lexical selection or choice of words, and internal monitoring of speech. In most traditional models, conduction aphasia is linked to damage to white matter fibers connecting superior temporal regions to LIFG. If repetition is preserved, we get two more variants of fluent aphasia. If comprehension is impaired, the aphasic disorder is known as *transcortical sensory aphasia*—not to be confused with Wernicke's aphasia, often accompanied by poor repetition. If instead comprehension is intact, the deficit is limited to naming, and the patient is said to be anomic. Patients with *anomic aphasia* have difficulties with supplying the words for the ideas they intend to express, in particular content words, such as nouns and verbs, in both speech and writing. Their expressive language may present adequate grammatical structure (e.g., the presence of function words), but thoughts are often conveyed with great effort.

The traditional classification of aphasic disorders has been widely discussed and criticized. It is considered an imperfect aid for research, diagnosis, and treatment, but it remains a necessary reference point for developing newer neurolinguistic models and better classifications. An area of current research is the functional neuroanatomy of aphasia, stimulated by clinical applications of M/EEG and fMRI, and the development of new methods for mapping language functions during surgery, such as for tumor resection. Recent advances come from studies of stroke patients. Stroke can be ischemic (blood vessels get blocked; thus blood flow to cortical areas is interrupted) or hemorrhagic (blood vessels rupture, and blood invades neural tissue); in both cases, brain cells are damaged. Around 30 percent of all cases of stroke result in acute aphasia.[4] Damage to core areas of speech and language networks (e.g., LIFG or left temporal regions) is likely to cause severe or even global aphasia. Damage to the dorsal stream connecting temporal, parietal, and frontal language regions is accompanied by nonfluent aphasia, or motor speech impairment, while ventral stream lesions can compromise comprehension. Naming, repetition, grammar, and other language processes rely on interactions between the two streams and therefore can be damaged by lesions to different parts of the core network.[5] These conclusions are supported by studies of brain tumor patients.[6]

Emerging classifications of aphasia increasingly take into consideration the causal origin and functional evolution of the disorder. While brain injury results in sudden damage to language function, neurodegenerative diseases, such as Alzheimer's disease and frontotemporal degeneration, can cause a gradual, irreversible loss of language skills. *Primary progressive aphasia* (PPA) may result from the deterioration of cortical tissue in speech and language networks. It often begins as a subtle loss of performance in certain language tasks, such as word finding and naming, and may develop into severe forms of nonfluent or even global aphasia.[7] As for aphasic syndromes resulting from traumatic brain injury or stroke, the functional and behavioral profiles of PPA will depend on the location and extent of damage to language networks or other brain regions at a particular stage of the disease. PPA presents at least three variants: (1) semantic PPA, where patients may experience difficulties in naming or word comprehension as a result of abnormal changes to temporal lobe function; (2) nonfluent/agrammatic PPA, where patients have difficulties organizing and articulating speech in terms of phonology and grammar; and (3) logopenic (from Greek, meaning “lack of words”) PPA, where patients often show poor word finding, naming, and repetition skills but their semantic, syntactic, and motor speech abilities are spared.

> Emerging classifications of aphasia increasingly take into consideration the causal origin and functional evolution of the disorder.

Language Reorganization after Brain Damage

Neurodegenerative disease currently offers little hope of restoration of function, but the prospects are sometimes considerably better after (localized) brain lesions. The brain can respond to damage in different ways. Two principles are particularly important in this context.[8] The first is *degeneracy*: the idea that the same function can be performed or approximated by different systems or in different ways, a bit like reaching the same destination from home via two different routes: if one is not accessible (suppose it is interrupted by roadwork), the other can be used. If some cortical structures are lesioned, information may be rerouted through either spared regions of the left-hemispheric language network or corresponding (homologous) right-hemispheric regions, or nonlanguage areas. Another principle is *functional reserve*: the idea that brain networks have residual capacities that are normally not used but could be mobilized in special circumstances. Degeneracy and functional reserve allow the brain to react to damage by compensating for loss of function.[9]

Studies of patients with left temporo-parietal or left frontal stroke have underscored the dynamic nature of the reorganization process. In general, the brain’s response to stroke unfolds in three phases: (1) the *acute phase*, shortly after the stroke, where the destructive effects of the lesions on brain function are most evident; (2) the *subacute phase*, when reorganization is underway and some functions are gradually or partially regained; and (3) the *chronic phase*, when performance stabilizes to a level similar to or lower than the normal one. Damage to different structures in the language network produces very different effects in the acute phase. The pattern is complex and may be simplified as follows.[10] Left temporo-parietal stroke tends to reduce overall activity in the language network, also in more distant regions such as LIFG. Instead, frontal stroke often depletes local (frontal) activity, while responses in temporo-parietal cortex may be relatively preserved. Functionally, this suggests that activity in frontal regions depends on inputs from temporo-parietal regions, while the reverse is not necessarily the case.

The reorganization of language in the brain may differ between temporo-parietal and frontal stroke patients. There is in both cases increased engagement of areas contiguous with the lesion (perilesional cortex) and of spared cortex in language regions on the left hemisphere, in particular during the subacute phase. This is consistent with local remapping of functions and functional reserve. There is intrinsic residual capacity within the language network that can be deployed in response to damage; this holds for temporal-parietal and frontal areas alike. In temporo-parietal stroke patients, there is little or no functional remapping to homologue regions of the right brain. Instead, domain-general networks, engaged in multiple tasks across cognitive domains, are recruited to compensate for loss of functions in left temporo-parietal areas. In contrast, in frontal stroke patients, we see a rapid compensatory response on multiple fronts simultaneously: in intact language areas, in right-hemispheric homologues of the PFC, and in domain-general systems. These results do not tell us what the brain is doing functionally to recover from stroke, but they show that it is using several strategies simultaneously: restoration within spared language areas of the left hemisphere and compensation via domain-general systems. In stroke patients, these strategies dominate over recruitment of regions or networks unrelated functionally to speech and language.

The Puzzle of Agrammatism

Some nonfluent aphasic patients present a deficit that may be roughly described as impaired grammar. This has been called *agrammatism*, and it too is a complex disorder of productive and receptive language. Agrammatic speech lacks free-standing grammatical morphemes (e.g., function words, such as prepositions, pronouns, and auxiliaries) and inflectional morphemes or agreement features (such as tense). Traditionally, it is contrasted with *paragrammatism*, in which grammatical structure is preserved, but lexical or functional elements are wrongly selected or substituted with incorrect elements. Agrammatism was thought to be a manifestation of a motor speech disorder, particularly in Broca's aphasia, before the idea of damage to a syntactic component of the language system began to be taken seriously and explored systematically. This led to exciting research programs, with significant contributions from theoretical linguistics.[11]

> There is intrinsic residual capacity within the language network that can be deployed in response to damage.

One big puzzle of agrammatism is whether grammar is impaired in both speech production and comprehension. Early research purported to show that agrammatic patients have comprehension deficits: they understand syntactically complex yet semantically constrained sentences ("The apple that the boy is eating is sweet"), where interpretations are suggested by factual knowledge, but they have difficulty with sentences that require syntactic processing ("The cow that the monkey is scaring is yellow").[12] Later work has shown that agrammatic speech does not necessarily imply a comprehension deficit and that agrammatic patients are able to judge whether sentences are grammatical. Thus, in agrammatism, syntactic knowledge may not be lost, but syntactic processes might be weakened, slowed, or cut off from systems of phonology and semantics.

More recent studies on the neural bases of impaired grammar have indicated that agrammatism may result from frontal cortical lesions, while paragrammatism is associated with posterior temporal damage. These two syndromes are more likely to arise with left- than right-hemispheric injuries. This evidence points once again to the distributed nature of linguistic and grammatical representations and processes in the brain.[13] Furthermore, grammar is connected to systems of phonology and semantics. These interfaces subserve the language system's capacity to dynamically reorganize itself in response to frontal or temporal damage. For example, in ERP experiments, agrammatic patients show N400 effects where control participants show P600 effects in sentences containing grammatical errors.[14] These results suggest that the semantic system compensates in real time for a loss or reduction of grammatical processing skills, consistent with the notion of multiple-route plasticity (see above).

Debates now focus on whether agrammatic speech is a reflection of the deficit or the brain's adaptation to it. The deficit itself is a slowing down or weakening of lexical retrieval and syntactic processing, and the brain's reaction consists in simplifying or skipping those difficult processes, which results in the characteristic pattern of agrammatic speech. Another area of discussion is whether the disorder affects syntax as such or its interfaces with phonology and semantics, especially at the discourse level. Indeed, some of the known symptoms of agrammatism include impaired production or comprehension of sentences where syntax and discourse semantics interact—with relative pronouns, tense, or other grammatical devices crucial for expressing meaning beyond the sentence level. Agrammatism is now reframed as a deficit of how phonology, morphology, and semantics are organized for spoken language production and comprehension. These developments challenge the idea of an autonomous system of grammar in the brain which can be damaged selectively and independently.[15]

9
Neurogenetics of Language

Language scientists have long been interested in what we call the genetic component of language: any property of language, or of our capacity to acquire and use languages, that precedes learning or exposure to environmental input.[1] This is a legitimate concern that has, however, fueled much speculation. All children share a capacity to learn language, and all languages share some properties (formal structures and operations) that do not seem to be learnable based on input alone. Is it possible that such properties of languages or the learning capacity as such are genetically specified? Could it be that the requisite information is encoded in one or more genes and that that is what explains the fact that only humans, and no other species, have language?

To address the genetic bases of language, we need to understand what genes are and what they do. Genes are stretches of DNA that play specific causal roles in synthesizing proteins—the molecules that constitute the organism structurally and functionally. Some genes specify instructions for building a brain that can learn and process information. That is how genetics is relevant to language science: through neurobiology or, more precisely, through neurogenetics. Neurogenetics is the study of how specific genes inform the way a (human) brain is built, maintained, and repaired—and to some extent also how it functions in typical and atypical circumstances. Neurogenetics may not explain why languages are the way they are—for example, why they share certain basic properties and why they differ in others. But it can elucidate how a brain that can learn to use any human language is constructed at the molecular and cellular levels. It can also illuminate what can go wrong during the construction process so that the development of language or cognition may be disrupted or delayed.

Genes, Brain Development, and Brain Function

The road from genes to brain, cognition, and behavior is long and tortuous. It has a relatively simple origin (genetic information is carried in one molecule, DNA, by one code, built out of a mere four nucleotide bases) but a stunningly complex destination (brains and minds). Also, only a small fraction of all human DNA carries genetic information, and the number of protein-coding genes is limited to 20,000 to 25,000. Each gene is surrounded by bits of DNA that serve regulatory functions, including instructions on when and where to turn on or off certain genes (more below). We inherit two copies of each gene, one from each parent. Each gene comes in different variants that produce different effects in the organism (e.g., on eye color), in interaction with other genes, with regulatory elements in DNA, and with environmental factors.[2]

Genes specify instructions to build proteins, which in turn make up an organism's structure and functions. A key process here is *gene expression*: a sequence of molecular events that read out individual genes in DNA and produce specific effects in cells. Each cell in an organism, including neurons, contains the same DNA and therefore the same genes. But not all genes are expressed everywhere in the body or at the same time during development, and genes are often expressed multiple times at multiple locations. In general, genes produce specific cellular effects depending on where and when they are expressed. In the brain, gene expression determines, through protein synthesis, a range of processes leading to the correct formation and function of brain circuits, including language networks—for example, how neuronal precursor cells proliferate and differentiate to become specific neuron types, where these should migrate in the cortex during development, and how they should wire up with other neurons. Even learning, memory, and online information processing, which modify synaptic connections between neurons, are mediated by proteins and genes. All of these processes require triggering by internal actors (DNA) or external events (inputs). These are not limited in time to development but play out continuously throughout life.

Genetics aspires to explain morphological variation in organisms (their phenotype) in terms of genetic variation (or genotype). Brain anatomy differs across individuals in some measures, such as size and thickness of cortical areas. It is now possible to assess how structural variation in the brain depends on common genetic variants (also known as *gene polymorphisms*) by combining structural imaging (MRI) with genetics in large data sets. Recent studies have found that over three hundred regions of DNA can influence cortical structure in humans. Some genes are expressed early in development, even before birth: those affect how the cortex folds up and the surface of cortical regions. Other genes are expressed later in life, and through adulthood, and determine cortical thickness. Several DNA stretches affect the way language regions are formed, including subportions of the LIFG and temporal cortex.[3] But the same genes direct the growth of other brain regions, and in some cases even of nonneural structures in the organism. There are no genes specifically and exclusively dedicated to regulating the development of speech, language, or reading networks in the brain.

> In the brain, gene expression determines, through protein synthesis, a range of processes leading to the correct formation and function of brain circuits, including language networks.

FOXP2: A Speech and Language Gene?

The nexus between genes, brain, cognition, and behavior can be studied through common polymorphisms in typical or atypical populations or through rare mutations in individuals. In 2001, a rare gene mutation was described that resulted in a severe developmental speech and language disorder.[4] For years, it had been known that the members of a family living in the British Isles could inherit deficits in sequencing the complex mouth and face movements that enable fluent speech. This deficit, known as childhood apraxia of speech (CAS), can be accompanied by impaired spoken language production or comprehension. It was found that all affected members of the family carried a rare point mutation of gene FOXP2. This missense mutation (amino acid substitution) in one copy of FOXP2 is sufficient to undermine the functions of the encoded protein, which is a transcription factor that drives the expression of other genes. This is a cascade of effects that ultimately results in a reduced ability to control the fine movements required by speech production.

The press saluted the discovery with sensationalist headlines, such as “Scientists Unlock Mysteries of Speech” and “First Language Gene

Discovered."[5] Simon Fisher, one of the scientists involved in the original FOXP2 study, was eager to clarify: "We don't think this is *the* speech gene. It influences the ability to speak clearly. The mutation doesn't remove the capacity for speech completely." Over the past decade, neurogeneticists have learned a lot about FOXP2 and its functions. Human and animal studies show that it supports the development and function of neuronal circuits relevant to motor skills and vocal behaviors, besides being expressed in other tissues such as lung, heart, even bone.[6] Neuroimaging experiments in individuals carrying a FOXP2 mutation show abnormal structural features and activation patterns in language tasks across cortical and subcortical regions of high FOXP2 expression—parts of the striatum, the cerebellum, and the inferior frontal gyrus (Broca's region).[7]

In addition to research on rare mutations, studies on common polymorphisms have identified several genes that may influence the organization and function of speech and language networks in the brain. FOXP2 is just one of many genes expressed in language-relevant regions, such as the LIFG and temporal cortex, especially during development.[8] The CNTNAP2 gene, for instance, shows high expression in early development in both prefrontal cortex and superior temporal cortex, both part of the core language network. Indeed, some variants of CNTNAP2 have been associated with poorer performance in language tasks in children with specific language impairment (SLI), a developmental deficit affecting a child's productive and receptive language skills but not speech articulation specifically, in contrast to CAS.[9] So while rare mutations can cause rare disorders (FOXP2 and CAS, documented in 0.1 percent of children), common gene polymorphisms can be reflected in more prevalent, but still atypical, phenotypes (CNTNAP2 and SLI, diagnosed in 7 percent of children around age 6). CNTNAP2 is one of many genes whose expression patterns are regulated by FOXP2, which again underscores the wide potential role of the latter gene in human cortical development and eventually in speech or language functions.

We have already noted that the links between genes, brain, and (typical or atypical) speech or language behavior are bound to be subtle and intricate. Another caveat is that the direct effects of most common gene polymorphisms on brain structure are usually small. If they produce measurable impacts on behavior, such as on performance in speech or language tasks, it is because of the domino effect that they trigger on the function of neuronal circuits, often protracted over several years in infancy and childhood. Such cascading effects on speech or language skills are larger in atypical groups. The links between common gene variants and cognition or behavior are otherwise difficult to establish in typical populations. For instance, small-sample studies reported associations between common variants of FOXP2 and task-based activity in LIFG and between other gene polymorphisms and functional asymmetry of superior temporal regions, all part of core language networks. But these findings have not been replicated with larger participant samples.[10] So far, there is very little evidence for the effects of common polymorphisms on variability in neural anatomy or function in the general population. Neurogeneticists now believe that either the impact of FOXP2 on brain structures is confined to rare instances of disruptive mutations or that the effects of common FOXP2 variants on the brain are too elusive to be detected with current imaging methods.

Genetic Risk Factors for Developmental Dyslexia

Developmental dyslexia is defined by specific difficulties in the acquisition of reading and writing skills, despite typical cognition, perception, and early educational opportunities. It is diagnosed in about 5 to 10 percent of children in industrialized countries. It has been associated with deficits of aspects of phonological and orthographic processing and with dysfunction of the perisylvian language network. The severity of dyslexic disorders can be modulated by the degree of transparency of the phonology-orthography interface in a given language (how words are pronounced vs. how they are written). One leading hypothesis is that the auditory-phonological deficit in dyslexia originates from microstructural anomalies of the temporal cortex. Research in neurogenetics is gaining new insights on the alleles (gene variants) associated with either developmental dyslexia or variability in reading, writing, and spelling skills in the general population.[11]

Several candidate dyslexia genes have been recently identified on at least five chromosome regions.[12] People who carry these “susceptibility alleles” are at risk of developing a form of dyslexia in their childhood. There is evidence linking some of these alleles to anomalies in neuronal migration or in wiring during cortical development, eventually resulting in anomalies of temporal lobe structure and functions.[13] It has also been shown that disrupting the expression of specific genes, associated with dyslexia in humans (e.g., DCDC2), leads to major impairments of auditory functions in mice.[14] Studies have investigated the genetic origin of the cortical malformations that can disrupt the function of neuronal circuits critical for reading. Such anomalies can be produced artificially in animals by blocking the expression of dyslexia susceptibility genes, such as DYX1C1, KIAA0319, DCDC2, and ROBO1.[15]

Especially studies based on animal models can test alternative causal hypotheses about how the expression of genes implicated in dyslexia in humans affects the correct development of neuronal circuits. Studies in humans have used sophisticated statistical methods to test associations of common gene polymorphisms with structural properties of key language and literacy regions of the brain, including IFG and the middle and superior temporal cortex.[16] Two of the most important polymorphisms linked to language and literacy in a recent genome-wide association meta-analysis (rs59197085 on gene CCDC136/FLNC, and rs5995177 on RBFOX2) were statistically associated with reduced cortical thickness in frontal, temporal, and parietal regions involved in language and literacy. As noted for FOXP2, however, it is at present difficult to establish robust links between genetic variation in the general population and phenotypic variation in typical reading, writing, and spelling skills. Identifying the genetic risk factors of developmental dyslexia seems more feasible, especially by combining genome-wide association data, brain imaging, and manipulations in animal models. It should be emphasized that genetics provides only a partial picture of risk factors. Several environmental variables may increase or decrease the risk of developing dyslexia. A key question is how social, economic, educational, and other variables interact with susceptibility genes. It will be crucial to distinguish interactions where environmental factors can exacerbate genetic bias from those where they can have a protective effect, offsetting some of the negative impact of genetic or environmental risk factors.[17]

10
Neuroethology of Language

Like most other cognitive functions, language results from complex interactions between genes and the environment. Genes direct the growth of language networks in the brain. The product is an embodied brain, whose primary remit is not to passively receive and analyze sensory inputs, but to act adaptively in the natural and social world. Speech and language networks in the brain supply neural mechanisms for compositional communication: only humans exchange information cooperatively through open-ended systems of signals with systematic, predictable relationships between form and meaning. Communication is hugely important for individual and collective adaptive behavior in humans.

Yet several other species communicate, if not quite compositionally (e.g., primates), and even those that do not still show complex vocal behaviors—songbirds, for example. Some aspects of human linguistic communication might therefore be comparable to similar skills in other species. One aim of neuroethology is to understand the brain bases of behavior using a comparative methodology. Neuroethology can help us fractionate human speech and language into meaningful neural and behavioral building blocks and trace their origins in the evolution of avian, mammalian, and primate brains.

Human Evolution, Speech, and Language

It is all too easy to be deceived by our peculiar perspective and believe that humans are unlike any other species. As a matter of definition and fact, each species is special, but all this diversity and apparent uniqueness cannot disguise the deep evolutionary continuity that holds together the natural world. Charles Darwin (1809–1882) was among the first to think clearly about human language and communication in terms of evolutionary relatedness and continuity. His vision is encapsulated in this remarkable quote from *The Descent of Man* (1871): “The lower animals differ from man solely in his almost infinitely larger power of associating together the most diversified sounds and ideas.” “Almost infinitely larger” here reminds us that what looks like a qualitative difference may be a matter of degree: the human ability to associate forms and meanings sits on the far end of a continuum of perceptual and cognitive functions in animals.

This view implies that there could be animal models of aspects of human speech, language, or communication. By “animal models,” we mean species that show one or more perceptual, motor, or cognitive skills that correspond to the capacities that constitute speech and language in humans. This correspondence can be of two kinds: it can be a case of *homology*, where humans and other species share one or more traits because those traits were inherited from a common ancestor, or it can be a case of *analogy*, where different species show similar traits with similar functions that nevertheless evolved independently in the two species. For example, bird and bat wings are analogous as wings and homologous as forelimbs. A neuroethology of language should identify and study instances of homology versus analogy between speech- or language-relevant traits in humans and other animals. What might these traits be?

Two types of capacities and behaviors seem to be constitutive of human speech and language and attested in other species. The first is the set of cognitive, motor, and sensory skills that ethologists refer to as *vocal learning*: the ability to acquire individual sounds and sound sequences or patterns, typically by imitation, to recognize and produce freely those sounds, and to modify their structure. Vocal learning is not a common trait and is shared with considerable variation by a number of species, including cetaceans, pinnipeds, bats, elephants, and some birds—hummingbirds, songbirds, and parrots. In primates, vocal learning appears to be limited:[1] humans are probably the only living primate species showing sophisticated vocal learning skills, exhibited so clearly in early spoken language acquisition and bilingualism (chapters 5 and 6). The question is whether vocal learning in these species evolved separately, or whether instead there is a common origin and therefore deep homology linking species from birds to humans.[2]

The second capacity, semantic communication as mediated by sensory signals, such as vocalizations and gestures, is shared between humans and other primates and likely other mammals too. In humans, this is supported by what linguists call *compositionality*—the idea that combinatorial representations of sound, or other perceptual signals (e.g., gestures), and meaning are linked together in predictable, systematic ways. Syntax, recursive hierarchical structures of the form [X [Y Z] . . .] (chapter 1), is part of what enables the mapping between form and meaning. Darwin sensed that this ability was different, or “almost infinitely larger,” in humans. Now we know that other primates can associate together vocal or visual signals and meanings, yet without the flexibility afforded by syntax. We do not know whether syntax and compositionality have emerged recently in our species’ evolutionary history or whether they have deeper roots in primate signaling and communication.[3]

Neuroethology of Vocal Learning

Basic forms of vocal plasticity have been found in species that are not proficient vocal learners, such as mice and frogs. Scientists have come around to the view that vocal learning is not a discrete, all-or-none trait but a complex skill that has multiple interacting components. Vocal learning is also different from motor learning (the ability to learn to execute and modify novel complex movements) and auditory learning (the ability to learn to recognize new sounds and sequences or associations). That is also why infants can understand spoken language before they can produce it (auditory learning sets on earlier than vocal learning) and why in most primates and in some other species, auditory learning skills are more advanced than vocal learning skills. Where does vocal learning in humans come from, then? It now seems established that the answer lies not in the organization of vocal organs in humans but in the brain: a vocal learning pathway, possibly accompanied by greater neuronal density in forebrain regions, is present in several species showing imitative, voluntary, and controlled vocal learning of song or speech.[4] This new pathway may have evolved several times independently in different species, building on existing and common pathways for auditory learning, motor learning, and innate vocal production.

Vocal learning is only analogous, not homologous, across species of vocal learners: it likely has evolved multiple times independently in birds, mammals, and humans. Evolutionary models of its emergence are therefore bound to be different across species: different selective pressures may have acted on different traits and on different genes in different organisms. But there may be similarities too. After all, evolution acts only on a limited number of structures, and analogy implies only that a trait was not inherited from a common ancestor. For example, sexual selection seems to have favored the emergence of vocal learning at least in some species, such as songbirds, perhaps with predators selecting against it:[5] distinctive singers are easier to identify and target as prey. This hypothesis could explain both why vocal learning emerged and why it is relatively rare. But can it also explain the emergence of vocal learning in humans? Unlike birdsong, speech serves mainly semantic purposes, not sexual selection ones. Language could not have made our ancestors easier targets for predators in the same way as birdsong probably did. Any plausible evolutionary story for the emergence of speech must take that into account. Even so, different evolutionary pressures in songbirds and humans may well have acted on some of the same genes. FOXP2 (chapter 9) is well conserved in vertebrates. FOXP2 mutations matching those producing apraxia of speech in humans disrupt vocal learning in mice and zebra finches.[6] FOXP2 is involved in vocal learning in humans as well as songbirds, pointing to partly shared genetic mechanisms.

Vocal learning may be analogous in songbirds and humans, but that does not make birdsong and speech, let alone birdsong and language, analogous too. Studies have suggested that birdsong is syntactically organized and that some songbird species can even learn recursive structures in songs.[7] But syntax is a formal system mapping sound to meaning, and neither abstract structure, such as categories of basic elements that can be freely combined, nor interfaces with meaning have been demonstrated in birdsong. For example, the black-capped chickadee’s mating songs show a degree of combinatorial structure,[8] and their calls are modulated to signal predator size.[9] But these two skills are only loosely analogous to human syntax and referential semantics, and in songbirds they never actually integrate to produce a compositional system of syntactically structured referential signals.[10] What about nonhuman primates?

Neuroethology of Semantic Communication

Human language is a fairly recent evolutionary trait, but it emerged in the context of an existing biological substrate, the primate brain, capable not only of perceiving the world and acting in it, but also of learning, planning, reasoning, communication, and much more. The basic architecture of perception and action systems is, by and large, the same or very similar in humans and other primates, particularly in chimpanzees, our closest relatives. Chimpanzees and macaques possess homologues of the ventral and dorsal pathways connecting temporal and auditory cortex to the inferior frontal cortex, a clue that the basic dual-stream architecture that subserves speech and language in humans is ancient and conserved in some primates.[11] The more we learn about the brains of apes and monkeys, the more evident it is that, on a coarse neuroanatomical scale, there may be more similarities than differences between us and them. But these results make language more, not less, puzzling. Where does our unique capacity for abstract syntax and compositional semantics come from, if not from the brain's architecture, which appears to be largely shared with other primates lacking those skills?

Many primate species use calls or gestures to signal external events, such as the presence of specific threats in the environment, from obstacles to movement to potential land or airborne predators.[12] Producing or perceiving such communicative signals activates temporal and frontal areas in macaques and chimpanzees, including left-hemispheric homologues of human LIFG.[13] The communicative signals of some monkey and ape species are sequences of vocal sounds with specific meanings. Yet the organization of the elements in a sequence is linear, not hierarchical, and, moreover, the meaning of a sequence does not depend in systematics way on the meaning of individual sounds.[14] That is, the communicative signals of nonhuman primates have only holistic or idiomatic meanings, whereas phrases or sentences in human language can be interpreted holistically or compositionally (e.g., "to kick the bucket").[15]

Compositionality sets us apart from other primates. Our brains may be similar, but the communicative signals we exchange are not. How can this seeming paradox be resolved? There are two possible candidate explanations. One is the *neuro*ethological hypothesis that there may be fine neuroanatomical or physiological differences between us and nonhuman primates that account for behavioral or cognitive differences relevant for communication. There are multiple potentially explanatory neural differences. One is the vocal learning circuits discussed above: perhaps the emergence of vocal learning in humans, in the context of an already sophisticated primate brain, was sufficient to yield hierarchical syntax, as it were "for free." But as observed earlier, it is one thing to be able to produce and recognize structurally complex vocalizations; it is quite another to be able to map them to complex meanings. So even on this type of proposal, compositionality remains a puzzle. Another view is that the anatomical difference that is unique to humans does not involve frontal regions, such as vocal production or vocal learning circuits, but rather posterior temporal and parietal cortex, implicated in high-level semantics.[16]

> Compositionality sets us apart from other primates. Our brains may be similar, but the communicative signals we exchange are not. How can this seeming paradox be resolved?

The second hypothesis is a neuro*ethological* one. The ecologies and the behaviors of humans and nonhuman primates are strikingly different not just today (an easy but flawed comparison) but for at least the past 2 million years. Syntax and compositional semantics may not be a matter of brainpower after all. We know that primates are highly intelligent animals. It may just be that early humans occupied ecological niches that required specific behavioral adaptations—for example, the use of vocal or gestural signals with complex form and meaning. This shared capacity of the primate brain may be triggered and exploited to different degrees in different primates: holistic signals suffice for apes and monkeys, but compositional signals are very useful in the human ecology. What is special to our brain, relative to other primates, may then be limited to our unique vocal learning circuit, possibly with interfaces to shared systems of cognition.

11
The Future of Neurolinguistics

Scientists are often interested in anticipating trends in their own fields. Staying ahead of the curve is a key ingredient of scientific success, but it is notoriously difficult to achieve. Both neuroscience and linguistics are relatively fragmented and ebullient fields at present, which makes it even harder to predict developments at their intersection. Instead of trying to guess what neurolinguistics might look like in X years, I mention a few emerging trends that are likely to shape the field in the near future. So this chapter is more an exercise in extrapolation than prediction, assuming that current tendencies will continue and perhaps expand.

An important development in recent neurolinguistic research is an active effort to move beyond the traditional approach to mapping, focused primarily on pinning down linguistic processes in brain space and time (chapters 2 and 3), toward using new signal domains in experimental data to infer the neurophysiological mechanisms behind language use. The goal is no longer to address "where" and "when" questions but also eventually to answer "how" questions. The trick here is to find aspects of neural data that do not just correlate with properties of stimuli, but that effectively carry information about them. In M/EEG data, oscillations may be one such type of signal. Neuronal networks oscillate at multiple frequencies, reflecting slower or faster synchronous firing of neurons. Different frequency bands may carry information about specific temporal features of the stimulus—for example, about the slower and faster rhythmic properties of the speech signal.[1] In fMRI data, distributed activation patterns across the cortex have also been shown to carry information about high-level properties of semantic representations, such as their form and content.[2] These are just two recent examples, and much more is under way in several labs around the world.

Increasingly subtle measures of brain dynamics and newer approaches to analyzing neural data are opening up opportunities for major empirical progress in neuroscience. But especially in neurolinguistics, theory and computational modeling are still required to provide the formal frameworks in which brain activity can be interpreted as evidence for or against the existence of particular mechanisms (chapter 4). Importantly, these emerging approaches do not obliterate older data types but instead raise pressing new questions, such as how one can still use ERP and BOLD fMRI data to study the internal organization of the language system and how robust effects obtained with those measures, such as the N400 or P600, can be explained in mechanistic terms.

A recent tectonic shift in the language sciences, with many repercussions also for neurolinguistics, concerns our idea of language as a cognitive and computational system. For decades, psycholinguists and neurolinguists embraced a view of language as essentially "words plus grammar," as befits the generative model sketched in chapter 1. This was a tremendous leap forward, starting in the late 1950s, when much aphasiology was largely focused on single words and only occasionally considered sentences or discourse. But somewhat ironically, the generative account of language as a system of semantically interpretable syntactic and logical structures was never fully exploited; for example, we still do not know what role exactly compositionality plays in the human language processing architecture. We do have the capacity for deriving sentence meanings based on just "words plus grammar," but when and how is that capacity exercised in actual sentence processing? It is clear now that the picture of language suggested by generative syntax and compositional semantics is only a partial guide to language in the brain.

Newer experimental studies in psycholinguistics and neurolinguistics take seriously the view of human language as a complex cognitive system for situated communication. In this emerging picture, word meanings may be accessed via several modalities, simultaneously or alternatively, while the meanings of complex expressions are not constrained just by syntax and logic, but also by independent principles of cognition and action, such as reasoning and pragmatics. Experiments with more ecologically valid conditions, where multimodal information processing and interaction become possible, are trending in neurolinguistics, and so are studies targeting the complex relationship between systems of meaning and grammar in the brain.[3]

A recent turn of events in the information sciences, which is beginning to have an impact on neurolinguistics, is a new understanding of computation in distributed systems, such as artificial and biological neural networks. The emphasis in these models, of which deep learning is but one example, is no longer on transforming strings of symbols (inputs) into other strings of symbols (outputs) based on prespecified rules, but on learning the transformation functions incrementally based on data, usually with feedback or rewards. Artificial neural networks are set up with different architectures (number of layers and units, for example), different input data, and different training regimes, typically to solve optimization problems: the aim is to learn the "best" function mapping inputs to outputs.

These systems are very useful in several engineering domains, but it is unclear whether they are realistic models of how minds and brains work. On the pro side, they may provide models of brain-like distributed computation, which explain how new external events change the way information flows in the network: indeed, some aspects of languages may be acquired by learning to process them through iterated use.[4] On the con side, the brain does not just learn: it grows so that certain functions can develop (chapters 5 to 7). Moreover, it has a complex evolved architecture that far surpasses that of current neural networks (chapters 9 and 10). This prior structure is what makes learning and processing in the brain possible and computationally more efficient than brute-force function optimization. Some neurolinguists think that neural nets will revolutionize the field. Others believe that a lot of biological detail will have to be built into such systems to turn them into plausible models of cognitive brain function.

In addition, both the social structure and the public perception of neurolinguistics are rapidly changing. A more diverse community is now leading the field, which has huge benefits for the science we do. For example, neurolinguistics is practiced in countries where understudied languages are spoken, which is good news if we want our theories to hold not just for English and a handful of other (mostly European) languages. Among the general public, there is growing awareness of speech or language disorders, developmental deficits, and brain damage (chapters 8 and 9). It is important to continue researching these conditions to ensure that those who are affected by them receive the best evidence-based care in the clinic and full understanding in society at large.

> Research on language in the brain can tell us to what extent we are (among other things) thinking machines.

Finally, the impact of neurolinguistics on the wider cultural landscape is likely to grow in the years to come. Language may be one of the few aspects of the human mind that stands good chances of being mechanizable in algorithmic terms. Research on language in the brain can tell us to what extent we are (among other things) thinking machines. Language may also be a lower-hanging fruit in the old philosophical tree of the mind-body problem: How can mental processes arise in neurobiological machinery? This problem may be easier to address, and may therefore deserve greater efforts in the short term, for language than for, say, consciousness, or for many other higher cognitive capacities that have intrigued philosophers for centuries.

Glossary

mbiguity

he property of a linguistic expression of having different alternative meanings or different alternative grammatical analyses

nomaly

he property of a linguistic expression of having deviant meaning or being grammatically ill formed

phasia

neurological disorder characterized by a reduced ability to produce or understand speech; “disphasia” if mild

praxia

neurological disorder characterized by a reduced ability to execute skilled movements; “dispraxia” if mild

rchitecture

he functional organization of an information processing system, often specified in terms of components or modules

xon

he long filamentous part of a neuron along which electric impulses are conducted from the neuron’s body to other neurons

aseline

measurement of the variable of interest (behavior or brain activity) prior to or independent of an experimental intervention, and against which measurements of the same variable in response to a stimulus manipulation are compared

hronometry

he experimental study of the timing of mental processes, such as their latency relative to a stimulus and their duration

lause

. well-formed string of words containing a subject and a predicate and expressing a thought or proposition

ompositionality

ı formal theories of semantics, the principle that the meaning of a complex expression is a function of its constituent parts and of the way those parts are syntactically combined

omputation

. rule-governed process of transformation of interpretable strings of symbols into other interpretable strings of symbols

onnectivity

he pattern of anatomical connections between different parts of the brain; *anatomical connectivity* is distinct from *functional connectivity* (temporal correlations between active parts of the brain) and *effective connectivity* (causal interactions between active parts of the brain)

ontent word

. word that has semantic content and can refer to objects, events, or properties, such as nouns, verbs, adjectives, and adverbs in English

orpus callosum

. large bundle of nerve fibers underneath the cortex connecting the left and right hemispheres of the brain

ortex

he thin outer layer of the cerebrum, constituted by folded gray matter

evelopment

he emergence during infancy, childhood, and adolescence of adult skills and competencies, driven by learning and maturation processes

iscourse

connected series of spoken, written, or signed sentences

issociation

n neuropsychology, the fact that some capacities or functions may be preserved while others are reduced or destroyed by brain damage

istinctive features

asic properties of phonemes, traditionally described in binary terms: each phoneme is assumed to either have or lack a particular feature, and each is characterized by a unique combination of features

yslexia

congenital or acquired disorder involving difficulties in reading written words, letters, or other symbols; "alexia" if severe and if resulting from brain damage; "agraphia" if restricted to writing difficulties

lectrocorticography (ECoG)

neurophysiological method for recording brain currents from a number of electrodes placed on the surface of the cortex

lectroencephalography (EEG)

neurophysiological method for recording brain currents from a number of electrodes placed on the surface of the head

vent-related potentials (ERPs)

type of EEG signal time-locked to specific stimuli or responses

xperimental design

he structure of an experiment specifying how the variables controlled by the researcher are related to each other and to measured variables

igurative

he use of an expression such that its meaning differs from the literal or the compositional one, for example, in metaphor

unction word

word that contributes to grammatical structure, such as articles, pronouns, auxiliaries, conjunctions, and prepositions in English

unctional magnetic resonance imaging (fMRI)

n imaging technique that uses high-intensity magnetic fields to measure changes in brain metabolism and function

unctional near-infrared spectroscopy (fNIRS)

n imaging technique that uses near-infrared light to measure changes in brain metabolism and function

ray matter

he darker brain and spinal cord tissue, consisting mainly of neuronal bodies and dendrites

ierarchy

he relation of syntactic dependency between one constituent and another constituent embedded in it

olism

he idea that the meaning of an expression as a whole is independent of the meanings of the constituent parts

:omonymy

. relationship between two or more words with identical form (spelling or pronunciation or both) but different and unrelated meanings

liom

.n expression whose conventional meaning is not derived from the meanings of the constituent parts

ıference

he process of deriving conclusions from premises or data

ıverse problem

he problem of reconstructing from a set of observations the causal or formal structure of the system that produced them

anguage switching

he process of alternating between one or more languages or varieties during a conversation by the same speaker

atency

he time at which a brain response occurs relative to a stimulus

exicon

he inventory of all the words and other basic units of form and meaning (morphemes) of a language

ocalization

he process of tracing mental functions to their bases in the brain

Iagnetoencephalography (MEG)

. neurophysiological method for recording magnetic fields from the brain using superconductive sensors on the surface of the head

Iaturation

he genetically constrained process of growth and organization of anatomical structures in animals, including the nervous system

Iechanism

system of components and interactions between them that gives rise to a specific function or behavior in a biological system

Ientalism

he view that language is a mental phenomenon to be understood in terms of psychological and ultimately neurobiological processes

Ietabolism

he chemical-biological processes that occur in an organism in order to carry out energy-demanding tasks, such as processing information

Iorphology

he study of the internal structure of words

Iultimodal

ıvolving multiple sensory modalities, such as vision and hearing

euron

type of brain cell specialized in receiving and transmitting electrical impulses

ʼrthography

he conventional spelling and writing system of words in a language

ʼuantifier

word or phrase that can be combined with a noun phrase to specify the relative amount that is being considered, for example, *few* in “few dogs”

honetics

he study of the physical and physiological bases of speech sounds and of how they are produced and perceived

honology

he study of basic categories of speech sounds and their relations in specific languages and across languages

hrase

. well-formed string of one or more words that can function as a syntactic constituent in a clause or sentence

lasticity

he capacity of parts of the brain to undergo structural or functional changes in response to experience or other events, such as lesions

olarity

he positive or negative sign of electrical currents or magnetic fields

olysemy

he property of a linguistic expression of having several related senses

ositron emission tomography (PET)

.n imaging technique that uses radioactive substances to measure metabolic processes in the brain

ragmatics

he study of principles of language use and communication

.ecursion

he application of a formal operation to the result of applications of the same operation

eference

he semantic relationship between a linguistic expression and the entities it stands for

ensitive period

period of increased sensitivity to particular types of stimuli, which typically results in faster or better learning; also referred to as "critical period"

entence

well-formed string of words containing one or more clauses

imultaneous bilingual

person who has acquired two or more languages at the same age

ynapse

he junction between two neurons, consisting of a small gap across which electric impulses pass by diffusion of neurotransmitters

yntax

he study of the internal structure of well-formed multiword strings, such as phrases, clauses, and sentences

ime-locked activity

rain activity lined up in time to specific events, such as a stimulus

ransitivity

he property of a verb to take direct objects

/hite matter

he paler brain and spinal cord tissue, consisting mainly of axonal fibers and enveloping myelin sheets

Acknowledgments

I am grateful to the five anonymous reviewers who commented on the book's proposal. Their advice helped me organize the book's structure and content in the early stages of the writing process, when it was most needed. I also wish to thank the two anonymous reviewers who provided detailed feedback on intermediate drafts of the book. Gregory Hickok deserves special mention and my gratitude for his careful expert reading of the manuscript. Finally, I want to thank my students at the Norwegian University of Science and Technology for pushing me, sometimes unwittingly, to try out new ways of presenting neurolinguistics to inquisitive readers beyond the narrow circle of professional researchers.

Notes

Chapter 1

1. D. P. McCabe and A. D. Castel. “Seeing Is Believing: The Effect of Brain Images on Judgments of Scientific Reasoning,” *Cognition* 107 (2008): 343–352; D. S. Weisberg, F. C. Keil, J. Goodstein, E. Rawson, and J. R. Gray, “The Seductive Allure of Neuroscience Explanations,” *Journal of Cognitive Neuroscience* 20 (2008): 470–477; P. Legrenzi and C. Umiltà, *Neuromania: On the Limits of Brain Science* (New York: Oxford University Press, 2011).
2. W. J. M. Levelt, *A History of Psycholinguistics: The Pre-Chomskyan Era* (New York: Oxford University Press, 2012; P. Eling, “Neurolinguistics, History of,” in *International Encyclopedia of the Social and Behavioral Sciences*, 2nd ed., ed. James Wright, 16: 678–689 (Amsterdam: Elsevier, 2015); G.-J. Rutten, *The Broca-Wernicke Doctrine: A Historical and Clinical Perspective on Localization of Language Functions* (Berlin: Springer, 2017).
3. Eling, “Neurolinguistics, History of,” 680–681.
4. Eling, “Neurolinguistics, History of,” 681.
5. Levelt, *A History of Psycholinguistics*, 53–58; Eling, “Neurolinguistics, History of,” 682; Rutten, *The Broca-Wernicke Doctrine*, 3–4.
6. F. J. Gall, *Sur les fonctions du cerveau et sur celles de chacune de ses parties* (Ballière, 1822–1825).
7. Levelt, *A History of Psycholinguistics*, 56–59.
8. P. Broca, “Remarques sur le siège de la faculté du langage articulé, suivies d’une observation d’aphémie (perte de la parole),” *Bulletin et mémoires de la Société Anatomique de Paris* 6 (1861b): 330–357.
9. P. Broca, “Perte de la parole, ramollissement chronique et destruction partielle du lobe antérieur gauche du cerveau,” *Bulletin de la Société Anthropologique de Paris* 2 (1861a): 235–238; Broca, “Remarques sur le siège de la faculté du langage articulé”; P. Broca, “Nouvelle observation d’aphémie produite par une lésion de la moitié postérieure des deuxième et troisième circonvolutions frontales,” *Bulletin et mémoires de la Société Anatomique de Paris* 6 (1861c): 398–407.
10. Broca, “Nouvelle observation d’aphémie produite.”
11. Levelt, *A History of Psycholinguistics*, 88.
12. Broca, “Perte de la parole.”
13. C. Wernicke, C. (1874). “Der aphasische symptomencomplex: Eine psychologische studie auf anatomischer basis,” in *Wernicke’s Works on Aphasia: A Sourcebook and Review*, ed. G. H. Eggert, 91–145 (The Hague: Mouton, 1874).
14. Broca, “Remarques sur le siège de la faculté du langage articulé.”
15. Levelt, *A History of Psycholinguistics*, 70–73.
16. Levelt, *A History of Psycholinguistics*, 72–73.
17. Levelt, *A History of Psycholinguistics*, 79.
18. L. Lichtheim, “Über Aphasie,” *Deutsches Archiv für klinische Medizin* 36 (1885): 204–268. Also see L. Lichtheim, “On Aphasia,” *Brain* 7 (1885): 433–484.
19. N. Geschwind, “The Organization of Language and the Brain,” *Science* 170 (1970): 940–944.
20. F. de Saussure, *Cours de linguistique générale* (1916). Paris: Payot. A sign links “a concept and an acoustic image” (two mental entities). This idea agrees well with earlier notions used by neurologists and biologists. For example, we find similar ideas in Broca (“Remarques sur le siège

de la faculté du langage articulé") and C. Darwin, *The Descent of Man, and Selection in Relation to Sex* (1871). London: John Murray. See chapter 10.

21. Saussure, *Cours de linguistique générale*.

22. Eling, "Neurolinguistics, History of," 683.

23. "Any description and classification of aphasic syndromes must begin with the question of what aspects of language are impaired," is the programmatic statement from R. Jakobson, "Two Aspects of Language and Two Types of Aphasic Disturbances," in *Fundamentals of Language*, ed. R. Jakobson and M. Halle (The Hague: Mouton, 1956). For further discussion, see also R. Jakobson, "Towards a Linguistic Typology of Aphasic Impairments," in *Disorders of Language*, ed. A. V. S. de Reuck and M. O'Connor (Boston: Little, Brown, 1964), 21–41, and R. Jakobson, "Toward a Linguistic Classification of Aphasic Impairments," *Selected Writings II* (The Hague: Mouton, 1970).

24. Levelt, *A History of Psycholinguistics*, 405: "Jakobson remained an exception in aphasiology until the 'cognitive revolution.' The field suffered from a persistent lack of linguistic sophistication in both method and theory."

25. N. Chomsky, *Syntactic Structures* (The Hague: Mouton, 1957). See also his papers from the 1955–1960 period.

26. N. Chomsky, *Aspects of the Theory of Syntax* (Cambridge, MA: MIT Press 1965).

27. H, Gardner, *The Mind's New Science: A History of the Cognitive Revolution* (New York: Basic Books, 1987); G. A. Miller, "The Cognitive Revolution: A Historical Perspective," *Trends in Cognitive Sciences* 7 (2003): 141–144.

28. Levelt, *A History of Psycholinguistics*, 18.

29. J. Cohen-Cole, *The Open Mind: Cold War Politics and the Sciences of Human Nature* (Chicago: University of Chicago Press, 2014).

30. G. A. Miller and K. O. McKean, "A Chronometric Study of Some Relations between Sentences," *Quarterly Journal of Experimental Psychology* 16 (1964): 297–308; Y. Grodzinsky, "The Neurology of Syntax: Language Use without Broca's Area," *Behavioral and Brain Sciences* 23 (2000): 1–21.

31. E. C. Trager, "The Field of Neurolinguistics," *Studies in Linguistics* 15 (1961): 70–71. See also the early works by Alexander Luria and others.

32. E. H. Lenneberg, *Biological Foundations of Language* (New York: Wiley, 1967).

33. H. A. Whitaker, Editorial. *Brain and Language* 1 (1974): iii–iv.

34. M. Kutas and S. A. Hillyard, "Reading Senseless Sentences: Brain Potentials Reflect Semantic Incongruity," *Science* 207 (1980): 203–205; L. Osterhout and P. J. Holcomb, "Event-Related Brain Potentials Elicited by Syntactic Anomaly," *Journal of Memory and Language* 31 (1992): 785–806; P. Hagoort, C. Brown, and J. Groothusen, "The Syntactic Positive Shift (SPS) as an ERP Measure of Syntactic Processing," *Language and Cognitive Processes* 8 (1993): 439–483.

35. M Mather, J. T. Cacioppo, and N. Kanwisher, "Introduction to the Special Section: 20 Years of fMRI—What Has It Done for Understanding Cognition?" *Perspectives on Psychological Science* 8 (2013): 41–43.

36. M. A. Just, P. A. Carpenter, T. A. Keller, W. F. Eddy, and K. R. Thulborn, "Brain Activation Modulated by Sentence Comprehension," *Science* 274 (1996): 114–116.

Chapter 2

1. This is true in most circumstances, but people with speech disorders, for example, do experience delays between intention and production. The same can be said for second-language learners and some aphasic patients.
2. J. Hirschberg and C. D. Manning, “Advances in Natural Language Processing,” *Science* 349 (2015): 261–266.
3. Meyer, D. E., A. M. Osman, D. E. Irwin, and S. Yantis, “Modern Mental Chronometry,” *Biological Psychology* 26, nos. 1–3 (1988): 3–67.
4. G. A. Miller and K. O. McKean, “A Chronometric Study of Some Relations between Sentences,” *Quarterly Journal of Experimental Psychology* 16 (1964): 297–308.
5. S. J. Luck and E. S. Kappenman, eds., *The Oxford Handbook of Event-Related Potential Components* (New York: Oxford University Press, 2011).
6. J. Bickle, “Revolutions in Neuroscience: Tool Development,” *Frontiers in Systems Neuroscience* 10 (2016): 24; D. Parker, “Kuhnian Revolutions in Neuroscience: The Role of Tool Development,” *Biology and Philosophy* 33, no. 3 (2018): 17.
7. W. J. Levelt, *Speaking: From Intention to Articulation* (Cambridge, MA: MIT Press, 1993); G. Hickok and D. Poeppel, “The Cortical Organization of Speech Processing,” *Nature Reviews Neuroscience* 8 (2017): 393; D. B. Fry, *The Physics of Speech* (Cambridge: Cambridge University Press, 1979).
8. P. J. Monahan, “Phonological Knowledge and Speech Comprehension,” *Annual Review of Linguistics* 4 (2018): 21–47.
9. D. Poeppel, C. Phillips, E. Yellin, H. A. Rowley, T. P. Roberts, and A. Marantz, “Processing of Vowels in Supratemporal Auditory Cortex,” *Neuroscience Letters* 221 (1997): 145–148.
10. T. H. Heinks-Maldonado, D. H. Mathalon, M. Gray, and J. M. Ford, “Fine-Tuning of Auditory Cortex during Speech Production,” *Psychophysiology* 42 (2005): 180–190.
11. D. Poeppel and P. J. Monahan, “Feedforward and Feedback in Speech Perception: Revisiting Analysis by Synthesis,” *Language and Cognitive Processes* 26 (2011): 935–951.
12. C. Phillips, T. Pellathy, A. Marantz, E. Yellin, K. Wexler, D. Poeppel, M. McGinnis, and T. Roberts, “Auditory Cortex Accesses Phonological Categories: An MEG Mismatch Study,” *Journal of Cognitive Neuroscience* 12 (2000): 1038–1055; C. Phillips, “Levels of Representation in the Electrophysiology of Speech Perception,” *Cognitive Science* 25 (2001): 711–731; M. T. Diaz and T. Y. Swaab, “Electrophysiological Differentiation of Phonological and Semantic Integration in Word and Sentence Contexts,” *Brain Research* 1146 (2007): 85–100.
13. V. Van Wassenhove, K. W. Grant, and D. Poeppel, “Visual Speech Speeds Up the Neural Processing of Auditory Speech,” *Proceedings of the National Academy of Sciences* 102 (2005): 1181–1186.
14. M. Kutas and K. D. Federmeier, “Thirty Years and Counting: Finding Meaning in the N400 Component of the Event-Related Brain Potential (ERP),” *Annual Review of Psychology* 62 (2011): 621–647.
15. M. Kutas and S. A. Hillyard, “Reading Senseless Sentences: Brain Potentials Reflect Semantic Incongruity,” *Science* 207 (1980): 203–205; M. Kutas and S. A. Hillyard, “Brain Potentials during Reading Reflect Word Expectancy and Semantic Association,” *Nature* 307 (1984): 161.
16. D. Embick, and D. Poeppel, “Towards a Computational(ist) Neurobiology of Language: Correlational, Integrated and Explanatory Neurolinguistics,” *Language, Cognition and Neuroscience* 30 (2015): 357–366.

17. Kutas and Federmeier, “Thirty Years and Counting”; G. Baggio and P. Hagoort, “The Balance between Memory and Unification in Semantics: A Dynamic Account of the N400,” *Language and Cognitive Processes* 26 (2011): 1338–1367.

18. L. Pylkkänen and A. Marantz, “Tracking the Time Course of Word Recognition with MEG,” *Trends in Cognitive Sciences* 7 (2003): 187–189.

19. J. J. Van Berkum, “Understanding Sentences in Context: What Brain Waves Can Tell Us,” *Current Directions in Psychological Science* 17 (2008): 376–380; L. Pylkkänen, “Composition of Complex Meaning: Interdisciplinary Perspectives on the Left Anterior Temporal Lobe,” in *Neurobiology of Language*, ed. G. Hickok and S. L. Small, 621–631 (Orlando, FL: Academic Press, 2016); G. Baggio, *Meaning in the Brain* (Cambridge, MA: MIT Press, 2018).

20. L. Osterhout and P. J. Holcomb, “Event-Related Brain Potentials Elicited by Syntactic Anomaly,” *Journal of Memory and Language* 31 (1992): 785–806; P. Hagoort, C. Brown, and J. Groothusen, “The Syntactic Positive Shift (SPS) as an ERP Measure of Syntactic Processing,” *Language and Cognitive Processes* 8 (1993): 439–483.

21. The answer is *was*. This ERP result is from L. Osterhout and P. J. Holcomb, “Event-Related Brain Potentials Elicited by Syntactic Anomaly,” *Journal of Memory and Language* 31 (1992): 785–806.

22. P. Hagoort, “How the Brain Solves the Binding Problem for Language: A Neurocomputational Model of Syntactic Processing,” *NeuroImage* 20 (2003); S18-S29; A. D. Friederici, “Towards a Neural Basis of Auditory Sentence Processing,” *Trends in Cognitive Sciences* 6 (2002), 78–84.

23. A. Kim and L. Osterhout, “The Independence of Combinatory Semantic Processing: Evidence from Event-Related Potentials,” *Journal of Memory and Language* 52 (2005): 205–225; G. R. Kuperberg, “Neural Mechanisms of Language Comprehension: Challenges to Syntax,” *Brain Research* 1146 (2007): 23–49; O. Michalon and G. Baggio, “Meaning-Driven Syntactic Predictions in a Parallel Processing Architecture: Theory and Algorithmic Modeling of ERP Effects,” *Neuropsychologia* 131 (2019): 171–183.

24. S. Crain and M. Steedman, “On Not Being Led Up the Garden Path: The Use of Context by the Syntactic Processor,” in *Natural Language Parsing: Psychological, Computational and Theoretical Perspectives*, eds. David R. Dowty, Arnold Zwicky, and Lauri Karttunen (Cambridge: Cambridge University Press, 1985).

25. Friederici, “Towards a Neural Basis”; Hagoort, “How the Brain Solves the Binding Problem”; K. Steinhauer and J. E Drury, “On the Early Left-Anterior Negativity (ELAN) in Syntax Studies,” *Brain and Language* 120 (2012): 135–162.

26. For a recent study, see L. A. Fromont, K. Steinhauer, and P. Royle, “Verbing Nouns and Nouning Verbs: Using a Balanced Design Provides ERP Evidence against ‘Syntax-First’ Approaches to Sentence Processing,” *PLoS One* 15 (2020): e0229169.

Chapter 3

1. J. A. Fodor, "Diary," *London Review of Books* 21 (1999): 68–69, https://www.lrb.co.uk/v21/n19/jerry-fodor/diary.
2. S. Dehaene and L. Cohen, "Cultural Recycling of Cortical Maps," *Neuron* 56 (2007): 384–398.
3. N. K. Logothetis, J. Pauls, M. Augath, T. Trinath, and A. Oeltermann, "Neurophysiological Investigation of the Basis of the fMRI Signal," *Nature*,412 (2001): 150; N. K. Logothetis, "The Ins and Outs of fMRI Signals," *Nature Neuroscience* 10 (2007): 1230.
4. H. H. Clark, *Using Language* (Cambridge: Cambridge University Press, 1996); P. Hagoort, "The Neurobiology of Language beyond Single-Word Processing," *Science* 366 (2019): 55–58.
5. G. Baggio, *Meaning in the Brain* (Cambridge, MA: MIT Press, 2018).
6. Wernicke's area, according to standard, classical definitions, is approximately the posterior half or third of the STG. Functionally, it is a larger portion of temporal cortex if one defines it as the region that, when damaged, will yield Wernicke's aphasia. On this view, Wernicke's area would be a bigger region than shown in the figure.
7. E. F. Chang, J. W. Rieger, K. Johnson, M. S. Berger, N. M. Barbaro, and R. T. Knight, "Categorical Speech Representation in Human Superior Temporal Gyrus," *Nature Neuroscience* 13 (2010): 1428; A. Flinker, E. F. Chang, N. M., Barbaro, M. S. Berger, and R. T. Knight, "Centimeter Language Organization in the Human Temporal Lobe," *Brain and Language* 117 (2011): 103–109; H. G. Yi, M. K. Leonard, and E. F. Chang, "The Encoding of Speech Sounds in the Superior Temporal Gyrus," *Neuron* 102 (2019): 1096–1110.
8. J. P. Rauschecker and S. K. Scott, "Maps and Streams in the Auditory Cortex: Nonhuman Primates Illuminate Human Speech Processing," *Nature Neuroscience* 12 (2009): 718.
9. P. W. Hullett, L. S. Hamilton, N. Mesgarani, C. E. Schreiner, and E. F. Chang, "Human Superior Temporal Gyrus Organization of Spectrotemporal Modulation Tuning Derived from Speech Stimuli," *Journal of Neuroscience* 36 (2016): 2014–2026; L. S. Hamilton, E. Edwards, and E. F. Chang, "A Spatial Map of Onset and Sustained Responses to Speech in the Human Superior Temporal Gyrus," *Current Biology* 28 (2018): 1860–1871; K. Jasmin, C. F. Lima, and S. K. Scott, "Understanding Rostral–Caudal Auditory Cortex Contributions to Auditory Perception," *Nature Reviews Neuroscience* 20 (2019): 425–434; D. Poeppel, "The Analysis of Speech in Different Temporal Integration Windows: Cerebral Lateralization as 'Asymmetric Sampling in Time,'" *Speech Communication* 41 (2003): 245–255; K. Okada, F. Rong, J. Venezia, W. Matchin, I. H. Hsieh, K. Saberi, J. T. Serences, and G. Hickok, "Hierarchical Organization of Human Auditory Cortex: Evidence from Acoustic Invariance in the Response to Intelligible Speech," *Cerebral Cortex* 20 (2010): 2486–2495.
10. J. Binder, "The New Neuroanatomy of Speech Perception," *Brain* 123 (2000): 2371–2372; J. R. Binder, "The Wernicke Area: Modern Evidence and a Reinterpretation," *Neurology* 85 (2015): 2170–2175; J. R. Binder, "Current Controversies on Wernicke's Area and Its Role in Language," *Current Neurology and Neuroscience Reports* 17 (2017): 58; Gregory Hickok reviews the evidence here: https://youtu.be/ShtQVAsfhSQ.
11. J. M. Rodd, M. H. Davis, and I. S. Johnsrude, "The Neural Mechanisms of Speech Comprehension: fMRI Studies of Semantic Ambiguity," *Cerebral Cortex* 15 (2005): 1261–1269; P. Hagoort, G. Baggio, and R. M. Willems, "Semantic Unification," in *The Cognitive Neurosciences*, 4th ed., ed. Michael S. Gazzaniga, 819–836 (Cambridge, MA: MIT Press, 2009).
12. G. Hickok and D. Poeppel, "The Cortical Organization of Speech Processing," *Nature Reviews Neuroscience* 8 (2007): 393; D. Ben Shalom and D. Poeppel, "Functional Anatomic Models of Language: Assembling the Pieces," *Neuroscientist* 14 (2008): 119–127; Hagoort et al., "Semantic

Unification"; D. Poeppel, K. Emmorey, G. Hickok, and L. Pylkkänen, "Towards a New Neurobiology of Language," *Journal of Neuroscience* 32 (2012): 14125–14131.

13. M. MacSweeney, R. Campbell, B. Woll, M. J. Brammer, V. Giampietro, A. S. David, G. A. Calvert, and P. K. McGuire, "Lexical and Sentential Processing in British Sign Language," *Human Brain Mapping* 27 (2006): 63–76; D. Poeppel, K. Emmorey, G. Hickok, and L. Pylkkänen, "Towards a New Neurobiology of Language," *Journal of Neuroscience* 32 (2012): 14125–14131.

14. E. F., Lau, C. Phillips, and D. Poeppel, "A Cortical Network for Semantics: (De)Constructing the N400," *Nature Reviews Neuroscience* 9 (2008): 920.

15. M. A. L., Ralph, E. Jefferies, E., K. Patterson, and T. T. Rogers, "The Neural and Computational Bases of Semantic Cognition," *Nature Reviews Neuroscience* 18 (2017): 42–45; L. Pylkkänen, "The Neural Basis of Combinatory Syntax and Semantics," *Science* 366 (2019): 62–66.

16. S. B. Eickhoff, S. Heim, K. Zilles, and K. Amunts, "A Systems Perspective on the Effective Connectivity of Overt Speech Production," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 367 (2009): 2399–2421; N. Janssen and C. C. R. Mendieta, "The Dynamics of Speech Motor Control Revealed with Time-Resolved fMRI," *Cerebral Cortex* 30, no. 1 (2019): 241–255; N. T. Sahin, S. Pinker, S. S. Cash, D. Schomer, and E. Halgren, "Sequential Processing of Lexical, Grammatical, and Phonological Information within Broca's Area," *Science* 326 (2009): 445–449; P. Hagoort and W. J. Levelt, "The Speaking Brain," *Science* 326 (2009): 372–373.

17. M. A. Just, P. A. Carpenter, T. A. Keller, W. F. Eddy, and K. R. Thulborn, "Brain Activation Modulated by Sentence Comprehension," *Science* 274 (1996): 114–116; D. Embick, A. Marantz, Y. Miyashita, W. O'Neil, and K. L. Sakai, "A Syntactic Specialization for Broca's Area," *Proceedings of the National Academy of Sciences* 97 (2000): 6150–6154.

18. M. Catani, D. K. Jones, and D. H. Ffytche, "Perisylvian Language Networks of the Human Brain," *Annals of Neurology* 57 (2005): 8–16; D. Saur, B. W. Kreher, S. Schnell, D. Kümmerer, P. Kellmeyer, M. S. Vry, R. Umarova, M. Musso, V. Glauche, S. Abel, and W. Huber, "Ventral and Dorsal Pathways for Language," *Proceedings of the National Academy of Sciences* 105 (2008): 18035–18040; A. D. Friederici, *Language in Our Brain: The Origins of a Uniquely Human Capacity* (Cambridge, MA: MIT Press, 2017).

19. N. Mesgarani, C. Cheung, K. Johnson, and E. F. Chang, "Phonetic Feature Encoding in Human Superior Temporal Gyrus," *Science* 343 (2014): 1006–1010. See also E. F. Chang, J. W. Rieger, K. Johnson, M. S. Berger, N. M. Barbaro, and R. T. Knight, "Categorical Speech Representation in Human Superior Temporal Gyrus," *Nature Neuroscience* 13 (2010): 1428.

20. E. Zaccarella, L. Meyer, M. Makuuchi, and A. D. Friederici, "Building by Syntax: The Neural Basis of Minimal Linguistic Structures," *Cerebral Cortex* 27 (2017): 411–421; A. D. Friederici, N. Chomsky, R. C. Berwick, A. Moro, and J. J. Bolhuis, "Language, Mind and Brain," *Nature Human Behaviour* 1 (2017): 713–722; E. Zaccarella, M. Schell, and A. D. Friederici, "Reviewing the Functional Basis of the Syntactic Merge Mechanism for Language: A Coordinate-Based Activation Likelihood Estimation Meta-Analysis," *Neuroscience and Biobehavioral Reviews* 80 (2017): 646–656.

21. T. M. Snijders, T. Vosse, G. Kempen, J. J. Van Berkum, K. M. Petersson, and P. Hagoort, "Retrieval and Unification of Syntactic Structure in Sentence Comprehension: An fMRI Study Using Word-Category Ambiguity," *Cerebral Cortex* 19 (2008), 1493–1503; Z. Zhu, P. Hagoort, J. X. Zhang, G. Feng, H. C. Chen, M. Bastiaansen, and Z. Wang, "The Anterior Left Inferior Frontal Gyrus Contributes to Semantic Unification," *NeuroImage* 60 (2012): 2230–2237.

22. M. Ben-Shachar, D. Palti, and Y. Grodzinsky, "Neural Correlates of Syntactic Movement: Converging Evidence from Two fMRI Experiments," *NeuroImage* 21 (2004), 1320–1336.

23. C. Rogalsky and G. Hickok, “The Role of Broca’s Area in Sentence Comprehension,” *Journal of Cognitive Neuroscience* 23 (2011): 1664–1680.

24. Y. Grodzinsky and A. Santi, “The Battle for Broca’s Region,” *Trends in Cognitive Sciences* 12 (2008): 474–480; R. M. Willems and P. Hagoort, “Broca’s Region: Battles Are Not Won by Ignoring Half of the Facts,” *Trends in Cognitive Sciences* 13 (2009): 101.

25. Pylkkänen, “The Neural Basis of Combinatory Syntax and Semantics.”

26. P. Hagoort and S. C. Levinson, “Neuropragmatics,” in *The Cognitive Neurosciences*, 667–674 (Cambridge, MA: MIT Press, 2014); P. Hagoort and P. Indefrey, “The Neurobiology of Language beyond Single Words,” *Annual Review of Neuroscience* 37 (2014): 347–362; Baggio, *Meaning in the Brain*.

27. A. M. Rapp, D. E. Mutschler, and M. Erb, “Where in the Brain Is Nonliteral Language? A Coordinate-Based Meta-Analysis of Functional Magnetic Resonance Imaging Studies,” *NeuroImage* 63 (2012): 600–610; I. C. Bohrn, U. Altmann, and A. M. Jacobs, “Looking at the Brains behind Figurative Language—A Quantitative Meta-Analysis of Neuroimaging Studies on Metaphor, Idiom, and Irony Processing,” *Neuropsychologia* 50 (2012): 2669–2683.

Chapter 4

1. B. C. Van Fraassen, *Scientific Representation: Paradoxes of Perspective* (New York: Oxford University Press, 2010).
2. K. R. Popper, "Science as Falsification," *Conjectures and Refutations* 1 (1963): 33–39; I. Lakatos, "Criticism and the Methodology of Scientific Research Programmes," *Proceedings of the Aristotelian Society* 69 (1968): 149–86; I. Lakatos, "Falsification and the Methodology of Scientific Research Programmes," in *Can Theories Be Refuted?*, ed. Sandra G. Harding, 205–259 (Dordrecht: Springer, 1976); D. Deutsch, *The Beginning of Infinity: Explanations That Transform the World* (London: Penguin UK, 2011).
3. D. Marr, *Vision: A Computational Investigation Into the Human Representation and Processing of Visual Information* (Cambridge, MA: MIT Press, 2010).
4. R. Jackendoff, *Foundations of Language: Brain, Meaning, Grammar, Evolution* (New York: Oxford University Press, 2002); P. W. Culicover and R. Jackendoff, *Simpler Syntax* (New York: Oxford University Press, 2005), and "The Simpler Syntax Hypothesis," *Trends in Cognitive Sciences* 10 (2006): 413–418; G. R. Kuperberg, "Neural Mechanisms of Language Comprehension: Challenges to Syntax," *Brain Research* 1146 (2007): 23–49; G. Baggio, *Meaning in the Brain* (Cambridge, MA: MIT Press, 2018).
5. G. Lakoff, *Women, Fire, and Dangerous Things* (Chicago: University of Chicago Press, 2008); H. Kamp, J. Van Genabith, and U. Reyle, "Discourse Representation Theory," in *Handbook of Philosophical Logic*, eds. Dov Gabbay and Franz Guenthner 125–394 (Berlin: Springer, 2011); L. Horn, "Implicature," in *Routledge Companion to Philosophy of Language*, ed. Delia Graff Fara and Gillian Russell, 75–88 (New York: Routledge, 2013); R. Stalnaker, *Context* (New York: Oxford University Press, 2014).
6. W. J. Levelt, *A History of Psycholinguistics: The Pre-Chomskyan Era* (New York: Oxford University Press, 2013).
7. P. Warren, *Introducing Psycholinguistics* (Cambridge: Cambridge University Press, 2012); G. Altmann, "Ambiguity, Parsing Strategies, and Computational Models," *Language and Cognitive Processes* 3 (1988): 73–97; G. T. Altmann, "The Language Machine: Psycholinguistics in Review," *British Journal of Psychology* 92 (2001): 129–170.
8. F. Ferreira, K. G., Bailey, and V. Ferraro, "Good-Enough Representations in Language Comprehension," *Current Directions in Psychological Science* 11 (2002): 11–15; M. H. Christiansen and N. Chater, *Creating Language: Integrating Evolution, Acquisition, and Processing* (Cambridge, MA: MIT Press, 2016).
9. For a discussion, see I. Bornkessel-Schlesewsky and M. Schlesewsky, "Sentence Processing: Toward a Neurobiological Approach," in *The Oxford Handbook of Neurolinguistics*, ed. Niels O. Schiller and Greig I. de Zubicaray, 676–709 (New York: Oxford University Press, 2019).
10. A. D. Friederici, *Language in our Brain: The Origins of a Uniquely Human Capacity* (Cambridge, MA: MIT Press, 2017).
11. Some points of contact are discussed in A. D. Friederici, N. Chomsky, R. C. Berwick, A. Moro, and J. J. Bolhuis, "Language, Mind and Brain," *Nature Human Behaviour* 1 (2017): 713–722.
12. P. Hagoort, "On Broca, Brain, and Binding: A New Framework," *Trends in Cognitive Sciences* 9 (2005): 416–423, and "MUC (Memory, Unification, Control): A Model on the Neurobiology of Language beyond Single Word Processing," in *Neurobiology of Language*, ed. Gregory Hickok and Steven L. Small, 339–347 (Orlando, FL: Academic Press, 2016).
13. H. D. Xiang, H. M.Fonteijn, D. G. Norris, and P. Hagoort, "Topographical Functional Connectivity Pattern in the Perisylvian Language Networks," *Cerebral Cortex* 20 (2009), 549–

560.

14. E. C. Ferstl, J. Neumann, C. Bogler, and D. Y. Von Cramon, "The Extended Language Network: A Meta-Analysis of Neuroimaging Studies on Text Comprehension," *Human Brain Mapping* 29 (2008): 581–593; I. C. Bohrn, U. Altmann, and A. M. Jacobs, "Looking at the Brains behind Figurative Language—A Quantitative Meta-Analysis of Neuroimaging Studies on Metaphor, Idiom, and Irony Processing," *Neuropsychologia* 50 (2012), 2669–2683; A. M. Rapp, D. E. Mutschler, and M. Erb, "Where in the Brain Is Nonliteral Language? A Coordinate-Based Meta-Analysis of Functional Magnetic Resonance Imaging Studies," *NeuroImage* 63 (2012): 600–610; P. Hagoort and P. Indefrey, "The Neurobiology of Language beyond Single Words," *Annual Review of Neuroscience* 37 (2014): 347–362; Baggio, *Meaning in the Brain*.

15. R. Jackendoff, *Foundations of Language: Brain, Meaning, Grammar, Evolution* (New York: Oxford University Press, 2002); Baggio, *Meaning in the Brain*.

16. G. Hickok and D. Poeppel, "Dorsal and Ventral Streams: A Framework for Understanding Aspects of the Functional Anatomy of Language," *Cognition* 92, nos. 1–2 (2004): 67–99; G. Hickok and D. Poeppel, "The Cortical Organization of Speech Processing," *Nature Reviews Neuroscience* 8 (2007): 393.

17. D. Poeppel, K. Emmorey, G. Hickok, and L. Pylkkänen, "Towards a New Neurobiology of Language," *Journal of Neuroscience* 32 (2012): 14125–14131.

18. E. Jonas and K. P. Kording, "Could a Neuroscientist Understand a Microprocessor?" *PLoS Computational Biology* 13 (2017): e1005268. For an early discussion of computational models in neurolinguistics, see M. A. Arbib and D. Caplan, "Neurolinguistics Must Be Computational," *Behavioral and Brain Sciences* 2 (1979): 449–460 (see also the commentary section).

19. On composition, see A. E. Martin and L. A. Doumas, "A Mechanism for the Cortical Computation of Hierarchical Linguistic Structure," *PLoS Biology* 15 (2017): e2000663; A. E. Martin and L. A. Doumas, "Predicate Learning in Neural Systems: Using Oscillations to Discover Latent Structure," *Current Opinion in Behavioral Sciences* 29 (2019): 77–83; A. E. Martin and G. Baggio, "Modeling Meaning Composition from Formalism to Mechanisms," *Philosophical Transactions of the Royal Society B* 375 (2019): 20190298.

Chapter 5

1. G. F. Marcus, S. Pinker, M. Ullman, M. Hollander, T. J. Rosen, F. Xu, and H. Clahsen, "Overregularization in Language Acquisition," *Monographs of the Society for Research in Child Development* 57 (1992): i, iii, v–vi, 1–178.

2. R. Ruhland and P. van Geert, "Jumping into Syntax: Transitions in the Development of Closed Class Words," *British Journal of Developmental Psychology* 16 (1998): 65–95.

3. For example, see the Speech and Language Developmental Milestones of the US National Institutes of Health: www.nidcd.nih.gov/health/speech-and-language.

4. According to a recent estimate, "The average English-speaking adult has learned 12.5 million bits of information [less than 1.5 megabytes], the majority of which is lexical semantics"; F. Mollica and S. T. Piantadosi, "Humans Store about 1.5 Megabytes of Information during Language Acquisition," *Royal Society Open Science* 6 (2019): 181393.

5. The fact that this is not how computers and programming work suggests, again, that the computing metaphor is not quite right as a model of the mind.

6. P. K. Kuhl, "Early Language Acquisition: Cracking the Speech Code," *Nature Reviews Neuroscience* 5 (2004): 831–843; P. K. Kuhl, "Brain Mechanisms in Early Language Acquisition," *Neuron* 67 (2010): 713–727.

7. G. Dehaene-Lambertz and E. S. Spelke, "The Infancy of the Human Brain," *Neuron* 88 (2015): 93–109; A. D. Friederici, "Neurophysiological Markers of Early Language Acquisition: From Syllables to Sentences," *Trends in Cognitive Sciences* 9 (2005): 481–488; Kuhl, "Early Language Acquisition."

8. R. A. Thompson and C. A. Nelson, "Developmental Science and the Media: Early Brain Development," *American Psychologist* 56 (2001): 5; G. Dehaene-Lambertz, L. Hertz-Pannier, and J. Dubois, "Nature and Nurture in Language Acquisition: Anatomical and Functional Brain-Imaging Studies in Infants," *Trends in Neuroscience* 29 (2006): 367–373; G. Dehaene-Lambertz, L. Hertz-Pannier, J. Dubois, and S. Dehaene, "How Does Early Brain Organization Promote Language Acquisition in Humans?" *European Review* 16 (2008): 399–411.

9. Y. Zhang, P. K. Kuhl, T. Imada, M. Kotani, and Y. I. Tohkura, "Effects of Language Experience: Neural Commitment to Language-Specific Auditory Patterns," *NeuroImage* 26 (2005): 703–720; P. Kuhl and M. Rivera-Gaxiola, "Neural Substrates of Language Acquisition," *Annual Review of Neuroscience* 31 (2008): 511–534.

10. E. Bergelson and D. Swingley, "At 6–9 Months, Human Infants Know the Meanings of Many Common Nouns," *Proceedings of the National Academy of Sciences* 109 (2012): 3253–3258; E. Bergelson and R. N. Aslin, "Nature and Origins of the Lexicon in 6-Mo-Olds," *Proceedings of the National Academy of Sciences* 114 (2017): 12916–12921.

11. Z. Káldy and N. Sigala, "The Neural Mechanisms of Object Working Memory: What Is Where in the Infant Brain?" *Neuroscience and Biobehavioral Reviews* 28 (2004): 113–121; K. Grill-Spector, G. Golarai, and J. Gabrieli, "Developmental Neuroimaging of the Human Ventral Visual Cortex," *Trends in Cognitive Sciences* 12 (2008): 152–162; L. B. Smith, "It's All Connected: Pathways in Visual Object Recognition and Early Noun Learning," *American Psychologist* 68 (2013): 618.

12. A. Qiu, S. Mori, and M. I. Miller, "Diffusion Tensor Imaging for Understanding Brain Development in Early Life," *Annual Review of Psychology* 66 (2015): 853–876.

13. M. Frank, M., Braginsky, D. Yurovsky and V. Marchman, *Variability and Consistency in Early Language Learning: The Wordbank Project* (Cambridge, MA: MIT Press, 2021). See also http://wordbank.stanford.edu.

14. J. Franck, S. Millotte, A. Posada, and L. Rizzi, “Abstract Knowledge of Word Order by 19 Months: An Eye-Tracking Study,” *Applied Psycholinguistics* 34 (2013): 323–336.

15. See, among others, G. F. Marcus, S. Vijayan, S. B. Rao, and P. M. Vishton, “Rule Learning by Seven-Month-Old Infants,” *Science* 283 (1999): 77–80; J. Gervain, M. Nespor, R. Mazuka, R. Horiem, and J. Mehler, “Bootstrapping Word Order in Prelexical Infants: A Japanese–Italian Cross-Linguistic Study,” *Cognitive Psychology* 1 (2008): 56–74; J. Gervain, and J. F. Werker, “Prosody Cues Word Order in 7-Month-Old Bilingual Infants,” *Nature Communications* 4 (2013): 1490; S. Benavides-Varela and J. Gervain, “Learning Word Order at Birth: A NIRS Study,” *Developmental Cognitive Neuroscience* 25 (2017): 198–208.

16. S. R. Waxman and D. B. Markow, “Words as Invitations to form Categories: Evidence from 12- to 13-Month-Old Infants,” *Cognitive Psychology* 29 (1995): 257–302; B. Höhle, J. Weissenborn, D. Kiefer, A. Schulz, and M. Schmitz, “Functional Elements in Infants’ Speech Processing: The role of Determiners in the Syntactic Categorization of Lexical Elements,” *Infancy* 5 (2004): 341–353; Y. Kedar, M. Casasola, and B. Lust, “Getting There Faster: 18-and 24-Month-Old Infants’ Use of Function Words to Determine Reference,” *Child Development* 77 (2006): 325–338; A. Christophe, S. Millotte, S Bernal, and J. Lidz, “Bootstrapping Lexical and Syntactic Acquisition,” *Language and Speech* 51 (2008): 61–75.

17. R. Jackendoff, “A Parallel Architecture Perspective on Language Processing,” *Brain Research* 1146 (2007): 2–22.

18. E. Morgan, A. van der Meer, M. Vulchanova, D. E. Blasi, and G. Baggio, “Meaning before Grammar: A Review of ERP Experiments on the Neurodevelopmental Origins of Semantic Processing,” *Psychonomic Bulletin and Review* 27 (2020): 441–464.

Chapter 6

1. F. Grosjean, "Neurolinguists, Beware! The Bilingual Is Not Two Monolinguals in One Person," *Brain and Language* 36 (1989): 3–15.
2. F. Grosjean, *Studying Bilinguals* (New York: Oxford University Press, 2008).
3. E. L Newport, D. Bavelier, and H. J. Neville, "Critical Thinking about Critical Periods: Perspectives on a Critical Period for Language Acquisition," in *Language, Brain and Cognitive Development: Essays in Honor of Jacques Mehler*, ed. E. Dupoux, 481–502 (Cambridge, MA: MIT Press, 2001); M. S. Thomas and M. H. Johnson, "New Advances in Understanding Sensitive Periods in Brain Development," *Current Directions in Psychological Science* 17 (2008): 1–5.
4. E. I. Knudsen, "Sensitive Periods in the Development of the Brain and Behavior," *Journal of Cognitive Neuroscience* 16 (2004): 1412–1425; H. B. Uylings, "Development of the Human Cortex and the Concept of 'Critical' or 'Sensitive' Periods," *Language Learning* 56 (2006): 59–90.
5. Newport et al. "Critical Thinking about Critical Periods"; H. J. Neville, H. J., D. L. Mills, and D. S. Lawson, "Fractionating Language: Different Neural Subsystems with Different Sensitive Periods," *Cerebral Cortex* 2 (1992): 244–258; R. Slabakova, "Is There a Critical Period for Semantics?" *Second Language Research* 22 (2006): 302–338; J. Rothman, "Why All Counter-Evidence to the Critical Period Hypothesis in Second Language Acquisition Is Not Equal or Problematic," *Language and Linguistics Compass* 2 (2008): 1063–1088.
6. Thomas and Johnson, "New Advances in Understanding"; A. E. Hernandez, *The Bilingual Brain* (New York: Oxford University Press, 2013); S. Sulpizio, N. Del Maschio, D. Fedeli, and J. Abutalebi, "Bilingual Language Processing: A Meta-Analysis of Functional Neuroimaging Studies," *Neuroscience and Biobehavioral Reviews* 108 (2020): 834–853.
7. L. Osterhout, A. Poliakov, K. Inoue, J. McLaughlin, G. Valentine, I. Pitkanen, C. Frenck-Mestre, and J. Hirschensohn, "Second-Language Learning and Changes in the Brain," *Journal of Neurolinguistics* 21 (2008): 509–521.
8. D. Klein, K. Mok, J. K. Chen, and K. E. Watkins, "Age of Language Learning Shapes Brain Structure: A Cortical Thickness Study of Bilingual and Monolingual Individuals," *Brain and Language* 131 (2014): 20–24.
9. Sulpizio et al., "Bilingual Language Processing."
10. C. D. Martin, B. Dering, E. M. Thomas, and G. Thierry, "Brain Potentials Reveal Semantic Priming in Both the 'Active' and the 'Non-Attended' Language of Early Bilinguals," *NeuroImage* 47 (2009): 326–333.
11. J. Abutalebi and D. Green, "Bilingual Language Production: The Neurocognition of Language Representation and Control," *Journal of Neurolinguistics* 20 (2007): 242–275; Hernandez, *The Bilingual Brain*; C. Reverberi, A. K. Kuhlen, S. Seyed-Allaei, R. S. Greulich, A. Costa, J. Abutalebi, and J. D. Haynes, "The Neural Basis of Free Language Choice in Bilingual Speakers: Disentangling Language Choice and Language Execution," *NeuroImage* 177 (2018): 108–116.
12. A. S. Dick, N. L. Garcia, S. M. Pruden, W. K. Thompson, S. W. Hawes, M. T. Sutherland, M. C. Riedel, A. R. Laird, and R. Gonzalez, "No Evidence for a Bilingual Executive Function Advantage in the ABCD Study," *Nature Human Behaviour* 3 (2019): 692–701.
13. E. Bialystok, F. I. Craik, and G. Luk, "Bilingualism: Consequences for Mind and Brain," *Trends in Cognitive Sciences* 16 (2012): 240–250.

Chapter 7

1. UNESCO Institute for Statistics global databases, 2019.
2. S. Dehaene, *Reading in the Brain: The New Science of How We Read* (New York: Penguin, 2009).
3. Hemianopia is a loss of vision in half of the visual field. Because incoming light is projected by the lens of each eye such that visual field images are inverted, the left half of each retina receives light from the right visual field, and vice versa. The left half of each retina is connected to the left occipital (visual) cortex, and the right halves are wired up to the right occipital cortex. A right hemianopia results from damage to the left visual cortex, and vice versa for a left hemianopia. See N. Geschwind, "The Anatomy of Acquired Disorders of Reading," in *Selected Papers on Language and the Brain*, ed. Norman Geschwind, 1–17 (Berlin: Springer, 1974).
4. W. J. M. Levelt, *A History of Psycholinguistics: The Pre-Chomskyan Era* (New York: Oxford University Press, 2012.
5. Dehaene, *Reading in the Brain*.
6. The posterior lesion is most likely to produce a generalized impairment of reading, as this area is involved in processing all types of letter-like stimuli: if there is damage to the posterior fusiform gyrus, signals from occipital cortex cannot access abstract orthographic representations. The anterior lesion will instead damage only or primarily n-gram or word representations, sparing the patient's ability to recognize letters and engage in letter-by-letter reading.
7. S. Epelbaum, P. Pinel, R. Gaillard, C. Delmaire, M. Perrin, S. Dupont, S. Dehaene, and L. Cohen, "Pure Alexia as a Disconnection Syndrome: New Diffusion Imaging Evidence for an Old Concept," *Cortex* 44 (2008): 962–974.
8. J. V. Baldo, N. Kacinik, C. Ludy, S. Paulraj, A. Moncrief, V. Piai, B. Curran, T. Herron, and N. F. Dronkers, "Voxel-Based Lesion Analysis of Brain Regions Underlying Reading and Writing," *Neuropsychologia* 115 (2018): 51–59.
9. Dehaene, *Reading in the Brain*.
10. B. D. McCandliss, L. Cohen, and S. Dehaene, "The Visual Word Form Area: Expertise for Reading in the Fusiform Gyrus," *Trends in Cognitive Sciences* 7 (2003): 293–299.
11. S. Dehaene, S., F. Pegado, L. W. Braga, P. Ventura, G. Nunes Filho, A. Jobert, G. Dehaene-Lambertz, R. Kolinsky, J. Morais, and L. Cohen, "How Learning to Read Changes the Cortical Networks for Vision and Language," *Science* 330 (2010): 1359–1364.
12. S. Dehaene, L. Cohen, J., Morais, and R. Kolinsky, "Illiterate to Literate: Behavioural and Cerebral Changes Induced by Reading Acquisition," *Nature Reviews Neuroscience* 16 (2015): 234–244.
13. W. P. Morgan, "A Case of Congenital Word Blindness," *British Medical Journal* 2 (1896): 1378.
14. G. Silani, G., U. Frith, J. F. Demonet, F. Fazio, D. Perani, C. Price, C. D. Frith, and E. Paulesu, "Brain Abnormalities Underlying Altered Activation in Dyslexia: A Voxel Based Morphometry Study," *Brain* 128 (2005): 2453–2461; U. Kuhl, N. E. Neef, I. Kraft, G. Schaadt, L. Dörr, J. Brauer, I. Czepezauer, B. Müller, A. Wilcke, H. Kirsten, F. Emmrich, J. Boltze, A. D. Friederici, and M. A. Skeide, "The Emergence of Dyslexia in the Developing Brain," *NeuroImage* 211 (2020): 116633.

Chapter 8

1. M. C. Dewan, A. Rattani, S. Gupta, R. E. Baticulon, Y. C. Hung, M. Punchak, A. Agrawal, A. O. Adeleye, M. G. Shrime, A. M. Rubiano, and J. V. Rosenfeld, "Estimating the Global Incidence of Traumatic Brain Injury," *Journal of Neurosurgery* 130 (2018), 1080–1097; M. P. Lindsay, B. Norrving, R. L. Sacco, M. Brainin, W. Hacke, S. Martins, J. Pandian, and V. Feigin, "World Stroke Organization (WSO): Global Stroke Fact Sheet" (2019); S. T. Engelter, M. Gostynski, S. Papa, M. Frei, C. Born, V. Ajdacic-Gross, F. Gutzwiller, and P. A. Lyrer, "Epidemiology of Aphasia Attributable to First Ischemic Stroke: Incidence, Severity, Fluency, Etiology, and Thrombolysis," *Stroke* 37 (2006): 1379–1384.

2. W. J. M. Levelt, *A History of Psycholinguistics: The Pre-Chomskyan Era* (New York: Oxford University Press, 2012; D. Vandenborre, E. Visch-Brink, and P. Mariën, "The Development of Modern Approaches to Aphasia: A Concise Overview," *International Journal of Rehabilitation Research* 38 (2015): 189–194; N. F. Dronkers, M. V. Ivanova, and J. V. Baldo, "What Do Language Disorders Reveal about Brain–Language Relationships? From Classic Models to Network Approaches," *Journal of the International Neuropsychological Society* 23 (2017): 741–754.

3. H. H. J. Kolk, "Multiple Route Plasticity," *Brain and Language* 71 (2000): 129–131; H. Duffau, "The Error of Broca: From the Traditional Localizationist Concept to a Connectomal Anatomy of Human Brain," *Journal of Chemical Neuroanatomy* 89 (2018): 73–81.

4. Engelter et al., "Epidemiology of Aphasia."

5. J. Fridriksson, J., D. B. den Ouden, A. E. Hillis, G. Hickok, C. Rorden, A. Basilakos, G. Yourganov, and L. Bonilha, "Anatomy of Aphasia Revisited," *Brain* 141 (2018): 848–862.

6. M. C. Tate, G. Herbet, S. Moritz-Gasser, J. E. Tate, and H. Duffau, "Probabilistic Map of Critical Functional Regions of the Human Cerebral Cortex: Broca's Area Revisited," *Brain* 137 (2014): 2773–2782.

7. M. Mesulam, "Primary Progressive Aphasia," *Annals of Neurology* 49 (2001): 425–432.

8. J. D. Stefaniak, A. D. Halai, and M. A. L. Ralph, "The Neural and Neurocomputational Bases of Recovery from Post-Stroke Aphasia," *Nature Reviews Neurology* 16 (2019): 43–55.

9. D. Saur and G. Hartwigsen, "Neurobiology of Language Recovery after Stroke: Lessons from Neuroimaging Studies," *Archives of Physical Medicine and Rehabilitation* 93 (2012): S15–S25.

10. A. Stockert, M. Wawrzyniak, J. Klingbeil, K. Wrede, D. Kümmerer, G. Hartwigsen, C. P. Kaller, C. Weiller, and D. Saur, "Dynamics of Language Reorganization after Left Temporo-Parietal and Frontal Stroke," *Brain* 143 (2020): 844–861.

11. Y. Grodzinsky, *Theoretical Perspectives on Language Deficits* (Cambridge, MA: MIT Press, 1990). For an overview and further discussion, see J. Druks, *Contemporary and Emergent Theories of Agrammatism: A Neurolinguistic Approach* (London: Routledge/Psychology Press, 2016).

12. A. Caramazza and E. B. Zurif, "Dissociation of Algorithmic and Heuristic Processes in Language Comprehension: Evidence from Aphasia," *Brain and Language* 3 (1976): 572–582.

13. For a recent hypothesis on the division of labor between frontal and temporal regions, see W. Matchin and G. Hickok, "The Cortical Organization of Syntax," *Cerebral Cortex* 30 (2020): 1481–1498.

14. P. Hagoort, M. Wassenaar, and C. Brown, "Real-Time Semantic Compensation in Patients with Agrammatic Comprehension: Electrophysiological Evidence for Multiple-Route Plasticity," *Proceedings of the National Academy of Sciences* 100 (2003): 4340–4345.

15. For a discussion, see Druks, *Contemporary and Emergent Theories*.

Chapter 9

1. C. D. Darlington, "The Genetic Component of Language," *Heredity* 1 (1947): 269–286; L. Jenkins, "Language and Genetics," *Theoretical Linguistics* 5 (1978): 77–82; R. C. Berwick and N. Chomsky, "Why Only Us: Recent Questions and Answers," *Journal of Neurolinguistics* 43 (2017): 166–177.
2. For an excellent and readable introduction to human genetics, see S. Mukherjee, *The Gene: An Intimate History* (New York: Penguin Books, 2016).
3. K. L. Grasby et al., "The Genetic Architecture of the Human Cerebral Cortex," *Science* 367, no. 6484 (2020): eeaa6690. For background, see S. E. Fisher and S. C. Vernes, "Genetics and the Language Sciences," *Annual Review of Linguistics* 1, no. 1 (2015): 289–310; S. A Graham and S. E. Fisher, "Understanding Language from a Genomic Perspective," *Annual Review of Genetics* 49 (2015): 131–160.
4. C. S. Lai, S. E. Fisher, J. A. Hurst, F. Vargha-Khadem, and A. P. Monaco, "A Forkhead-Domain Gene Is Mutated in a Severe Speech and Language Disorder," *Nature* 413 (2001): 519–523.
5. The articles can be retrived at http://news.bbc.co.uk/2/hi/sci/tech/1575471.stm and http://news.bbc.co.uk/2/hi/science/nature/2192969.stm.
6. S. E. Fisher, "Human Genetics: The Evolving Story of FOXP2," *Current Biology* 29 (2019): R65–R67.
7. For early studies, see K. E. Watkins, F. Vargha-Khadem, J. Ashburner, R. E. Passingham, A. Connelly, K. J. Friston, R. S. Frackowiak, M. Mishkin, and D. G. Gadian, "MRI Analysis of an Inherited Speech and Language Disorder: Structural Brain Abnormalities," *Brain* 125 (2002): 465–478; F. Liégeois, T. Baldeweg, A. Connelly, D. G. Gadian, M. Mishkin, and F. Vargha-Khadem, "Language fMRI Abnormalities Associated with FOXP2 Gene Mutation," *Nature Neuroscience* 6 (2003): 1230–1237.
8. M. B. Johnson, Y. I., Kawasawa, C. E. Mason, Ž. Krsnik, G. Coppola, D. Bogdanović, D. H. Geschwind, S. M. Mane, M. W. State, and N. Šestan, "Functional and Evolutionary Insights into Human Brain Development through Global Transcriptome Analysis," *Neuron* 62 (2009): 494–509.
9. S. C. Vernes, D. F. Newbury, B. S. Abrahams, L. Winchester, J. Nicod, M. Groszer, M. Alarcón, P. L. Oliver, K. E. Davies, D. H. Geschwind, and A. P. Monaco, "A Functional Genetic Link between Distinct Developmental Language Disorders," *New England Journal of Medicine* 359 (2008): 2337–2345; S. A. Graham, P. Deriziotis, and S. E. Fisher, "Insights into the Genetic Foundations of Human Communication," *Neuropsychology Review* 25 (2015): 3–26; P. Deriziotis and S. E. Fisher, "Speech and Language: Translating the Genome," *Trends in Genetics* 33 (2017): 642–656.
10. P. Pinel, F. Fauchereau, A. Moreno, A. Barbot, M. Lathrop, D. Zelenika, D. Le Bihan, J. B. Poline, T. Bourgeron, and S. Dehaene, "Genetic Variants of FOXP2 and KIAA0319/TTRAP/THEM2 Locus Are Associated with Altered Brain Activation in Distinct Language-Related Regions," *Journal of Neuroscience* 32 (2012): 817–825; M. Hoogman, T. Guadalupe, M. P. Zwiers, P. Klarenbeek, C. Francks, and S. E. Fisher, "Assessing the Effects of Common Variation in the FOXP2 Gene on Human Brain Structure," *Frontiers in Human Neuroscience* 8 (2014): 473; S. A. Graham and S. E. Fisher, " Understanding Language from a Genomic Perspective," *Annual Review of Genetics* 49 (2015): 131–160; J. Uddén, A. Hultén, K. Bendtz, Z. Mineroff, K. S. Kucera, A. Vino, E. Fedorenko, P. Hagoort, and S. E. Fisher, "Toward Robust Functional Neuroimaging Genetics of Cognition," *Journal of Neuroscience* 39 (2019): 8778–8787.

11. S. E. Fisher and J. C. DeFries, "Developmental Dyslexia: Genetic Dissection of a Complex Cognitive Trait," *Nature Reviews Neuroscience* 3 (2002): 767–780.

12. J. Williams and M. C. O'Donovan, "The Genetics of Developmental Dyslexia," *European Journal of Human Genetics* 14 (2006): 681–689.

13. A. L. Giraud and F. Ramus, "Neurogenetics and Auditory Processing in Developmental Dyslexia," *Current Opinion in Neurobiology* 23 (2013): 37–42.

14. T. M. Centanni, A. B. Booker, F. Chen, A. M. Sloan, R. S. Carraway, R. L. Rennaker, J. J. LoTurco, and M . P. Kilgard, "Knockdown of Dyslexia-Gene Dcdc2 Interferes with Speech Sound Discrimination in Continuous Streams," *Journal of Neuroscience* 36 (2016): 4895–4906.

15. A. M. Galaburda, J. LoTurco, F. Ramus, R. H. Fitch, and D. D. Rosen, "From Genes to Behavior in Developmental Dyslexia," *Nature Neuroscience* 9 (2006): 1213–1217.

16. A. Gialluisi, T. Guadalupe, C. Francks, and S. E. Fisher, "Neuroimaging Genetic Analyses of Novel Candidate Genes Associated with Reading and Language," *Brain and Language* 172 (2017): 9–15.

17. N. Becker, M. Vasconcelos, V. Oliveira, F. C. D. Santos, L. Bizarro, R. M. D. Almeida, J. Fumagalli De Salles, and M. R. S. Carvalho, "Genetic and Environmental Risk Factors for Developmental Dyslexia in Children: Systematic Review of the Last Decade," *Developmental Neuropsychology* 42 (2017): 423–445.

Chapter 10

1. C. I. Petkov and E. Jarvis, “Birds, Primates, and Spoken Language Origins: Behavioral Phenotypes and Neurobiological Substrates,” *Frontiers in Evolutionary Neuroscience* 4 (2012): 12.
2. E. D. Jarvis, “Evolution of Vocal Learning and Spoken Language,” *Science* 366 (2019): 50–54.
3. M. D. Hauser, N. Chomsky, and W. T. Fitch, ‘The Faculty of language: What Is It, Who Has It, and How Did It Evolve?” *Science* 298 (2002): 1569–1579; S. W. Townsend, S. Engesser, S. Stoll, K. Zuberbühler, and B. Bickel, “Compositionality in Animals and Humans,” *PLoS Biology* 16 (2018); K. Zuberbühler, 2020. “Syntax and Compositionality in Animal Communication,” *Philosophical Transactions of the Royal Society B*, 375, no. 1789, 20190062.
4. E. D. Jarvis, “Learned Birdsong and the Neurobiology of Human Language,” *Annals of the New York Academy of Sciences* 1016 (2004): 749; S. Nowicki and W. Searcy, “The Evolution of Vocal Learning,” *Current Opinion in Neurobiology* 28 (2014): 48–53; S. C. Vernes and G. S. Wilkinson, “Behaviour, Biology and Evolution of Vocal Learning in Bat,” *Philosophical Transactions of the Royal Society B* 375 (2020): 20190061.
5. Jarvis, “Learned Birdsong.”
6. S. Haesler, K. Wada, A. Nshdejan, E. E. Morrisey, T. Lints, E. D. Jarvis, and C. Scharff, “FoxP2 Expression in Avian Vocal Learners and Non-Learners,” *Journal of Neuroscience* 24 (2004): 3164–3175; S. Haesler, C. Rochefort, B. Georgi, P. Licznerski, P. Osten, and C. Scharff, “Incomplete and Inaccurate Vocal Imitation after Knockdown of FoxP2 in Songbird Basal Ganglia Nucleus Area X,” *PLoS Biology* 5 (2007).
7. T. Q. Gentner, K. M. Fenn, D. Margoliash, and H. Nusbaum, “Recursive Syntactic Pattern Learning by Songbirds,” *Nature* 440 (2006): 1204–1207: “We thus demonstrate that starlings can recognize syntactically well-formed strings, including those that use a recursive centre-embedding rule. At least a simple level of recursive syntactic pattern processing is therefore shared with other animals. These results challenge the recent claim that recursion forms the computational core of a uniquely human narrow faculty for language.” For a critical discussion, see C. A. Van Heijningen, J. De Visser, W. Zuidema, and C. Ten Cate, “Simple Rules Can Explain Discrimination of Putative Recursive Syntactic Structures by a Songbird Species,” *Proceedings of the National Academy of Sciences* 106 (2009): 20538–20543.
8. M. S. Ficken and J. W. Popp, “Syntactical Organization of the Gargle Vocalization of the Black-Capped Chickadee,” *Parus atricapillus*. *Ethology* 91 (1992): 156–168.
9. C. N. Templeton, E. Greene, and K. Davis, “Allometry of Alarm Calls: Black-Capped Chickadees Encode Information about Predator Size,” *Science* 308 (2005): 1934–1937.
10. Compositionality of calls or songs is currently not attested in songbirds. For a claim that it might be, see S. Engesser, A. R. Ridley, and S. W. Townsend, “Meaningful Call Combinations and Compositional Processing in the Southern Pied Babbler,” *Proceedings of the National Academy of Sciences* 113 (2016): 5976–5981. However, there is also some evidence that their calls are combinatorial: meaningful yet constituted of meaningless elements, a bit like words and morphemes in relation to their constituent sounds. See S. Engesser, J. L. Holub, L. G. O’Neill, A. F. Russell, and S. W. Townsend, “Chestnut-Crowned Babbler Calls Are Composed of Meaningless Shared Building Blocks,” *Proceedings of the National Academy of Sciences* 116 (2019): 19579–19584.
11. J. P. Rauschecker and S. K. Scott, “Maps and Streams in the Auditory Cortex: Nonhuman Primates Illuminate Human Speech Processing,” *Nature Neuroscience* 12 (2009): 718–724; I. Bornkessel-Schlesewsky, M. Schlesewsky, S. L. Small, and J. P. Rauschecker, “Neurobiological

Roots of Language in Primate Audition: Common Computational Properties," *Trends in Cognitive Sciences* 19 (2015): 142–150; F. Balezeau, B. Wilson, G. Gallardo, F. Dick, W. A. Hopkins, A. Anwander, A. D. Friederici, T. D. Griffiths, and C. I. Petkov, "Primate Auditory Prototype in the Evolution of the Arcuate Fasciculus," *Nature Neuroscience* 23 (2020): 1–4.

12. R. M. Seyfarth and D. L. Cheney, "The Origin of Meaning in Animal Signals," *Animal Behaviour* 124 (2017): 339–346.

13. R. Gil-da-Costa, A. Martin, M. A. Lopes, M. Munoz, J. B. Fritz, and A. R. Braun, "Species-Specific Calls Activate Homologs of Broca's and Wernicke's Areas in the Macaque," *Nature Neuroscience* 9 (2006): 1064–1070; J. P. Taglialatela, J. L. Russell, J. A. Schaeffer, and W. D. Hopkins, "Communicative Signaling Activates 'Broca's' Homolog in Chimpanzees," *Current Biology* 18 (2008): 343–348.

14. K. Ouattara, A. Lemasson, and K. Zuberbühler, "Campbell's Monkeys Concatenate Vocalizations into Context-Specific Call Sequences," *Proceedings of the National Academy of Sciences* 106 (2009): 22026–22031.

15. K. Arnold, and K. Zuberbühler, "Call Combinations in Monkeys: Compositional or Idiomatic Expressions?" *Brain and Language* 120 (2012): 303–309; K. Zuberbühler and A. Lemasson, "Primate Communication: Meaning from Strings of Calls," in *Language and Recursion*, ed. Francis Lowenthal and Laurent Lefebvre, 115–125 (New York: Springer, 2014); P. Schlenker, E. Chemla, and K. Zuberbühler, "Semantics and Pragmatics of Monkey Communication," in *Oxford Research Encyclopedia of Linguistics*, ed. Mark Aronoff (Oxford: Oxford University Press, 2017); K. Zuberbühler, "Combinatorial Capacities in Primates," *Current Opinion in Behavioral Sciences* 21 (2018): 161–169.

16. Balezeau et al., "Primate Auditory Prototype."

Chapter 11

1. A. L. Giraud and D. Poeppel, "Cortical Oscillations and Speech Processing: Emerging Computational Principles and Operations," *Nature Neuroscience* 15 (2012): 511–517.
2. G. Baggio, P. Cherubini, D. Pischedda, A. Blumenthal, J. D. Haynes, and C. Reverberi, "Multiple Neural Representations of Elementary Logical Connectives," *NeuroImage* 135 (2016): 300–310.
3. J. Holler and S. C. Levinson, "Multimodal Language Processing in Human Communication," *Trends in Cognitive Sciences* 23 (2019): 639–652; F. Mollica, M. Siegelman, E. Diachek, S. T. Piantadosi, Z. Mineroff, R. Futrell, H. Kean, P. Qian, and E. Fedorenko, "Composition Is the Core Driver of the Language-Selective Network," *Neurobiology of Language* 1, no. 1 (2020): 104–134.
4. M. H. Christiansen and N. Chater, *Creating Language: Integrating Evolution, Acquisition, and Processing* (Cambridge, MA: MIT Press, 2016).

Further Readings

aggio, G. *Meaning in the Brain*. Cambridge, MA: MIT Press, 2018.

ehaene, S. *Reading in the Brain: The New Science of How We Read*. New York: Penguin Books, 2010.

mbick, D., and D. Poeppel. "Towards a Computational(ist) Neurobiology of Language: Correlational, Integrated and Explanatory Neurolinguistics." *Language, Cognition and Neuroscience* 30 (2015): 357–366.

riederici, A. *Language in Our Brain. The Origins of a Uniquely Human Capacity*. Cambridge, MA: MIT Press, 2017.

riederici, A. D., N. Chomsky, R. C. Berwick, A. Moro, and J. J. Bolhuis. "Language, Mind and Brain." *Nature Human Behaviour* 1 (2017): 713–722.

agoort, P., ed. *Human Language. From Genes to Brain and Behavior*. Cambridge, MA: MIT Press, 2019.

ickok, G., and S. L. Small, eds. *Neurobiology of Language*. Orlando, FL: Academic Press, 2015.

gram, J. C. *Neurolinguistics: An Introduction to Spoken Language Processing and Its Disorders*. Cambridge: Cambridge University Press, 2007.

emmerer, D. *Cognitive Neuroscience of Language*. London: Psychology Press, 2014.

evelt, W. J. M. *A History of Psycholinguistics: The Pre-Chomskyan Era*. New York: Oxford University Press, 2012.

ylkkänen, L. "The Neural Basis of Combinatory Syntax and Semantics." *Science* 366 (2019): 62–66.

temmer, B., and H. A. Whitaker. *Handbook of Neurolinguistics*. Amsterdam: Elsevier, 1998.

Index

The MIT Press Essential Knowledge Series

AI Assistants, Roberto Pieraccini
AI Ethics, Mark Coeckelbergh
Algorithms, Panos Louridas
Anticorruption, Robert I. Rotberg
Annotation, Remi H. Kalir and Antero Garcia
Auctions, Timothy P. Hubbard and Harry J. Paarsch
Behavioral Insights, Michael Hallsworth and Elspeth Kirkman
Biofabrication, Rita Raman
The Book, Amaranth Borsuk
Carbon Capture, Howard J. Herzog
Citizenship, Dimitry Kochenov
Cloud Computing, Nayan B. Ruparelia
Collaborative Society, Dariusz Jemielniak and Aleksandra Przegalinska
Computational Thinking, Peter J. Denning and Matti Tedre
Computing: A Concise History, Paul E. Ceruzzi
The Conscious Mind, Zoltan E. Torey
Content, Kate Eichhorn
Contraception: A Concise History, Donna J. Drucker
Critical Thinking, Jonathan Haber
Crowdsourcing, Daren C. Brabham
Cybersecurity, Duane Wilson
Cynicism, Ansgar Allen
Data Science, John D. Kelleher and Brendan Tierney
Death and Dying, Nicole Piemonte and Shawn Abreu
Deconstruction, David J. Gunkel
Deep Learning, John D. Kelleher
Extraterrestrials, Wade Roush
Extremism, J. M. Berger

Fake Photos, Hany Farid
fMRI, Peter A. Bandettini
Food, Fabio Parasecoli
Free Will, Mark Balaguer
The Future, Nick Montfort
GPS, Paul E. Ceruzzi
Haptics, Lynette A. Jones
Hate Speech, Caitlin Ring Carlson
Information and Society, Michael Buckland
Information and the Modern Corporation, James W. Cortada
Intellectual Property Strategy, John Palfrey
The Internet of Things, revised and updated edition, Samuel Greengard
Irony and Sarcasm, Roger Kreuz
Ketamine, Bita Moghaddam
Machine Learning, revised and updated edition, Ethem Alpaydın
Machine Translation, Thierry Poibeau
Macroeconomics, Felipe Larraín B.
Memes in Digital Culture, Limor Shifman
Metadata, Jeffrey Pomerantz
The Mind–Body Problem, Jonathan Westphal
MOOCs, Jonathan Haber
Neurolinguistics, Giosuè Baggio
Neuroplasticity, Moheb Costandi
Nihilism, Nolen Gertz
Open Access, Peter Suber
Paradox, Margaret Cuonzo
Phenomenology, Chad Engelland
Post-Truth, Lee McIntyre
Quantum Entanglement, Jed Brody
Recommendation Engines, Michael Schrage
Recycling, Finn Arne Jørgensen

Robots, John Jordan
School Choice, David R. Garcia
Science Fiction, Sherryl Vint
Self-Tracking, Gina Neff and Dawn Nafus
Sexual Consent, Milena Popova
Smart Cities, Germaine R. Halegoua
Spaceflight: A Concise History, Michael J. Neufeld
Spatial Computing, Shashi Shekhar and Pamela Vold
Supernova, Or Graur
Sustainability, Kent E. Portney
Synesthesia, Richard E. Cytowic
The Technological Singularity, Murray Shanahan
3D Printing, John Jordan
Understanding Beliefs, Nils J. Nilsson
Virtual Reality, Samuel Greengard
Visual Culture, Alexis L. Boylan
Waves, Frederic Raichlen

Giosuè Baggio is professor of psycholinguistics at the Norwegian University of Science and Technology in Trondheim. He has studied philosophy, logic, and cognitive neuroscience in Pavia, Strasbourg, Amsterdam, and Nijmegen. Before his current appointment, he worked at the Donders Institute for Brain, Cognition and Behaviour at the Max Planck Institute for Psycholinguistics, and at the International School for Advanced Studies. He is author of several scientific articles and of the book *Meaning in the Brain* (MIT Press, 2018).